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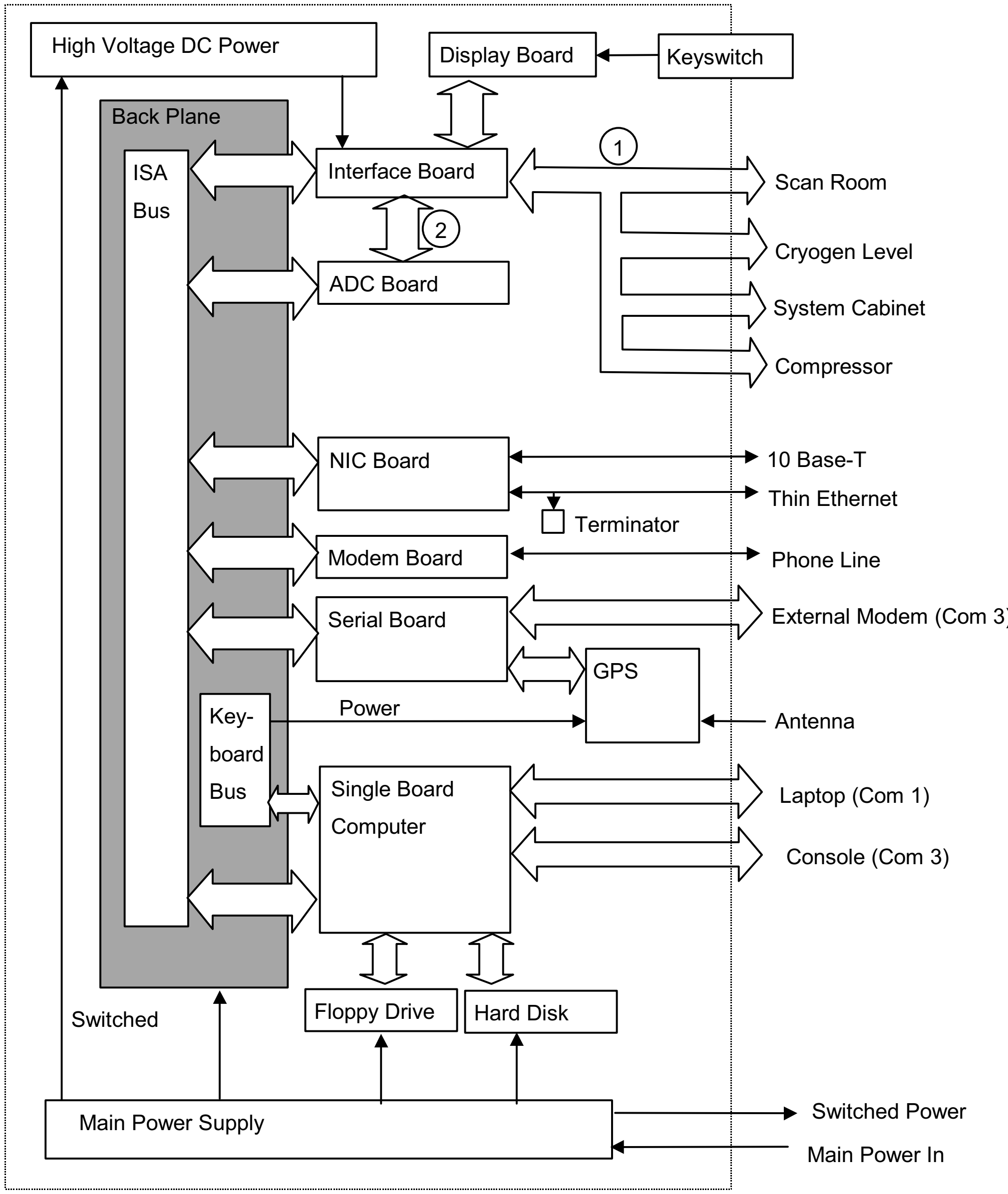
1- BLOCK DIAGRAM AND THEORY

1-1 Magnet Monitor Internal Block Diagram



MAGNET MONITOR INTERNAL BLOCK DIAGRAM
ILLUSTRATION 1-1

1-2 Scope

This section contains specific information related to the Magnet Monitor, “Smart Helium Meter” (SHeM). The information contained herein, when used in addition to that contained in the referenced documents, will provide personnel with sufficient information to Operate, Install, Troubleshoot, and maintain a Magnet Monitor.

IMPORTANT

This manual covers cylindrical magnets only. For the HFO magnet use direction 2277211.

Note

There are two different vendors for the Magnet Monitor. The original vendor was Granite. Granite was replaced by Advantech in October of 2001. This manual reflects the differences between the two different vendors.

1-3 Reference Documents

2229090-100	Magnet Monitor Operators Manual
2230681	Magnet Monitor Hardware Installation Manual
2252032	Magnet Monitor Proprietary Diskette, revision 1.6
2252032-*	Magnet Monitor Proprietary Diskette , revision *
2209000PSP	Magnet Monitor purchase specification

* indicates current revision of magnet monitor software

1-4 System Requirements

The Magnet Monitoring Kit works with the following magnets: S-II, S-III, S-III Magnashield, S-IV, S-V, S-X, S-XC, CX, Max and LCC. Magnet Monitoring requires a Laptop Computer for local viewing of Proprietary level parameters, as well as for configuring the Magnet Monitor. Once installed, set up and connected to the OnLine Center, a laptop is not absolutely necessary for Magnet Monitoring.

1-5 Service Objectives

The purpose of the Magnet Monitoring program is to transform the approach for servicing magnets from “reactive” to “proactive.” Magnet Monitoring has the following main objectives:

- Reduce cryogen consumption
- Provide proactive magnet servicing
- Provide remote access to magnet parameters to assist in problem diagnosis
- Improve cryogen fill schedule and efficiency
- Improve customer up-time
- Help support centers assist customers with chiller issues

1-6 Magnet Monitor Overview

The Magnet Monitor (MM) is a stand-alone computer, with the exception of two signal inputs from the Systems Cabinet, and of course the signal connections to the Magnet, it is independent from the MR System.

To be effective as a 24-hour/day monitor, it is required that the MM receives its main AC power from a different source than the MR system. In some environments it may be advantageous to use a UPS back-up supply, although a UPS is not currently a standard part of the installed MM.

When the MM is initially provided to a clinical site, its installed software is of the Non-Proprietary type. The Proprietary software is to be installed locally by the Field Service Engineer and only in situations where GE has the service contract for the applicable site. In a case where GE loses the service contract for a particular site, the Proprietary software must be removed, and replaced with the Non-Proprietary version.

The Non-Proprietary version of the software provides a reduced set of functional operation. This reduced capability is identified in section 1-6-1, and will not be separated here. With regard to the MM hardware, there is no difference between the two modes, other than the fact that certain measurements are not taken, or controls and functions not activated for the Non-Proprietary case.

Magnet Monitoring consists of hardware and software used in conjunction with the Magnet Monitor to acquire, analyze, and store magnet subsystem data. This data is then transferred via the PPADS System to the various support centers, where it can be reviewed and acted upon.

This section contains the objectives of the Magnet Monitoring program, lists the elements that are monitored, describes the Magnet Monitoring processes, and provides details about the components in the Magnet Monitoring Subsystem.

The Magnet Monitor is microprocessor-based equipment, utilizing a standard PC type computer as a central processing component.

There are several general capabilities built into the unit, and two basic software modes. The Non-Proprietary Software mode provides a subset of the full capability of the Magnet Monitor. For the list below, “(Non-Pro)” indicates a function that is available in the Non-Proprietary mode. All features and capabilities are available in the Proprietary mode.

1-6-1 Monitored Elements and Features

The Magnet Monitoring equipment monitors, or controls, the following parameters:

- Measure and record system parameters
- Helium Level (Non-Pro)
- Up to 4 cryogenic temperatures
    - Coldhead
    - Sleeve
    - Recondenser
    - Shield
- Helium vessel pressure (Non-Pro)
- Various Compressor parameters
    - Helium temperature high
    - Helium pressure low
    - Compressor on
    - 24 VDC present
    - Fuse off
    - Cooling water flow
    - Water temperature
    - Compressor run time
- Magnetic field detection
- Helium fill in progress
- RF present
- MR System Powered up
- Magnet heater duty cycle
- Control certain Magnet properties/functions
    - Helium vessel heater (Non-Pro)
    - Automatic Compressor Reset (Non-Pro)
    - Manual Compressor Reset (via laptop)

- Maintain internal data base
- Generate alarms for out-of-limit conditions
- Automatically call GE OnLine Center to report alarm conditions
- Receive calls from the OnLine Center and report status
- Allow local access with a laptop computer (GE Employees only)
- Review measurement status
- View graphic or tabulated data
- Change set limits
- Generate local compressor reset
- Generate cryogen level reading
- Allow logging of critical notes or observations by GE Personnel

The Magnet Monitor is designed to facilitate certain other features, some of which are not currently implemented (indicated by “Future”). The following list shows those features;

- Web browser compatibility (TCP/IP & HTML)
- Global Positioning System (GPS) compatibility (Future)
- Ethernet connectivity (Future)

1-6-2 Magnet Monitoring Processes

The Magnet Monitoring Subsystem has eight primary processes:

- Collect data from each of the sensors.
- Provide automatic compressor reset.
- Control helium vessel pressure for certain magnets.
- Notify the PPADS system when a parameter is outside of its acceptable limits.
- Notify the PPADS system when a fill is performed.
- Send data to the PPADS system to provide trend and historical data.
- Permit local and remote access to current and historic data.
- Permit performance test data transfer to the ASC using APA protocol.

1-6-3 Sensor Data Collection

The Magnet Monitoring Subsystem collects data from three areas.

- The magnet room, where sensors measure the vessel pressure as well as the first and second stage shield cooler temperatures and magnetic field.
- The system cabinet, where an RF Unblank signal and PDU Line Trigger signal are monitored.
- The shield cooler compressor, from which the shield cooler sensor monitors digital status signals (Leybold and Sumitomo compressors only) and measures the flow rate and temperature of the water going into the shield cooler.

1-6-4 Data Transfer to PPADS

After the Magnet Monitoring hardware and software is installed, the subsystem saves the parameter data and calls into the PPADS system every week to place the data in the database. Normal operating data, alarm information, and fill data are all transferred to the PPADS system. This data is gathered by the IRI system on the mainframe and is used for forecasting cryogen orders and deliveries as well as for developing trends of magnet performance.

You can receive this data by sending email to the PPADS System with the appropriate commands.

1-6-5 Alarm Notification

If a parameter goes outside of its low or high limits (alarm), then the Magnet Monitor contacts the PPADS System. The PPADS System transfers alarm data to the appropriate support center. For America’s pole the OLC engineers are assigned the dispatch. In Europe and Asia the OLC engineers are notified of a magnet monitor alarm via e-mail. The engineers would review the data and take the necessary action.
An OLC MR engineer may take any of the following actions:

- Notify the customer that power needs to be restored to the equipment at the site.
- Reset the compressor remotely (if the site has a Sumitomo or Leybold compressor).
- Split the dispatch and assign it to an FE.
- Notify the customer about water supply problems.
- Disable an individual parameter alarm (Alarm Holdoff) for up to nine days while parts are being ordered, the system is being serviced, or the parameter is recovering to normal condition.
- Request trend data from PPADS to see if a part is beginning to fail.
- Call a Field Engineer (FE) directly for more information.

Note

Anyone can receive notification of an alarm via e-mail, by subscribing to sites that have a Magnet Monitor. Reference the appendix on PPADS of this manual to learn how to subscribe.

1-6-6 Fill Notification

If the subsystem detects a fill, then the Magnet Monitor contacts the PPADS System. The PPADS System transfers a copy of the fill data to IRI and also generates a fill report for engineers subscribed to that site.

1-6-7 Magnet Monitoring Components

The Magnet Monitoring Subsystem contains the following components:

Primary Components

- MR Magnet Monitor
- Global Modem
- MR Magnet Monitor Proprietary Software
- RuO Buffer Amplifier
- Water Flow and Temperature Sensor
- Magnetic Reed Switch
- Key for Fill Switch
- TSCC (Twin Systems Cooling Cabinet) used on certain TWIN systems
- TGWC (Twin Gradient Water Cooler) used on certain TWIN systems

Auxiliary Components

- Compressor w/Diagnostic Port
- Helium Probe (p/o magnet subsystem)
- Pressure Transducer (p/o magnet subsystem)
- Shield Temperature Diodes (p/o magnet subsystem)
- Heater Element (p/o LCC magnet subsystem)
- Vacuum sensor(TWIN system)

Optional Components

- Un-interruptable Power Supply (UPS)
- RJ11 Multiplexer (Switch Box)
- Laptop Computer w/Windows 95 or higher

1-6-8 MR Magnet Monitor Software

The Magnet Monitoring software programs are loaded from a 3.5 inch diskette. The data collected by the programs is stored on an internal hard drive. This data is accessible locally using a laptop computer. For further information regarding Magnet Monitor software, refer to those sections.

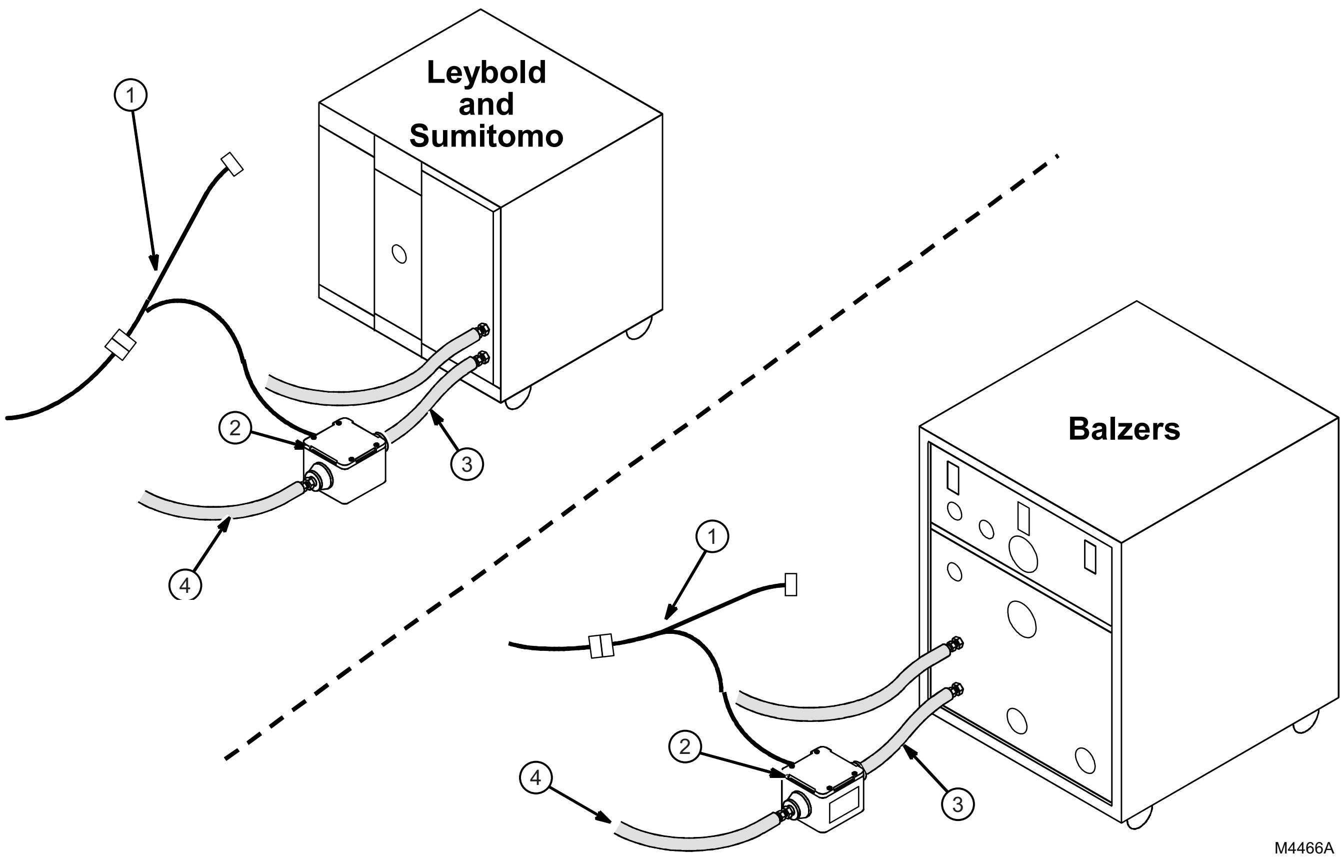
1-6-9 Water Flow and Temperature Sensors

The Water Flow and Temperature Sensors are combined in one package. This device provides a proportional, linear electrical output for both parameters while providing a visual indication for water flow. A new sensor was introduced in Q3, 2002. This sensor has a paddle wheel indicator for flow which cannot be accurately observed.

Specifications

Flow Range: 0-5 gpm or 6.6-17.0 LPM	Pressure Rating: Up to 250 psi or 0.65-15.5 KPA
Temperature Range: 0-100 Deg F or 4.5-24.0 Deg C	Accuracy: ± 4%

Water Flow and Temperature sensors can be used on Sumitomo, Leybold or Balzer compressors. See Illustration 1-3.



WATER FLOW AND TEMPERATURE SENSORS
ILLUSTRATION 1-3

TABLE 1-2
WATER FLOW AND TEMPERATURE SENSORS

No.	Column	Description
1	Compressor Interface	For Sumitomo and Leybold compressors only, via a 9

No.	Column	Description
	Cable	pin sub-D connector.
2	Water Flow & Temp sensor	Water flow is both electronic and visual. Water temperature is by thermocouple.
3	24" Hose	
4	Inlet Water Line	Using a ½ inch hose barb.

1-6-10 TSCC and TGWC Cabinets

The TSCC (Twin Systems Cooling Cabinet) is used to cool the cryo-cooler compressor and the gradient coil. At this time (December, 2002) there is no communication between the TSCC or TGWC and the magnet monitor.

1-6-11 Pressure Transducer

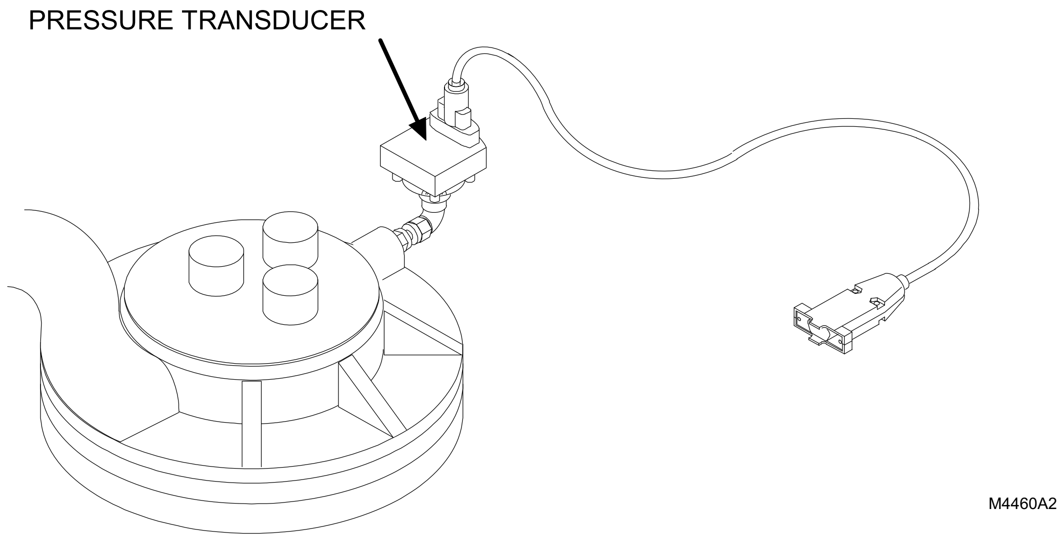
The Pressure Transducer monitors the helium vessel pressure.
<u>Specifications</u>
Range:15 psig
Excitation:12 VDC
Output:1-5 VDC

This transducer aids in assessing:

- Proper maintenance of vessel pressures before, during, and after a helium fill
- Boil-off issues
- Magnet quench
- Pressure control for LCC magnets

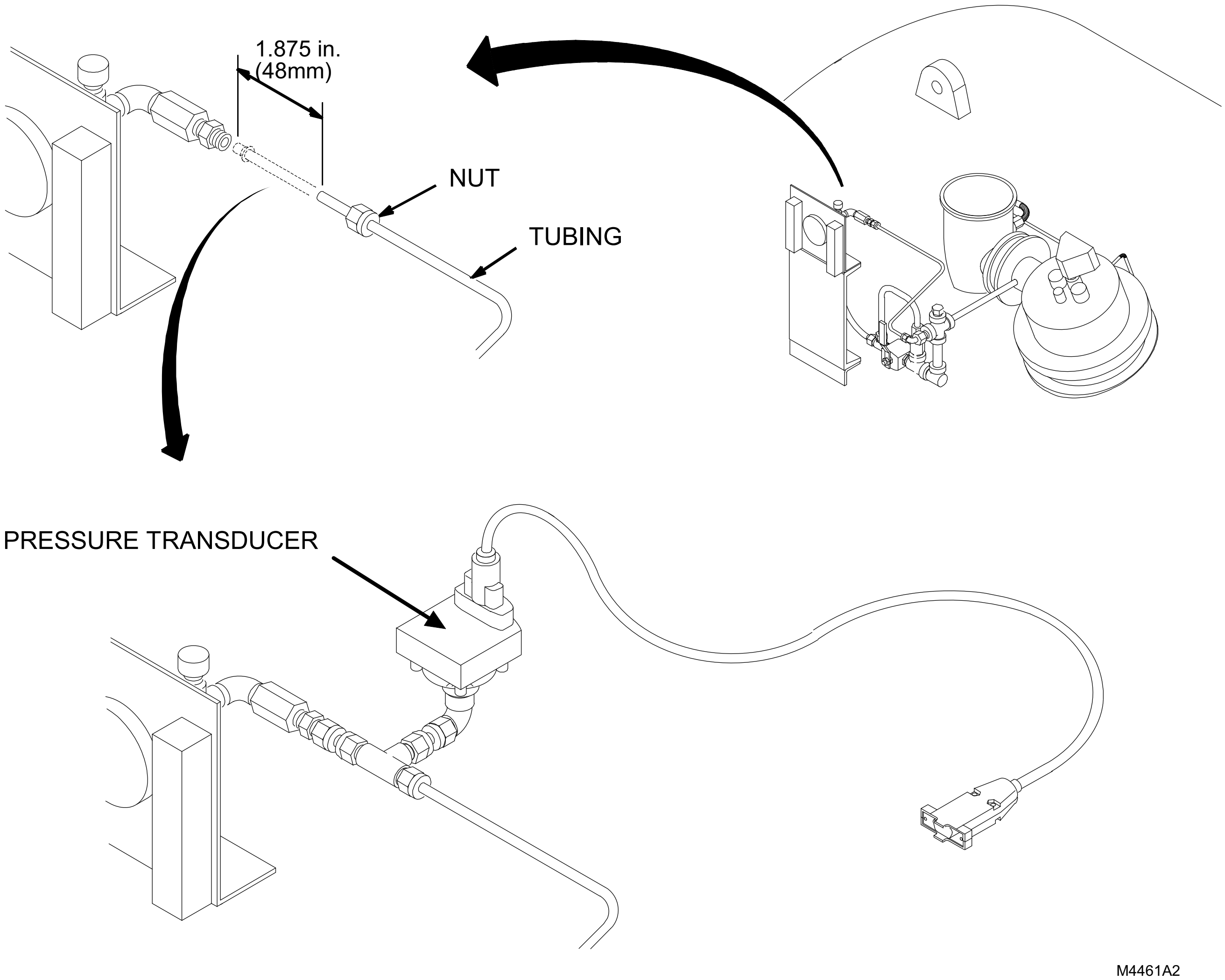
The location of the Transducer varies depending upon the magnet type.

For S-II, S-III, and some MR Max/0.5T Signa Magnets, the transducer fits on the existing capped-off fitting on the side of the service turret plenum. See Illustration 1-4.



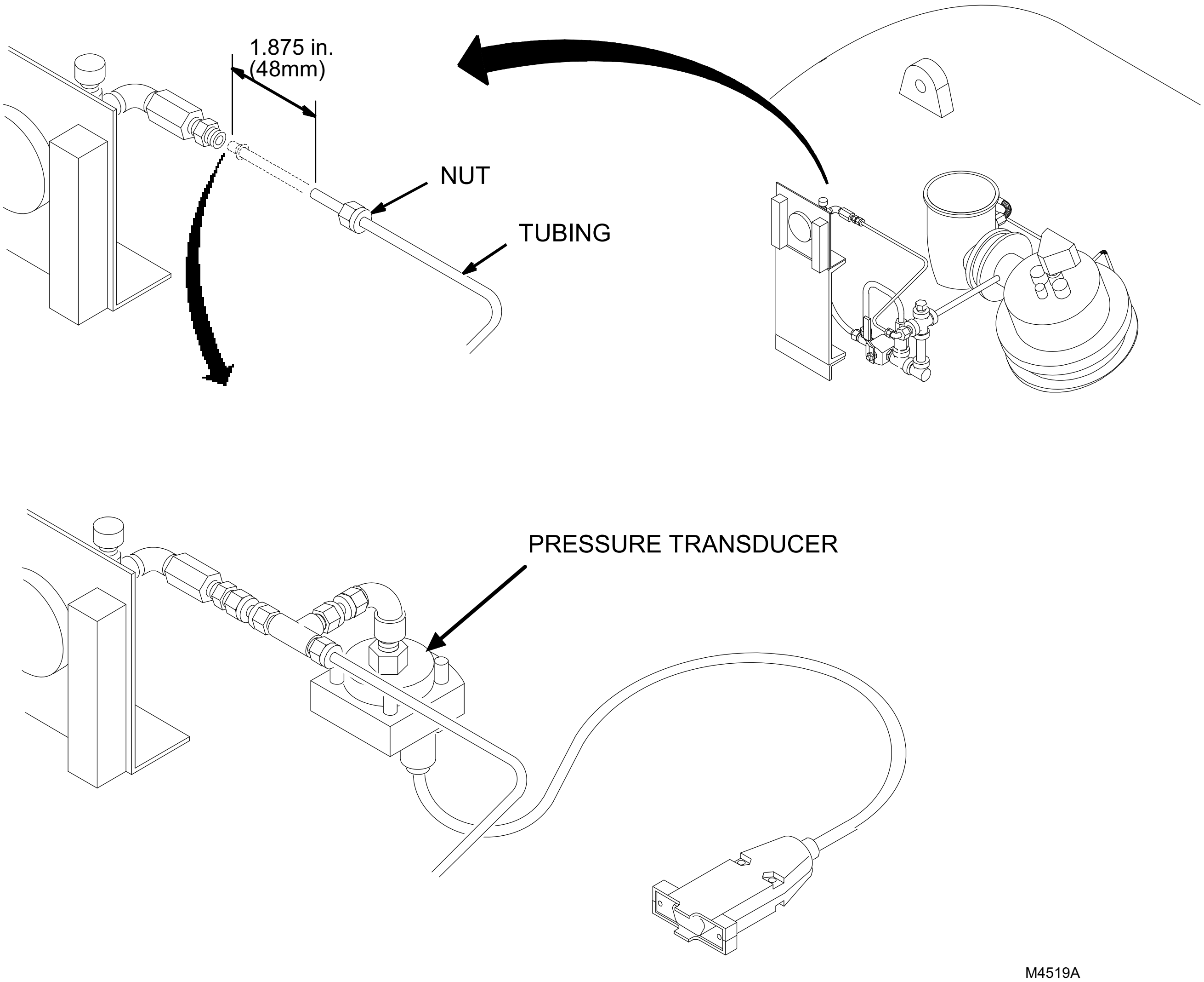
S-II, S-III, S-III MAGNASHIELD, AND 0.5T MR MAX (WITHOUT ALTERNATE VENTING) PRESSURE TRANSDUCER CONNECTIONS
ILLUSTRATION 1-4

For S-IV, S-V, S-X, and S-XC Magnets, the transducer fits on a union tee that is installed into the vent line near the mechanical pressure gauge. See Illustration 1-5.



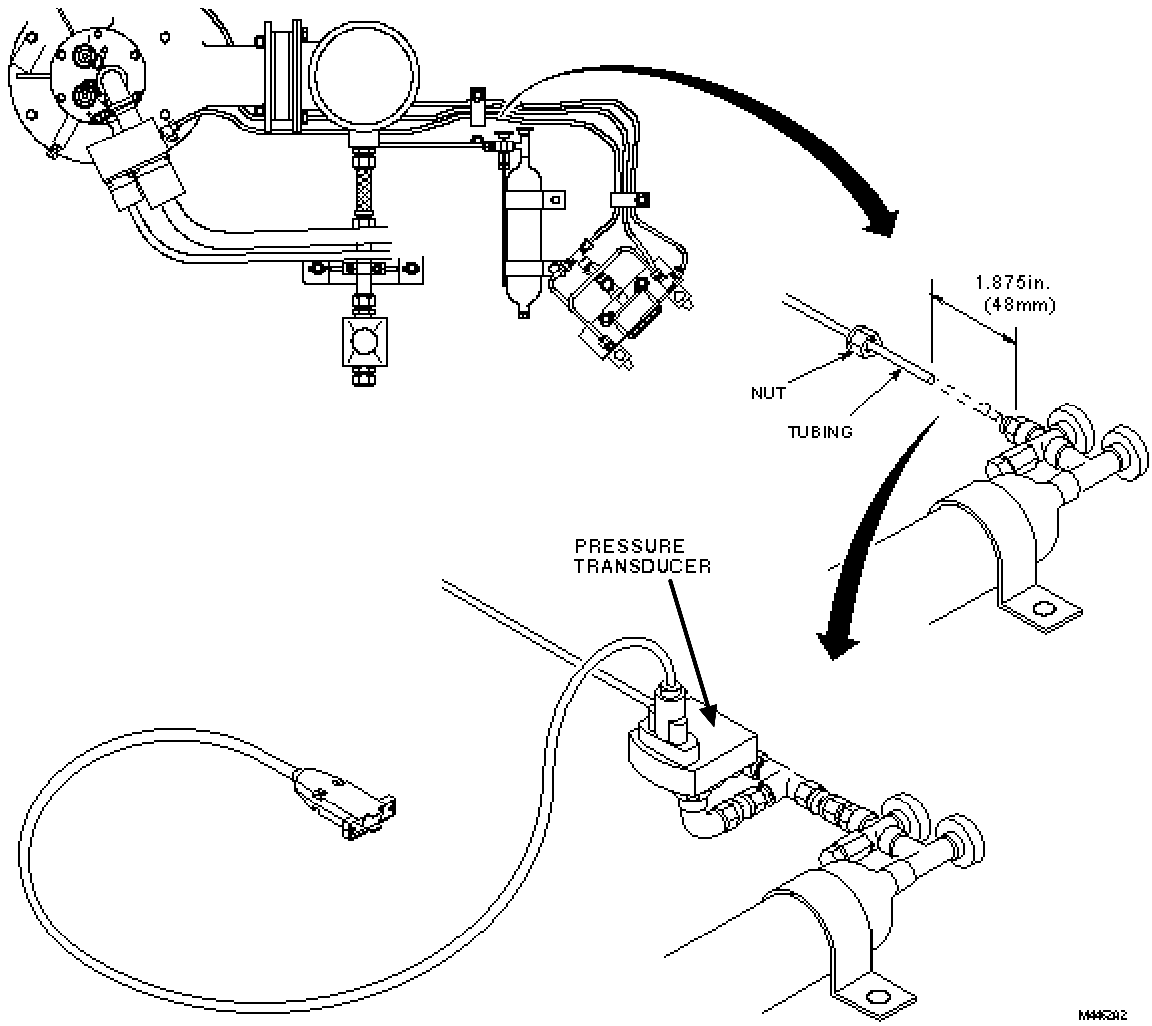
S-IV, S-V, S-X, AND S-XC UNION TEE AND PRESSURE TRANSDUCER CONNECTIONS
ILLUSTRATION 1-5

For C-X Magnets, the transducer fits on a union tee that is installed into the vent line near the mechanical pressure gauge. See Illustration 1-6.



C-X UNION TEE AND PRESSURE TRANSDUCER CONNECTIONS
ILLUSTRATION 1-6

For most MR Max/0.5T Signa Magnets, the transducer fits on a union tee that is installed into the vent line near the TAO valve. See Illustration 1-7.



0.5T UNION TEE AND PRESSURE TRANSDUCER CONNECTIONS
ILLUSTRATION 1-7

1-6-12 Reed Switch

The Magnetic Reed Switch is mounted on the rear pedestal behind the magnet. This switch provides an open contact when subjected to a magnetic field and a closed contact in the absence of a magnetic field. This switch helps the MR engineer determine if there is a loss of the magnetic field.

1-6-13 Key Switch

The Key Switch is mounted in the front panel of the Magnet Monitor. Field engineers and Air Products Technicians use this Switch when a helium fill is performed. Actuating the switch at the beginning of the fill causes the software to create and open a fill data file. This file stores the helium level, vessel pressure, and time/date. When the key is removed at the end of a fill, the file is sent to the PPADS system then on to the IBM Mainframe for cryogen administration purposes.
A key is distributed to each individual responsible for filling magnets. Return key to its storage area.

1-6-14 RJ11 Multiplexer (Switch Box)

On 3.x and 4.x Signa MR Systems, the RJ 11 Multiplexer (switch box) is used to connect multiple imaging systems onto a single phone line. On all other Signa MR Systems, this multiplexer is used to switch between the Magnet Monitor Modem and the InLink Modem on the Signa Computer in addition to connecting multiple imaging systems.

1-6-15 Laptop Computer

The Laptop allows the user to access the Login and Password prompts on the Magnet Monitor, as well as access to the Linux Operating System prompt. Login is required for access to the floppy drive as well as other software-supported features. It is allowable for the Laptop computer to be connected, and logged in, while maintaining communications capability with the OLC.
This connection is accomplished using a Null Modem serial cable between the Laptop and the Magnet Monitor.

1-6-16 Compressor Diagnostic Port

The Sumitomo and Leybold Compressors provide a Diagnostic Port, which is not available on Balzers or CTI shield coolers. This port is used to provide the following status signals and remote function:

- Helium pressure low
- Helium gas temperature high
- Control module On/Off
- Compressor On/Off
- Remote compressor reset

1-6-17 Helium Probe

The Helium Probe is mounted to the magnet cartridge inside the helium vessel. This probe acts like a variable resistor and the output voltage is proportional to the resistance of the probe.
The probe must receive the appropriate drive current for proper operation. For all GE manufactured magnets, the drive current is 75 milliamps. The constant current drive source is located in the Magnet Monitor. This current source is not field adjustable.

1-6-18 Helium Level Readings

The Magnet Monitor automatically takes a Helium Level reading every hour. Readings may be initiated manually by depressing the sample button located on the Magnet Monitor, or by activating a reading via the Laptop Magnet Monitor screens. The OLC can remotely sample the Helium Level.

1-6-19 Magnet Temperature Sensors

Ruthenium Oxide (RuO), SI410 or DT500 cryogenic sensors are located inside the magnet at various locations. These sensors each require a 10 microamp current source for excitation. There are four (4) such current sources within the Magnet Monitor. Each sensor is monitored every minute, the readings are available in the Current Value status screen. These current sources are not field adjustable. Verify accuracy by comparing readings to those of an independent instrument.

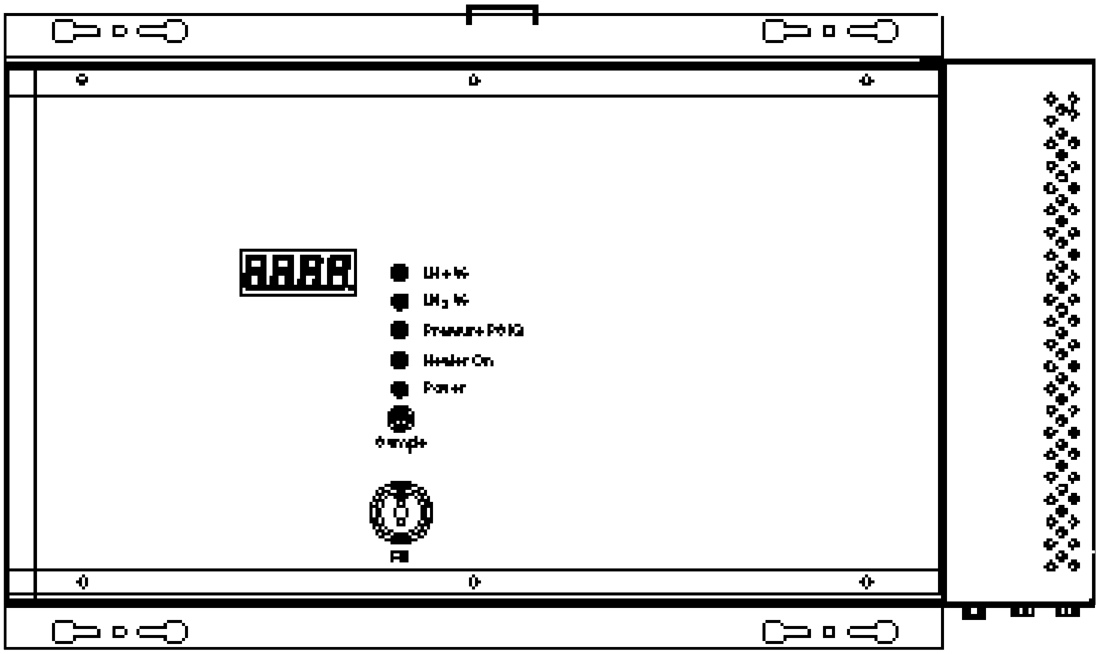
1-7 Magnet Monitor Configuration

Note

As of September, 2001 the Magnet Monitor Vendor changed from Granite to Advantech. Therefore there are 2 configurations for the Magnet Monitor.

1-7-1 Configuration for the Granite Unit

The Magnet Monitor is enclosed within an industrial type housing. Refer to Illustration 1-8 below. This unit weighs approximately 24 pounds, and is to be securely fastened to a wall located within the equipment room, in the orientation as shown below.

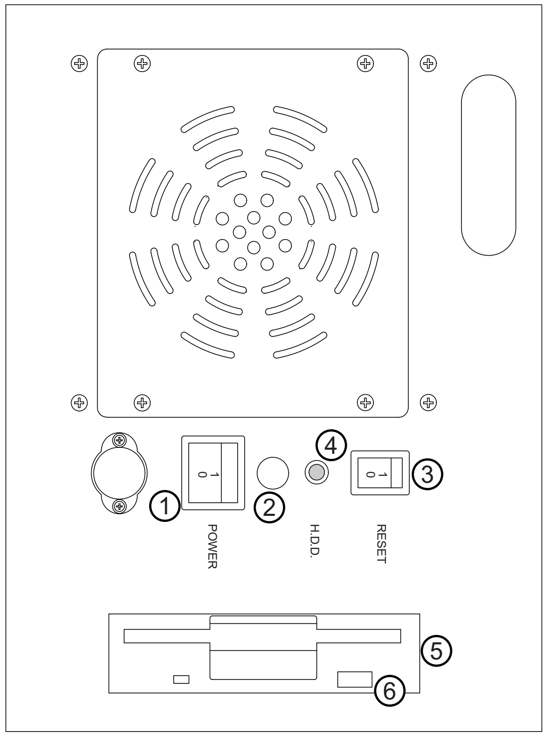


MAGNET MONITOR (TOP VIEW)
ILLUSTRATION 1-8

The local display system is comprised of a 4 digit 7-segment display, and several LEDs. The 7-segment display is for displaying Cryogen levels (in percent), and Helium vessel pressure (in PSIG). The LEDs indicate which parameter is being displayed at any given moment, with the exception of the “Heater” LED that indicates when the Helium vessel heater is activated.
The “Fill” switch is activated during the time cryogen fill is in progress. When in the fill mode, the cryogen levels are updated once per minute, otherwise this reading is taken automatically only once per hour.

1-7-1-1 Left Side Panel

The left side panel of the Magnet Monitor contains the following.



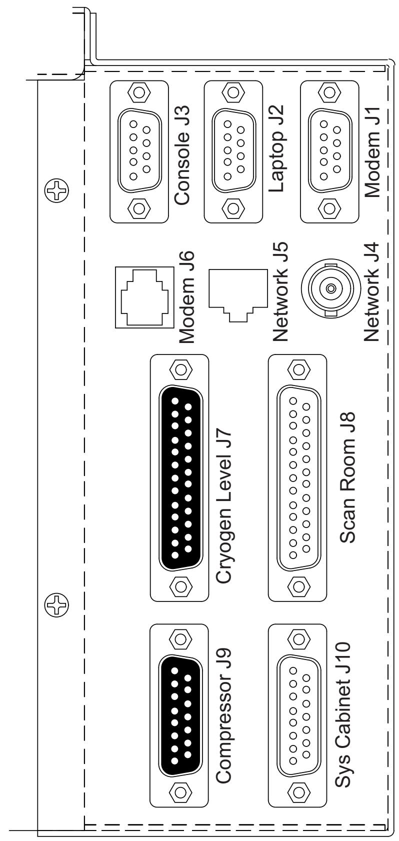
LEFT SIDE PANEL
ILLUSTRATION 1-9

TABLE 1-3
LEFT SIDE PANEL

No.	Description	Function
1	Power Switch	Turns power on or off.
2	Not used	Not used
3	[Reset] button	Resets the unit.
4	Green HDD light	Indicates when the internal hard-disk drive is being accessed.
5	Floppy disk drive	For software upgrades, DOS 1.44Mb
6	Green Floppy disk drive indicator light	Indicates when floppy drive is active

1-7-1-2 Connector Interface Panel

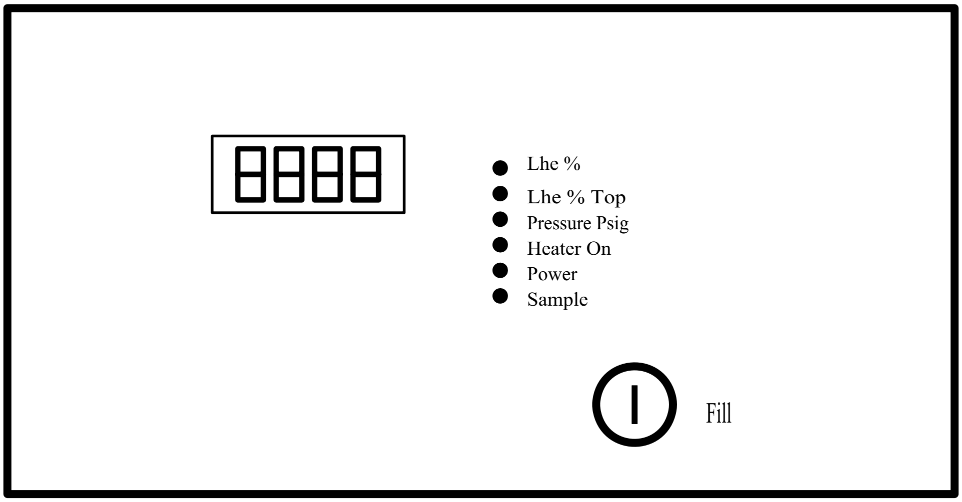
The lower right interface panel is used to interconnect to the Magnet subsystem, and other external interfaces. Refer to Illustration 1-10 below. Also refer to the Hardware Installation Manual for more specific connection information.



MAGNET MONITOR INTERFACE PANEL
ILLUSTRATION 1-10

1-7-2 Magnet Monitor Configuration, Advantech Unit only

The Magnet Monitor is enclosed within an industrial type housing. Refer to Illustration 1-11 below. This unit weighs approximately 24 pounds, and is to be securely fastened to a wall located within the equipment room, in the orientation as shown below.



MAGNET MONITOR (TOP VIEW)
ILLUSTRATION 1-11

The local display system is comprised of a 4 digit 7-segment display, and several LEDs. The 7-segment display is for displaying Cryogen levels (in percent), and Helium vessel pressure (in PSIG). The LEDs indicate which parameter is being displayed at any given moment, with the exception of the “Heater” LED that indicates when the Helium vessel heater is activated.
The “Fill” switch is activated during the time cryogen fill is in progress. When in the fill mode, the cryogen levels are updated once per minute, otherwise this reading is taken automatically only once per hour.

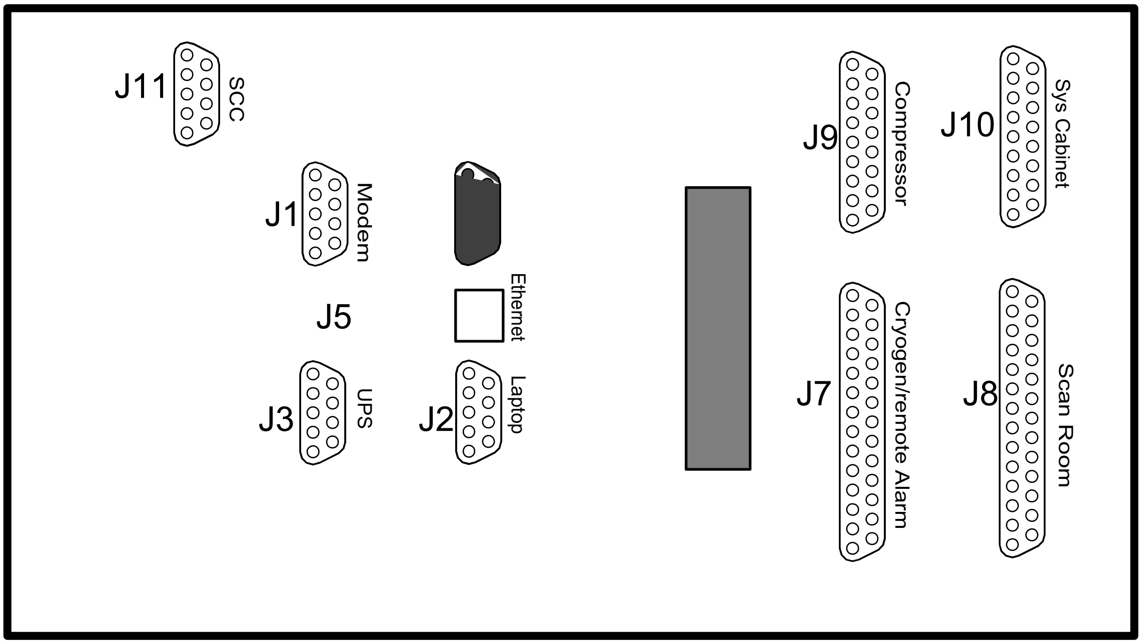
1-7-2-1 Left Side Panel

The left side panel of the Magnet Monitor contains the following.

Power switch to power on the unit
Reset button which will reset the unit
Green HDD light, indicates when the internal hard drive is being accessed
Floppy disk drive
Floppy LED light, indicates when the floppy drive is being accessed

1-7-2-2 Connector Interface Panel

The lower right interface panel is used to interconnect to the Magnet subsystem, and other external interfaces. Refer to Illustration 1-12 below. Also refer to the Hardware Installation Manual for more specific connection information.



MAGNET MONITOR INTERFACE PANEL
ILLUSTRATION 1-12

1-7-3 Interfaces to the Magnet and System Cabinet

After installation, the 4 D-Sub connectors at the lower right of the assembly provide all of the interface to the Magnet and System cabinet. For details on each connector, refer to the Hardware Installation Manual. Each pin is identified within that document.

System Cabinet Interface

Of these four connectors, one connects to the System cabinet, and picks up two signals there, which are utilized in the MM. One is the RF-Unblank signal from the IPG, and the other is a line trigger signal from the PDU.

The RF-Unblank is used to prevent the MM from activating a cryogen level reading during a scan, as well as to hold off making alarm decisions as long as a scan is in progress. The cryogen level readings are prevented due to the possibility that switching noise may create artifacts in the resultant images, and the alarm decisions are not made because the actual measured data during a scan may be corrupted by the RF or Gradient signals.
The line trigger signal is utilized as a means of determining if the System is powered up or not.
Both of these signals, RF-Unblank and Line Trigger, are monitored and displayed in the MM status screen as well.

Compressor Interface

Another of the four connectors is routed to the cryogenic compressor, here several status signals are monitored, and a reset line provides the ability for the MM to cause a reset of the compressor. Also, a water flow meter is attached to the cooling water inlet of the compressor. The water flow and its temperature are measured and displayed. The “Compressor Reset” signal and “Reset” control are the only two signals acted on. The other signals are monitored, displayed on the status screen, and logged for future reference. The details of these signals can be found in the Hardware Installation Manual.

Automatic Compressor Reset

The MM contains an automatic compressor reset feature. During normal operation, the MM monitors the “Compressor ON” signal from the cryogenic compressor. If this signal is found to be at a logic low (O Vdc), a reset signal is activated to the compressor. If after a pre-set delay, the signal is still not present (meaning that the compressor did not respond favorably to the reset signal), another reset signal is initiated. This process is repeated a prescribed number of times. If after this time the compressor is still not indicating a “Compressor ON” condition, all further attempts to reset are stopped and the OnLine Center is notified via Modem that this fault condition exists. If, for some reason, the OnLine Center is not contacted (i.e. no phone connection), the alarm will be logged for future reference and the MM will continue normal operation. Whenever the MM recognizes that the compressor is once again operating, it will re-activate its ability to generate a reset signal again and clear the alarm condition.

Magnet Interfaces (Two Cables)

The two remaining interface cables are connected to the Magnet, via the Penetration wall. One of the cables carries the cryogen level monitoring signals, as well as the heater signal, which is used to maintain pressure in some magnets (LCC). The Penetration wall is fitted with feed-through filters for both of these connections. The filters prevent extraneous noise leakage into the RF room.

The cryogen level signal is produced by applying a constant current source to the sensor (either 250 ma for certain Oxford Magnets, or 75 ma for most others), or by activating supply voltages to the “Pizza Board” (used only in Oxford fixed sites for Nitrogen readings). In either case, a voltage is produced that is proportional to the cryogen level. Cryogen levels are displayed in percent (%) of full on the MM status screen. Except during an RF-Unblank period, cryogen levels are automatically sampled once per hour. Depressing the sample button will also initiate a new reading, and placing the Fill Key in the “FILL” position will cause a level reading update every minute.

The heater control signal is a 12 Vdc supply voltage when activated, and drops to 0 volts when not activated. This signal is automatically activated whenever the magnet pressure drops below a prescribed limit, and is de-activated when the pressure crosses a prescribed upper limit. Generally these limits are set at 3.9 and 4.1 PSIG respectively.

The other cable is split three ways. It connects to the pressure transducer, a magnetic reed switch and the Coldhead sleeve (via a small “RuO Buffer amplifier” located on the top of the magnet).

A voltage is supplied to the pressure transducer, which in turn generates an output voltage proportional to the measured pressure; units of PSIG are used in the MM status screen.

The magnetic reed switch is located at the rear pedestal, it is closed when the magnetic field is down, and open when the field is up. This condition is sensed and displayed in the MM status screen. This is used to monitor for a quenched magnet condition.
The final set of connections at the magnet are used to monitor, up to, four cryogenic sensors (four sensing channels) located inside of the magnet at various places. Two of these channels are designed for Ruthenium Oxide sensors (RuO) and the other two are intended for SI410 or DT500 type sensors.

In all four cases, the measurements are a 4-wire configuration. Two wires conduct a stimulus current of 10 microamps to each sensor, and the other two wires sense the developed voltage.
For the two RuO channels, the developed signal is of a small magnitude (10mv to 25mv). For this reason the RuO Buffer amplifier is used to amplify these signals by a factor of 65. This results in an output range of 0.65 to 1.625 volts. Internally to the MM an additional gain of 2 is provided which increases these signals to the 1.3 to 3.3 volt range prior to application to the A/D converter.

The other two channels, used for SI410 or DT500 sensors, do not require further amplification.
Except during an RF-Unblank period, these temperature readings are sampled once per minute.

1-7-4 Magnet Monitor for Stand-Alone Magnet

The system cabinet signals are not absolutely required for the MM to operate. In their absence the MM will operate with disregard to RF-Unblank. The PDU line trigger signal is used to notify the MM that RF-Unblank must be considered.
This allows the use of the MM in cases where it is necessary to monitor and control a magnet that is either separate from an MR system, or where the system is powered off.

Note

When the Shem is connected to a working system it needs both signals RF-Unblank and System-Power-Cabinet connected properly to read the parameters. If RF-Unblank and System-Power-Cabinet are equal to 1 "on", the Shem will not acquire data. If one or both signals are zero the Shem will acquire data. Failure on not setting this cables properly might lead to wrong readings or magnet quench due halt of the Shem.

1-7-5 Data Collecting and Calculations

The following list of parameters are currently capable of being monitored by the MM. Some of these parameters may not be activated on certain magnet types.

TABLE 1-4
PARAMETERS MONITORED BY MAGNET MONITOR

No.	Description	Sampling Rate	Units	Comments
1	Compressor Water Flow	1/minute	GPM	Units can be changed to Liters
2	Compressor Water Temperature	1/minute	F	Units can be changed to Celsius
3	Helium Vessel Pressure	1/minute	PSIG	
4	Compressor Status Signals (Multiple)	1/minute	N/A	

No.	Description	Sampling Rate	Units	Comments
5	RF-Unblank	1/minute	N/A	
6	Line Trigger	1/minute	N/A	
7	Cryogen Level (Helium)	1/hour	%	Except during Fill, or when Sample button pressed
8	Cryogenic Temperatures (up to 4)	1/minute	K	
9	Magnetic Field detect (Reed switch)	1/minute	N/A	

Each parameter measured, not including the LOGIC type signals, is converted from a voltage into their respective units of measure. This is accomplished in software, by applying the reading to a pre-determined polynomial, having selectable coefficients, depending on the reading type.

The coefficients are stored in a configuration file (coef.cfg) (see Appendices D and E), and have been pre-determined for each sensor type. Each measured parameter can have up to 5 polynomials, with 5 coefficients, to provide the curve fitting necessary for accurate conversion. These coefficients will not require alteration unless a sensor type is changed to a different type, having a different transfer function. It is not recommended that field personnel alter these values in any way. To do so can cause erroneous readings, and defeats the need to maintain a common configuration among systems.

One anticipated need to alter these coefficients might be to display the readings in different units of measure. It is possible to create these additions, however qualified personnel must create them and be common with regard to software configuration management.

While in the Proprietary mode of operation, a database of readings is maintained within the MM. For each parameter Minute, Hour and Day values are retained, up to a maximum of 2048 data points per type. For example, Water Flow will have 3 data files associated with it (Minute, Hour and Day), and so on for each other parameter. This data is used for table and graphic displays in the available screens, and is also accessible to the OnLine Center as an FTPed file (although at this time the OLC has no specific plans to download those files).

1-7-6 Other External Interfaces

The MM is designed to provide three methods of external communication. Those are as follows:

- Modem
- Serial Port
- Ethernet

1-7-6-1 Granite Modem

The Granite MM is equipped with an internal modem for phone line connections within the USA. In this case, an external analog phone line is plugged directly into the receptacle on the Interface Panel. An additional serial port is accessible at the interface panel for the purpose of connecting an external modem. Both modems cannot be connected at the same time. When the external modem is required, as for all non-domestic installations, the internal modem must be removed. It is also necessary to change a jumper setting on the general I/O board, ISA SLOT 1. Refer to Appendix F for jumper settings. The Granite MM will recognize this action and adapt accordingly. For the external modem case, the Global Modem has been tested. Since the OnLine Center will occasionally need to make calls into the MM, be sure that the external modem is set up to automatically answer incoming calls.

1-7-6-2 Advantech Modem and External Modem Serial Port

The Advantech Magnet Monitor is equipped with an external modem (known as the Global Modem) for phone line connections. In this case, the modem is connected to the Magnet Monitor via a serial port, and the customer phone line is connected to the external modem. The phone line connection should pass through the UPS for transient protection.

1-7-7

Serial Port(s)

Aside from the serial port used for the external modem, there are two other serial interfaces. They are labeled “Console” and “Laptop”. Both are identical interfaces, and can be used interchangeably. Using a Null Modem serial cable, a Laptop computer can be connected here. These ports are configured for both PPP and TCP/IP connections, as well as normal serial RS-232 serial protocols.

The user can configure their Laptop as a Hyperlink terminal, or as a Null Modem Dial-up terminal. In the Hyperlink terminal mode, using the provided Linux Logins and passwords, it is possible to perform standard Linux commands, and thus communicate with the MM. A typical use of this method would be to view files, directories and so on. Using a Dial-up session, while logged in as a PPP user, the connection will allow the use of an Internet browser, such as Netscape or Microsoft Internet Explorer. Using a browser you can view and interact with the HTML screens provided by the MM.

Ethernet Interfaces

Ethernet Capabilities exist for “Beyond Modem’s” connection. Also there is Ethernet capability for the TWIN system. There is a cable between the Magnet Monitor and the Host computer. The connection method is RJ-45. This feature allows for viewing of the HTML screens on the Operator’s Console.

1-8 SOFTWARE OVERVIEW

1-8-1 Non-Proprietary Software

The MM is preloaded with a copy of the Non-Proprietary software. Its default configuration is set for operation with an LCC magnet type. For other magnet types, it is necessary to initiate a script file, via a laptop connection, to make the appropriate selections. Detailed information on this process will be found in the Hardware Installation Manual.
In the Non-Proprietary mode, the MM is capable of the following functions;

- Measure and display locally Cryogen pressure
- Measure and display locally Cryogen levels
- Control a heater used to regulate magnet pressure
- Monitor and provide automatic compressor reset
- Allow certain restricted Logins via the serial port, using a laptop computer

1-8-2 Proprietary Software

The MM can be loaded with a Proprietary version of the operating software. This software is Class M Proprietary, and must be handled as such.

The Proprietary software can only be installed and used by authorized GE personnel. The Proprietary software must be removed from any MM that is attached to a system not under GE service contract.

In the Proprietary mode, the MM is capable of the following functions;

- Measure and display locally and via HTML Cryogen pressure
- Measure and display locally and via HTML Cryogen levels
- Measure and display (via HTML) water flow
- Measure and display (via HTML) water temperature
- Measure and display (via HTML) up to 4 cryogenic temperatures
- Measure and display (via HTML) Compressor status
- Measure and display (via HTML) System Cabinet Status (RF-Unblank and Trigger)
- Monitor and display (via HTML) Magnet field status (Ramped/Quenched)
- Allow certain restricted and UN-restricted Logins via the serial port, using a laptop computer
- Provide tabulated and graphic displays of collected data (Min, Hour & Day) (HTML)
- Facilitate local “Lamp Test”, to verify local display is functional
- Maintain a local user log, user entered events and notes
- Control a heater used to regulate magnet pressure
- Monitor and provide automatic compressor reset
- Communicate via Ethernet connection (Future)
- Allow the use of an external or internal Modem
- Report all out-of-limit alarms to the OLC
- Allow manual compressor reset
- Maintain log of alarm events
- Support FTP file transfers
- Respond to OLC queries
- Facilitate FILL mode

2- PRO-ACTIVE SOFTWARE INSTALLATION

2-1 Archiving Insite Information

You can archive checkout information to a floppy provided software is active. In the event that a magnet monitor is replaced, the checkout information can be restored. This step eliminates the need for an Insite Checkout after magnet monitor replacement.

1. Insert a blank floppy into the magnet monitor floppy drive.

2. Connect a 9 pin female to 9 pin female Null-Modem cable between the Laptop connector on the magnet monitor and your laptop. Connect using Hyperterminal. See step 1 of section 2-2 for more information on using Hyperterminal to connect to the Magnet Monitor.

   Login name is **root** (case sensitive)
   Password is **mvserve** (case sensitive)

3. Mount the floppy by typing **mount –t msdos /dev/fd0 /mnt/floppy**

4. Copy the files by typing **cp /usr/g/insite/sclink.cfg /mnt/floppy**
   cp /shem/config/alarm.cfg /mnt/floppy
   cp /shem/config/param.cfg /mnt/floppy
   cp /shem/logs/.shemtype /mnt/floppy/shemtype (must be exact, file is not present on version 1.6 software)

5. Unmount the floppy by typing **umount /dev/fd0**

6. Remove floppy from the drive, label as **Insite Information** and store in a safe location.

2-2 Installing the Proprietary software

IMPORTANT

If you need to reset, or power down the Magnet Monitor, do the following steps to properly shut down the software.

- Hold the *Sample* button in for about four seconds. **halt** should appear on the display.
- Release the *Sample* button for less than 1 second, and then press the *Sample* button again. **oooo** (four lower case o's) should appear on the display.
  - Wait for at least **90 seconds**, then you may reset or power down the Magnet Monitor.

The following process describes how to install Proprietary software onto a Magnet Monitor. The process assumes that your Magnet Monitor currently has a Non-Proprietary version installed, and operating. You will need the Proprietary software diskette(2252032) for software installation.

1. Using a serial connection (9-pin female to 9-pin female Null Modem cable), and a laptop computer, open a Hyperterminal window (Hyperlink connection, refer to section 5 in the Hardware Installation Manual 2230681). If you do not have a copy of direction 2230681, do the following:

On the laptop in Windows 95:

   a. Open HyperTerminal by selecting Start/Programs/Accessories/hyperterminal
   b. Click on Hyperterminal
   c. Create a new connection named magnet monitor set-up
   d. Select direct to Com1, then OK
   e. Edit Bits per second to 115200, data bits to 8, parity to none
   f. Hit the enter key to get the Login prompt

Note

Some laptops may not have the hyperterminal executable available on their laptops. You will need to configure your laptop by doing the following steps:

Select Start-> Settings-> Control Panel-> Add/Remove Programs.
Select Windows setup on the top of the menu.
Select Communications, Details, Hyper Terminal, OK
Select Apply which is located on the lower right area of the window.
Select Start -> Find-> Files or Folders and type in Hypertrm, look in C:
Double click on Hypertrm.exe to begin Hyperterminal program.

2-2-1 Configuring the Magnet Monitor for LCC

The Advantech Magnet Monitor should have default settings for the HFO system. To verify the settings, connect the laptop to the Magnet Monitor as in step 2-1-2.

At the login prompt type: **setup**	Case sensitive password
At the password prompt type: **monitor**	Case sensitive password

The following will appear on the screen:
Do you wish to change the Configuration Type [y/n] **Y**

There are 2 choices for Configuration Type:

1. hfo
2. lcc

Enter your choice as a number from 1 to 2, or Q to quit: **2 [enter]**

If needed, you can change the Compressor Reset Values. The values mention below should not be changed.

Do you wish to change the Compressor Configuration values? [y/n] **Y**

Do you wish to Enable the Compressor Reset Function? [y/n] **Y**

Enter value for Startup Delay Time (minutes) [1:5] **5**

Enter value for Reset Delay Time (minutes) [1:30] **10**

Enter value for Recheck Delay Time (hours) [1:5] **1**

Enter value for Retry Count (count) [0:10] **2**

Will this Magnet Monitor be serially connected to a UPS [y, n] Select **Y** if this site has a UPS, majority of LCC magnets will not have a UPS.

The Magnet Monitor will now reset.

Hit [return] on the laptop.

2-2-2 Loading the Proprietary Software

1. At the Login prompt, type in **root**

At the password prompt, type in **mvserve**

Note

Some of the early Granite magnet monitors had a different login and password name. If the above login and password does not work try:

Login **ac79-E38** (case sensitive password)
Password **RvmL&Z2pFc** (case sensitive password)

2. Insert the Proprietary software floppy disk into the floppy drive on the Magnet Monitor.

CAUTION

VERIFY THAT YOU HAVE THE PROPER PROPRIETARY DISKETTE INSTALLED. FAILURE TO USE THE PROPER DISK WILL DESTROY THE HARD DRIVE ON THE MAGNET MONITOR. OBSERVE THE REVISION OF NON-PROPRIETARY SOFTWARE PRIOR TO LOADING PROPRIETARY SOFTWARE. THE FOLLOWING INFORMATION WILL APPEAR AFTER CONNECTING THE MAGNET MONITOR SERIAL PORT TO YOUR LAPTOP, AND ENTERING HYPERTERMINAL.

If the revision shown is: Magnet Monitor Non-Proprietary Software Rev. 2.3 Red Hat Linux release 6.1 (Cartman) Kernel Rev. 2.2.12-20
Then the Proprietary software that should be loaded is: **2252032-3 yellow floppy**

If the revision shown is: Magnet Monitor Non-Proprietary Software Rev. 2.0 Red Hat Linux release 6.1 (Cartman) Kernel Rev. 2.2.12-20
Then the Proprietary software that should be loaded is: **2252032-2 yellow floppy**

If the revision shown is: Magnet Monitor Non-Proprietary Software, Rev. 1.3 Red Hat Linux release 4.1 (Vanderbilt) Kernel Rev. 2.0.27
Then the Proprietary software that should be loaded is: **2252032 red floppy**

3. Mount the drive by typing the following:

mount –t msdos /dev/fd0 /mnt/floppy (case sensitive, don't ignore spaces)

4. Press [Enter].

5. Copy files to /tmp directory.

cp /mnt/floppy/* /tmp Press [Enter].

6. After the copy function is completed, unmount the floppy drive.

umount /dev/fd0 Press [Enter].

7. Remove floppy disk from drive.

8. Execute the install script.

/tmp/installp Press [Enter].

If the system replies the message **wrong version**, verify proper revision of Proprietary software is being used, refer to step 2 above.

9. When the install script completes, the Magnet Monitor will re-boot. This will cause a disconnect in your laptop connection.

2-2-3 Seting Time and Date

1. Once the Magnet Monitor has reset hit [Enter] on the laptop.
2. At the Login prompt, type in **root**

   At the password prompt, type in **mvserve**
3. At the prompt type in **date MMDDhhmmyy** where

   MM is the 2 digit month
   DD is the 2 digit day
   hh is the 2 digit hour in military time
   mm is the 2 digit minute
   yy is the 2 digit year

   Example is **date 0816132001** for August 16, 2001 at 1:20pm.

4. To write the date into firmware type **/sbin/clock --systohc** If you omit this step the magnet monitor will loose the time and date when rebooted.

5. Type **reboot** to reset the Magnet Monitor.

2-2-4 Restoring Insite Information

If you are replacing a magnet monitor and have previously archived the Insite information, you can restore it at this time.

1. Install the **Insite Information** floppy.
2. Re-connect using Hyperlink terminal again. Login as **root**, password **mvserve**.
3. Type **/MVServe/bin/check-out** [enter]

   The laptop screen will echo back that it is about to re-load the checkout information.
   Type **yes** to continue.
   The program will pull the information from the floppy onto the Magnet Monitor hard drive.
   It will take a few minutes for the data to transfer.
   The laptop screen will then prompt you to "hit any key to continue". Depress any key on the keyboard at this time. After another few minutes the Magnet Monitor will re-boot. The Magnet Monitor will then perform in the same state as the original.

IMPORTANT

If you have successfully completed Insite Information Restoration, do not run InitConfig. It will destroy all check-out information.

2-3 Online Center Install Check-out for Modem Connection (for broadband see sec 2-4)

The following steps will establish check-out with the OLC.

1. After the Magnet Monitor re-boots(re-booting can take up to 3 minutes), re-connect using the Hyperlink terminal again. Login as **root**, password **mvserve**, or login as **ac79-E38**, password **RvmL&Z2pFc**, depending on your login used in step 1 of section 2-1.

2. On your laptop, via the terminal window, type in the following command to initiate a script file: **cd /MVServe/bin**

3. Press [Enter].

4. **./InitConfig** (Slash is preceded by a decimal).

5. The Magnet Monitor is now ready for InSite installation (OnLine Center).

6. Press [Enter]. Answer all questions. The following is a list of typical answers:

Do you want to Continue with Configuring Magnet Monitor?
*********************** WARNING!!! ************************
If yes, you will remove any previous Insite Checkout information.
Enter (Y/N): y

Current Connectivity Configuration Modem
Use Default value? (Y/N) **Y**

Hospital Name: Putra Memorial
'Putra Memorial' correct? (Y/N) y
Dial Prefix: 9

'9' correct? (Y/N) y You may need to enter 9,1 if the phone line at your site requires a 1 before the 800 number

MAKE SURE YOU ENTER THE "COMMA" PRIOR TO A "1"

Tone or Pulse Dialing? (T/P): t
't' correct? (Y/N) y
Custom Modem Initialization
String: <enter modem string here if required>
'<enter modem string here if required>' correct? (Y/N) y
Select a modem type from the list:
(1) External_Motorola_Codex
(2) Internal_US_Robotics_336
(3) InitConfigCustom
(4) Robotectics33
(5) Internal_US_Robotics_288
(6) Internal_Hayes_Accura_144
(7) Generic_Global ⇦ **Select this for Global Multitec Modem**
(8) Generic ⇦ **Select this for Grainte Internal Modem**
(9) External_US_Robotics_288 **Use this for US Robotics V. See modem string below.**

Modem string for US Robotics **only** is:
AT &F1&C1&D2&A3&H1&R2&K1&M4 S13=1 S0=0 YO LO MO X3 &W

Pick selection (1-9):

Use current Country? US (Y/N) Select **N** if outside the States, and go through the list until you find the correct country.

Modem Initialization string
String 'AT &FO X4 &DO EO T MO Q2 SO=1 S7=90'
Do you want to use default string (Y/N) Select **Y** for the default or **N** if you want to change the default. The default should work for the majority of countries.

Do you want to continue with configuring Magnet Monitor's network?
********************************Warning!**********************************
If yes, you will remove any previous network configuration
Enter (Y/N) Select **N** if you do not have an Ethernet connection between Mag Mon and Host. Select **Y** if you do have the Ethernet connection. At this time the Ethernet connection is only used for the **Twin** system.
Magnet Monitor network address 3.44.56.78
Use default value (Y/N) **N**
Enter new IP address for Magnet Monitor **3.87.23.54** Example of network address provided by Hospital Network Administrator

Netmask 255.255.252.0
Use default value (Y/N) **Y**

Host IP address 3.44.56.78
Use default value (Y/N) **N**
Enter new IP address for Host **3.87.23.53** Example of network address provided by Network Admin

Magnet Monitor config changed.
Machine will reboot in one minute.
OK
[root@172 bin]#

7. Contact the Global Service Support Center in your GEMS service pole and work with the Pole OLC Magnet Support Group within the Service Support Center to complete an InSite Check-out (1-800-321-7937 in GEMS Am). You will be required to provide the following information.
   - Site/Hospital name
   - GE/HCS System ID and system type for customer system connectivity via the Magnet Monitor Computer (This information is obtained from the OLC)
   - The manufacturer of this scanner
   - Magnet Serial number
   - MAC Team leader
   - The model and computer (if required) of this scanner
   - Modem phone number

   They will place a call to your Magnet Monitor and initiate a file transfer. This transfer is sclink. When this file transfer ends, and Configlink is executed (by the OLC), your Magnet Monitor will re-boot once again.

10. Using a Dial-up connection and Null Modem cable, you may now use your laptop Internet browser to connect to the Magnet Monitor and view its current status and settings. Refer to Appendix B and C for these setups.

11. Once your browser is functional, select the Configuration screen, Alarms, and activate the Master Alarm (set Enable to yes). Be sure to select the "SAVE" button to save this change. The Magnet Monitor is now ready to automatically report alarms to the OLC. Refer to section 4 through 7 for proper set-up.

Note

To test if the Shem can dialout use the command **toucholc.pl** as shown, If dialout is not successful check: if the dialout code is correct, if the hospital is not blocking the dialout or if your country has a tool free number to access GE network.

[root@172 bin] **toucholc.pl <enter>**
Preparing touch files to be sent
.shemtype
system.cfg
sclink.cfg
Done (**First done means that Shem is preparing to send data)**
Sending Touch Notification Report to AutoSC
021206 11:25:54 AutoSC processing continues, please standby... Fri Dec 06 04:22:23 CST 2002
021206 11:25:58 AutoSC processing continues, please standby... Fri Dec 06 04:22:27 CST 2002
021206 11:26:02 AutoSC processing continues, please standby... Fri Dec 06 04:22:31 CST 2002

Done (**Second done means that Shem sent the data to ppads, so dialout is successful**)

Note

Allow 24 hours after InSite Check-out before running Communications Test (Refer to Illustration 4-15.)

2-4 Configuring magnet monitor for Beyond Modems

Do you want to Continue with Configuring Magnet Monitor?
************************ WARNING!!! *************************
If yes, you will remove any previous Insite Checkout information.
Enter (Y/N): y

Current Connectivity Configuration: 'Modem'
Use default value? (Y/N) **N**

Select from the following configurations:
(0) Not Configured
(1) Modem
(2) Beyond Modems
2
Beyond Modem Selected
'/etc/ppp/options.ttyS2.PRECHK': to present used to replace '/etc/ppp/options.tt
yS2'!
'/etc/hosts.PRECHK': to present used to replace '/etc/hosts'!
'/etc/sysconfig/network.PRECHK': to present used to replace '/etc/sysconfig/netw
ork'!
'/etc/ppp/ip-up.PRECHK': to present used to replace '/etc/ppp/ip-up'!
Hospital Name: 'Institut Mutualiste Montsouris'
Use default value? (Y/N) N
Enter proper hospital name

Do you want to Continue with Configuring Magnet Monitor's Network?
*********************** WARNING!!! ************************
'Y' to continue, 'N' to exit, 'X' to remove network config.
Enter (Y/N/X): **y**
MagMon IP Address: 10.250.24.91 **(Magnet Monitor IP Address)**
Enter proper Magnet Monitor Address

Netmask: 255.255.255.128 **(Magnet Monitor Net mask)**

'255.255.255.128' correct? (Y/N) **Enter proper MM Netmask**

Gateway: 10.250.24.126 **(Router IP address)**
'10.250.24.126' correct? (Y/N)   **Enter proper Gateway**

Host IP Address: 10.250.24.20 **(LX IP address)**
'10.250.24.20' correct? (Y/N)   **Enter proper Host Address**

FE Laptop Address: 10.250.24.6 **(FE Laptop Ip address)**
'10.250.24.6' correct? (Y/N) y

Attention: After checkout the Magnet Monitor will accept IP connection only for this 3 IP address, all others will be refused!

Magnet Monitor Config changed
Changes take effect after next reboot.

Magnet Monitor ready for 'Check Out'.
Machine will reboot in One Minute

Broadcast message from root (ttyS1) Fri Jul 19 12:13:10 2002...

The system is going DOWN for reboot in 1 minute !!

Broadcast message from root (ttyS1) Fri Jul 19 12:14:10 2002...

Contact the Global Service Support Center in your GEMS service pole and work with the Pole OLC Magnet Support Group within the Service Support Center to complete an InSite Check-out (1-800-321-7937 in GEMS Am). You will be required to provide the following information.

- Site/Hospital name
- GE/HCS System ID and system type for customer system connectivity via the Magnet Monitor Computer   (This information is obtained from the OLC)
- The manufacturer of this scanner
- Magnet Serial number
- MAC Team leader
- The model and computer (if required) of this scanner
- Modem phone number

They will place a call to your Magnet Monitor and initiate a file transfer. This transfer is sclink. When this file transfer ends, and Configlink is executed (by the OLC), your Magnet Monitor will re-boot once again.

Using a Dial-up connection and Null Modem cable, you may now use your laptop Internet browser to connect to the Magnet Monitor and view its current status and settings. Refer to Appendix B and C for these setups.

Once your browser is functional, select the Configuration screen, Alarms, and activate the Master Alarm (set Enable to yes). Be sure to select the “SAVE” button to save this change. The Magnet Monitor is now ready to automatically report alarms to the OLC. Refer to section 4 through 7 for proper set-up.

2-5 Subscribing to a Site

It is recommended that primary Field Engineers subscribe to their sites via E-mail. You will then be notified on E-mail if your site alarms for any reason, has a cryogen fill, or does a check-in with the PPADS database.

1. Go into Microsoft Exchange E-mail. Address E-mail to PPADSCMD.

2. In the body of the message type the following:
   begin
   subscribe 215341shem email (*where 215341shem is the cryo system ID of the site you wish to subscribe to*)
   end

3. For more information on PPADSCMD mail server commands, refer to Appendix G.

3- CONNECTIONS

3-1 Overview

There are two ways to locally access the Pro-Active software on the Magnet Monitor:

- Through a local serial connection using a laptop computer. (Refer to Appendices A, B and C.)
- Through an Ethernet network.

3-2 Local Connection – Laptop

Connect a 9 pin female to 9 pin female Null Modem cable between the serial port on your laptop to connector J2 on the Magnet Monitor.

Note

The "Hyperlink" connect type used in Appendix A is intended for installs and access to the Magnet Monitor operating system. The "Dial-up" and "Browser" described in Appendices B and C are used for local viewing of the MM HTML screens via a web browser.

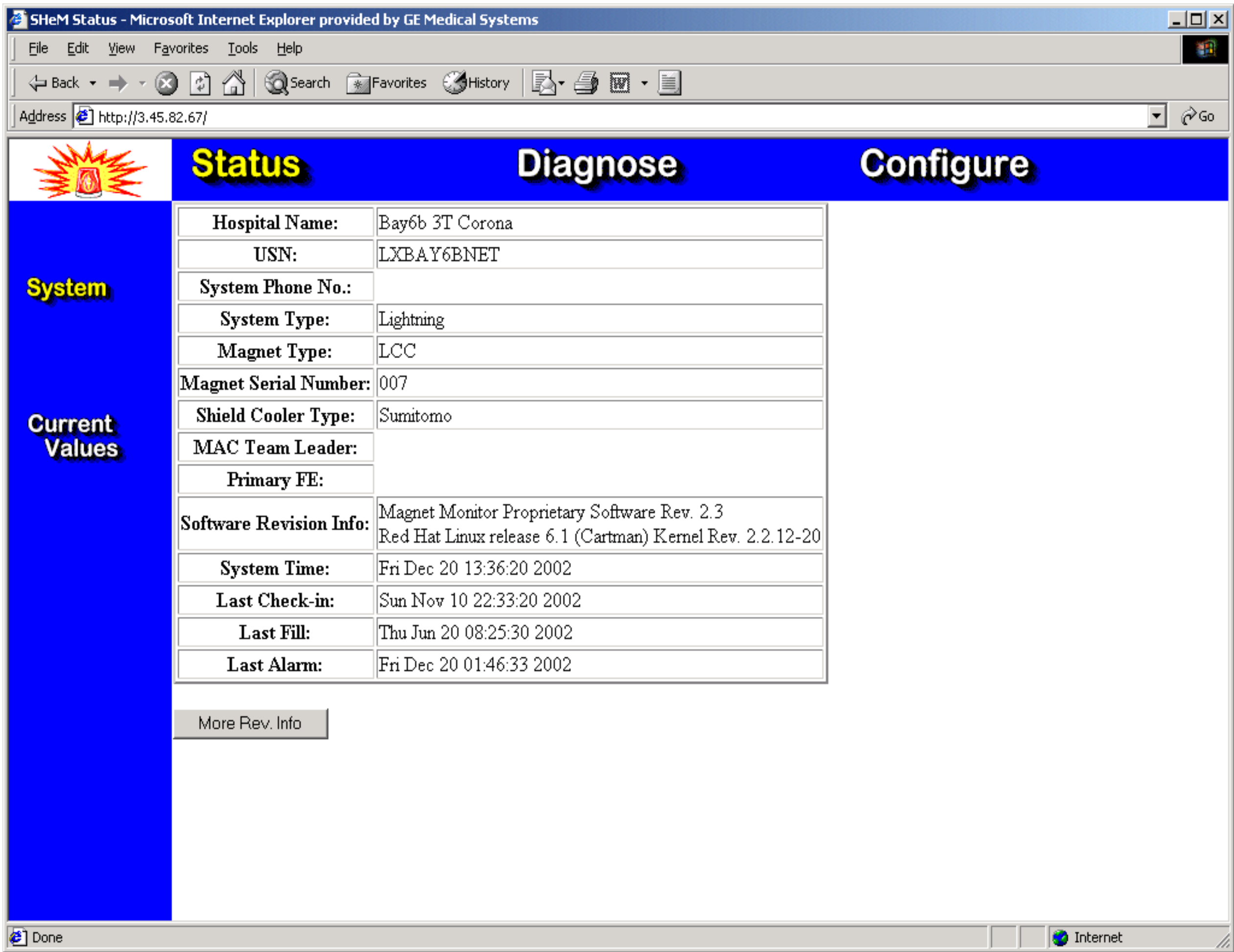
3-3 Ethernet Network Connection – RJ-45

For the TWIN system, the Ethernet connection exists between the Magnet Monitor and the Host computer. The magnet monitor can be accessed from the Common Service Desktop from Utilities menu. Select the option labeled **Magnet Monitor.** The following process can also be used.

1. Boot up the console to Signa level.
2. To execute communication, open a C-shell and type **su -** password is **operator**
3. At the prompt, type **insite_browser –n**
4. Select **edit** from the Netscape menu, then select **preferences.**
5. Click on the arrow that is on the left side of the "Advanced" menu tree.
6. Select **Proxies** from the Advanced menu tree, then select **Manual Proxy Configuration**.
7. Select **View**, then click in the **No Proxy for** text box.
8. Delete any data that is in the text box and type **MAGNET_MONITOR_IP**
9. Click **OK** to exit the Pro Dialog Box, then click **OK** to exit the preferences Dialog Box.
10. Select **File** from the Netscape menu, then select **open page**
11. Enter **MAGNET_MONITOR_IP** in the text box, and then select **open**
12. You should them be able to log into the magnet monitor.
13. Login name is **MMService,** password is **MagnetMonitor**

4- USER INTERFACE SCREEN DESCRIPTIONS

4-1 Overview



BROWSER WITH PRO-ACTIVE SCREEN
ILLUSTRATION 4-1

The Pro-Active software is accessed through screens displayed on an Internet browser.

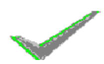
The Pro-Active software has three basic types of screens:

- Status screens which show currently set values and limits of monitored parameters and their alarm status.
- Diagnostic screens which provide tests for troubleshooting the Magnet Monitor.
- Configuration screens which are used to set up the Magnet Monitor and change monitored parameters.

The Pro-Active screens are organized like a table with the main screen type names as column headings and screens within a screen type as row headings. The selected screen is displayed in the "table cell" part. The screens are selected by clicking on the screen names listed in the column and row headings.

The upper left hand corner area has an alarm indicator. If any monitored parameter has its alarm enabled and an alarm condition is detected, the alarm indicator changes from a green check mark to a red flashing alarm light.

ALARM INDICATOR
ILLUSTRATION 4-2



CHECK INDICATOR
ILLUSTRATION 4-3

4-2 Accessing the Pro-Active Software

1. If you are using a laptop, connect it to the Magnet Monitor as described in "Local Connection – Laptop", section 3.2. Refer to Appendices A, B, and C for specific setup information.

2. Open the browser.

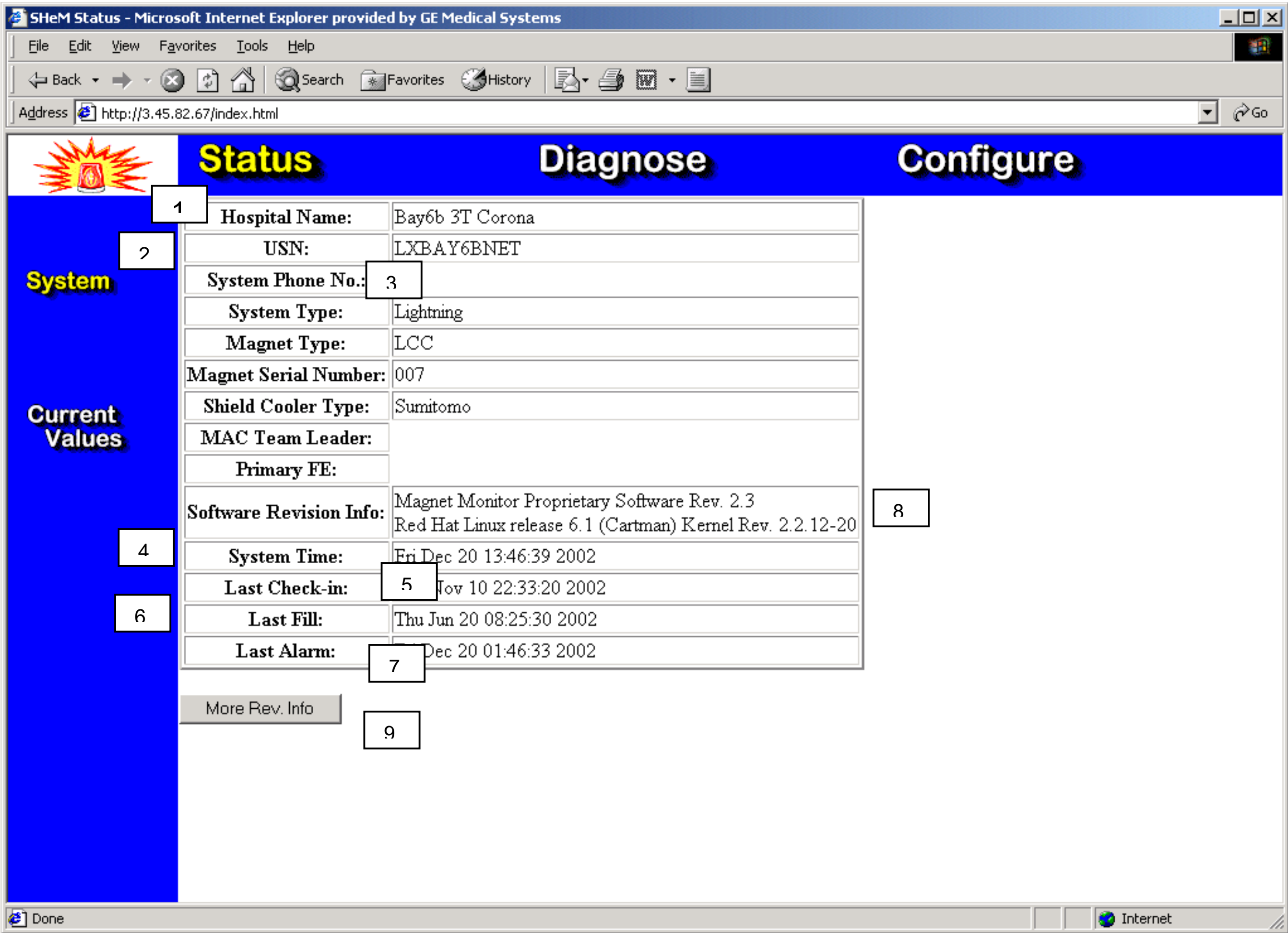
3. Click on [Open] and enter the Magnet Monitor's IP address (10.0.0.5).

4. Click on [Open] on the Open Location dialog.

5. The login name is **MMService** password is **MagnetMonitor** (case sensitive).

4-3 Status Screens

4-3-1 System Screen

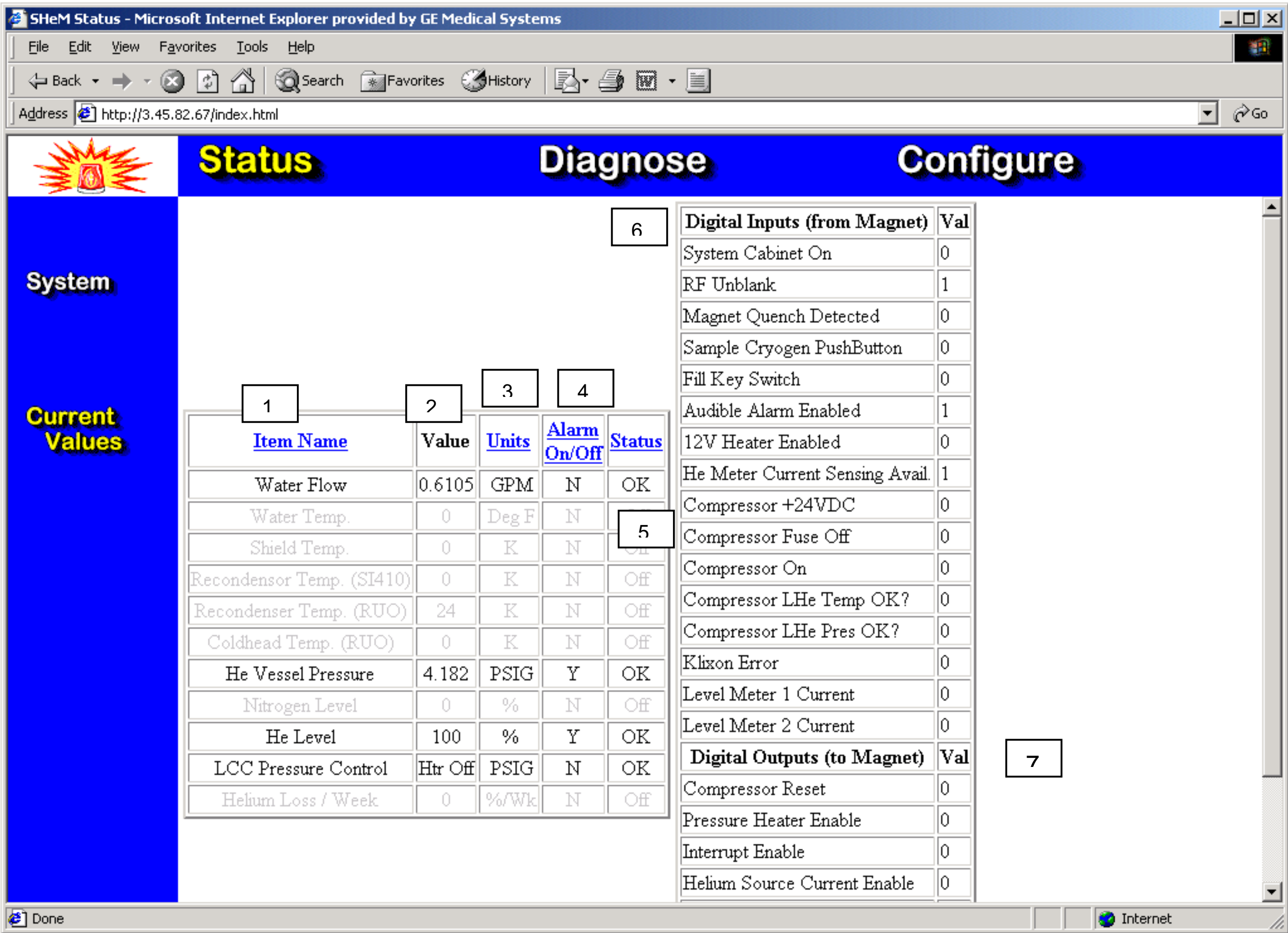


SYSTEM STATUS SCREEN
ILLUSTRATION 4-4

TABLE 4-1
SYSTEM STATUS SCREEN

No.	Field	Description
1	Site Name	Name of the hospital or geographic location.
2	USN	A unique name assigned to each specific unit.
3	System Phone No.	Phone number for PPADS; should not change.
4	System Time	Current time of the time zone where the unit is installed.
5	Last Check-in	Name of the periodic check-in log file.
6	Last Fill	Name of the fill log file.
7	Last Alarm	Name of the alarm log file.
8	Software revision	Revision that is on the magnet monitor
9	More Rev Info	Exact software revision

4-3-2 Current Values Screen



CURRENT VALUES SCREEN
ILLUSTRATION 4-5

TABLE 4-2
CURRENT VALUES SCREEN

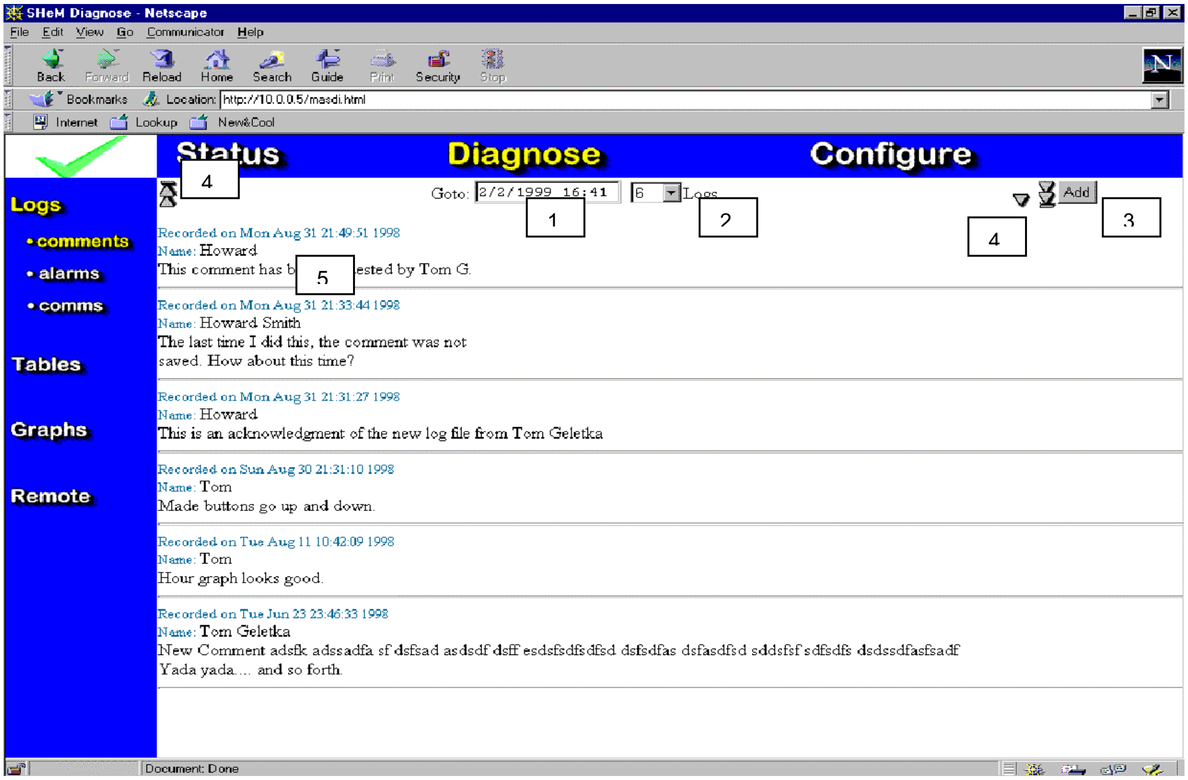
No.	Column	Description
1	Item Name	Name of the monitored parameter.
2	Value	Current monitored value.
3	Units	Units of measurement.
4	Alarm On/Off	Alarm On/Off setting. The display shows Y when the alarm for the monitored parameter is turned on and N when the alarm is turned off.
5	Status	Alarm status. The display shows HI or LOW when the parameter is in alarm state and OK when not in alarm state. **Note:** If the alarm is not enabled, OK will always be displayed.
6	Digital Input	Logic status of digital inputs. See table 4-3 for definitions of the status inputs.
7	Digital Output	Logic status of output functions.

TABLE 4-3
DESCRIPTION OF DIGITAL IMPUTS

Digital Input	**Value**	**Description**
System cabinet on	1	System cabinet has power
RF unblank	1	System is scanning or rf unblank cable is off
Magnet Quench Detected	1	Magnet reed switch is open, no magnetic field detected.
Sample cryogen pushbutton	1	The sample button was activated
Fill key switch	1	Magnet fill is activiated
Audible alarm enabled	0	Used for HFO only
Helium meter current sensing available	1	This magnet monitor is capable of sensing the proper output current to sample cryogens. This option works for the Advantech Magnet Monitor only.
Compressor 24VDC	1	Compressor has 24VDC
Compressor fuse off	1	Compressor fuse is OK
Compressor on	1	Compressor is running
Compressor Lhe Temp OK?	1	Compressor temperature is OK. Zero value means compressor has shut down due to high helium temperature
Compressor Lhe Pres OK?	1	Compressor pressure is OK. Zero means compressor has shut down due to low helium pressure
Klixon Error		Not used at this time
Level Meter 1 Current	1	Magnet Monitor has detected an error in the current being used to read helium level. The helium level may be inaccurate for the lower vessel, Advantech only
Level Meter 2 Current	1	Used for HFO only
Digital outputs		
Compressor reset	1	Reset signal being sent to the compressor
Pressure heater enable	1	Magnet heater is activated

4-4 Diagnose Screens

4-4-1 Logs Screen

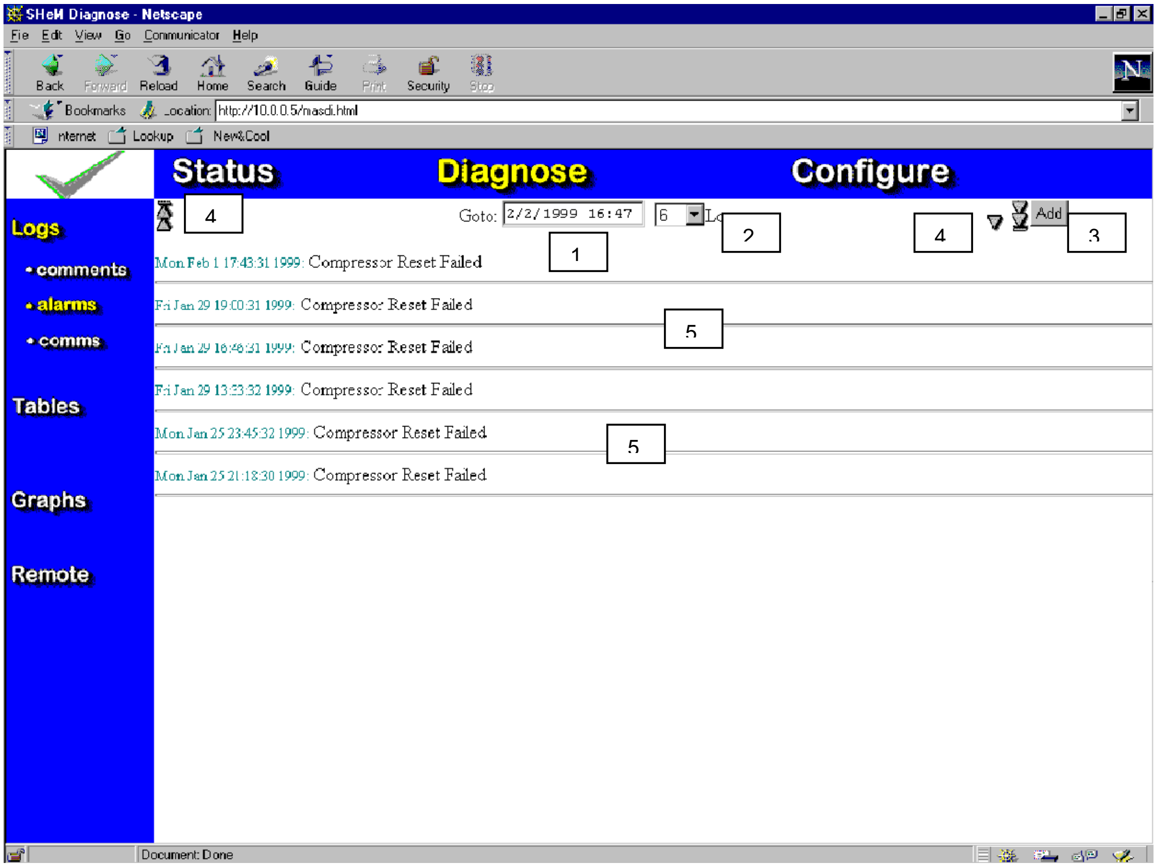


LOGS SCREEN
ILLUSTRATION 4-6

TABLE 4-4
LOGS SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Add	Select to add a logged entry.
4	Up/Down Arrows	As needed to scroll log entry.
5	Records	Actual example records.

Alarms Screen

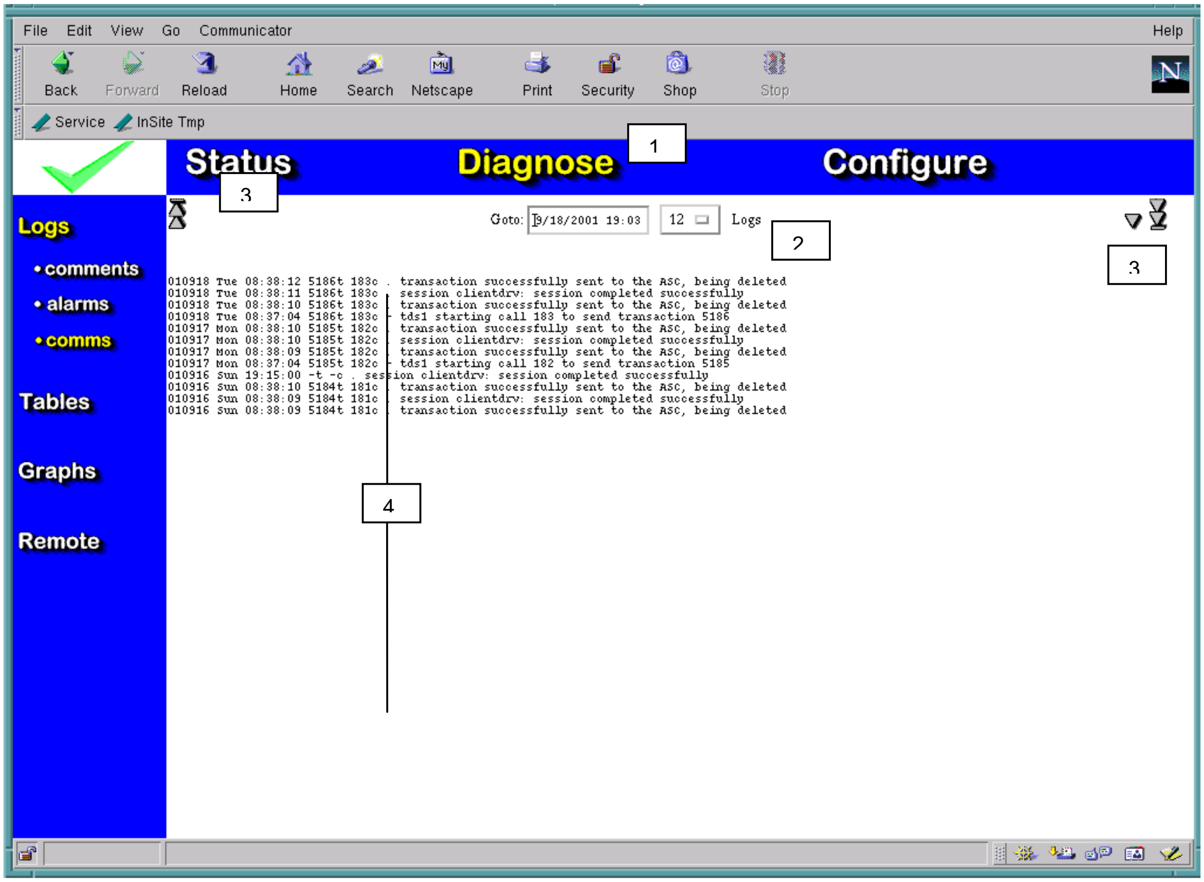


ALARMS SCREEN
ILLUSTRATION 4-7

TABLE 4-5
ALARMS SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Add	Select to add a logged entry.
4	Up/Down Arrows	As needed to scroll log entry.
5	Alarms	Recorded alarms.

Comms Screen



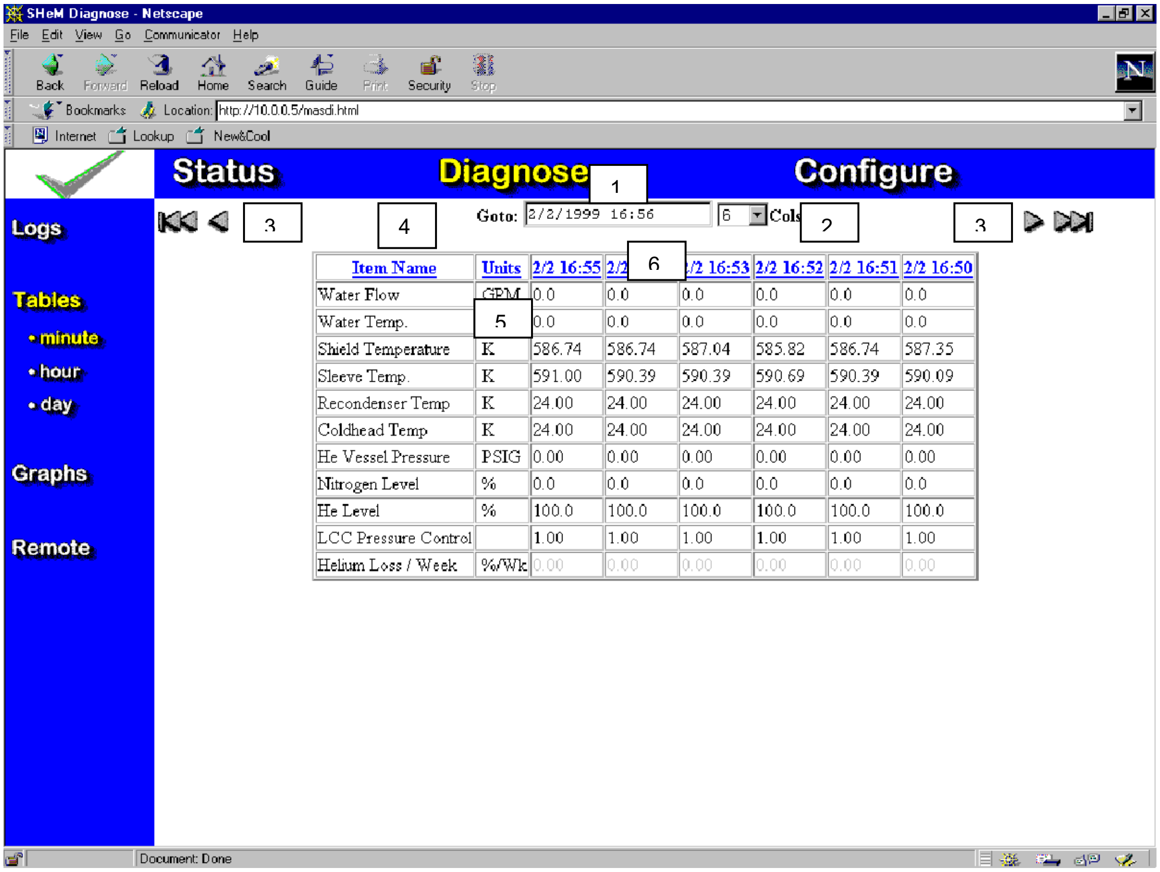
COMMS SCREEN
ILLUSTRATION 4-8

TABLE 4-6
COMMS SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Up/Down Arrows	As needed to scroll log entry.
4	Comm Information	Logged communications entries.

4-4-2 Tables Screens

Minute Screen

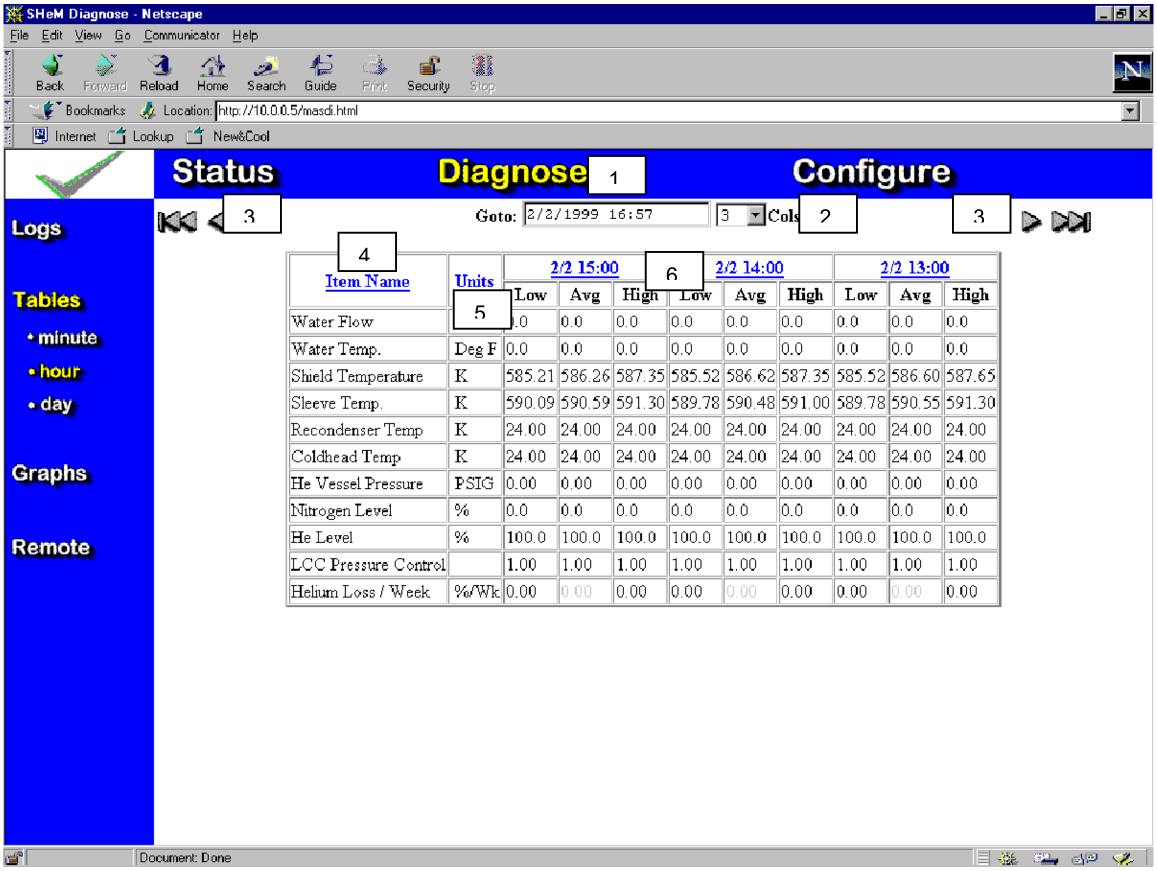


MINUTE SCREEN
ILLUSTRATION 4-9

TABLE 4-7
MINUTE SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Columns	Select to make greater/lesser columns of data available.
3	Right/left Arrows	As needed to scroll log entry.
4	Item Name	Parameter description.
5	Units	Units of measure.
6	Date/Time	Date and time data recorded.

Hour Screen

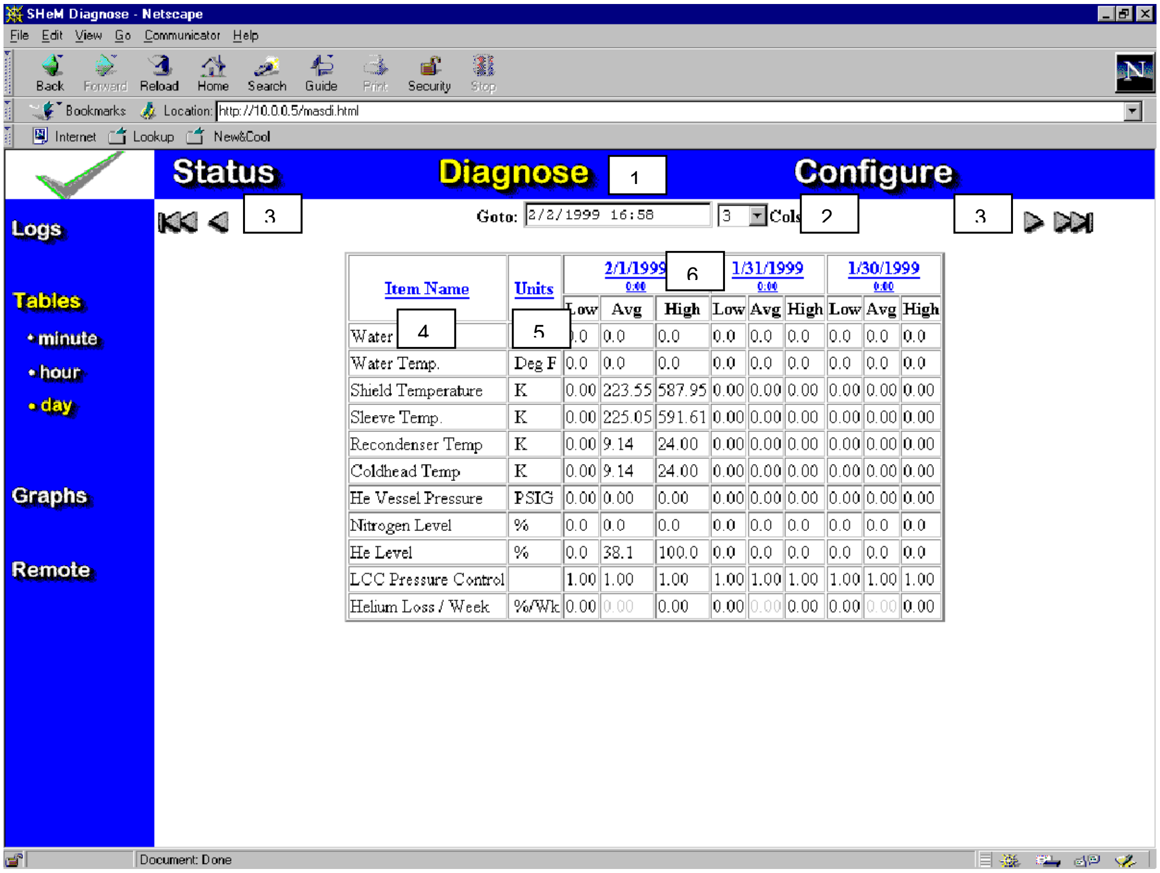


HOUR SCREEN
ILLUSTRATION 4-10

TABLE 4-8
HOUR SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Column	Select to make greater/lesser column available.
3	Right/Left Arrows	As needed to scroll column entry.
4	Item Name	Parameter description.
5	Units	Units of measure.
6	Date/Time	Date and time data recorded.

Day Screen

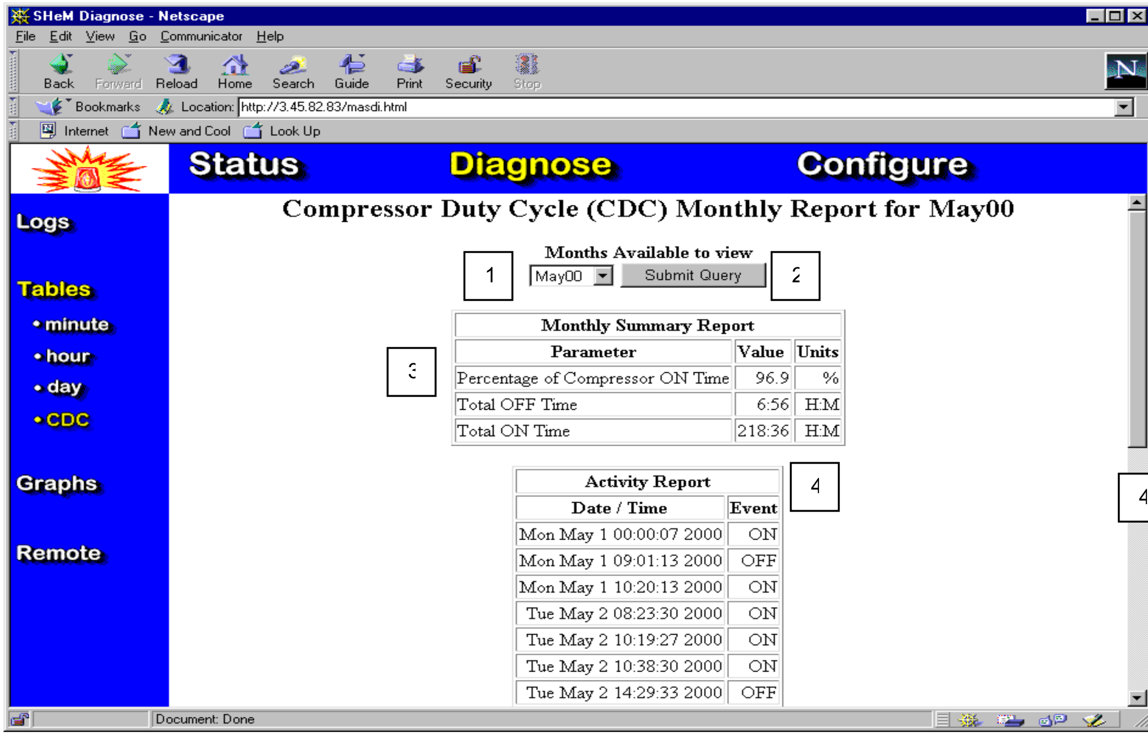


DAY SCREEN
ILLUSTRATION 4-11

TABLE 4-9
DAY SCREEN

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Column	Select to make greater/lesser column available.
3	Right/Left Arrows	As needed to scroll column entry.
4	Item Name	Parameter description.
5	Units	Units of measure.
6	Date/Time	Date and time data recorded.

Compressor Duty Cycle Screen



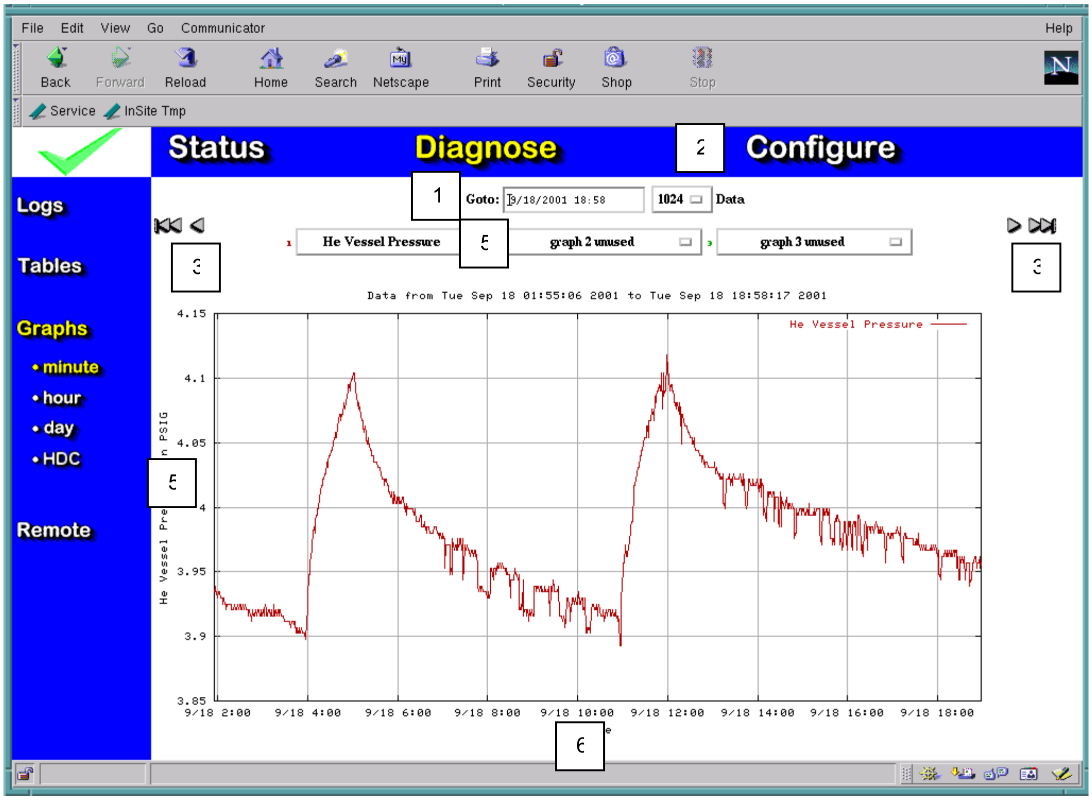
COMPRESSOR DUTY CYCLE SCREEN
ILLUSTRATION 4-12

TABLE 4-10
COMPRESSOR DUTY CYCLE SCREEN

No.	Field	Description
1	Month	Selects the month you wish to query.
2	Submit Query	Must be selected to activate query.
3	Percentage of compressor on time	Displays the percentage of time the compressor has been on for the month that was selected.
4	Activity report	Displays exact times the compressor was turned on or off. Two consecutive “ON” means the Magnet Monitor was reset.
5	Scroll bar	Allows you the ability to scroll down to view entire month of data if needed.

4-4-3 Graph Screens

Minute Graph

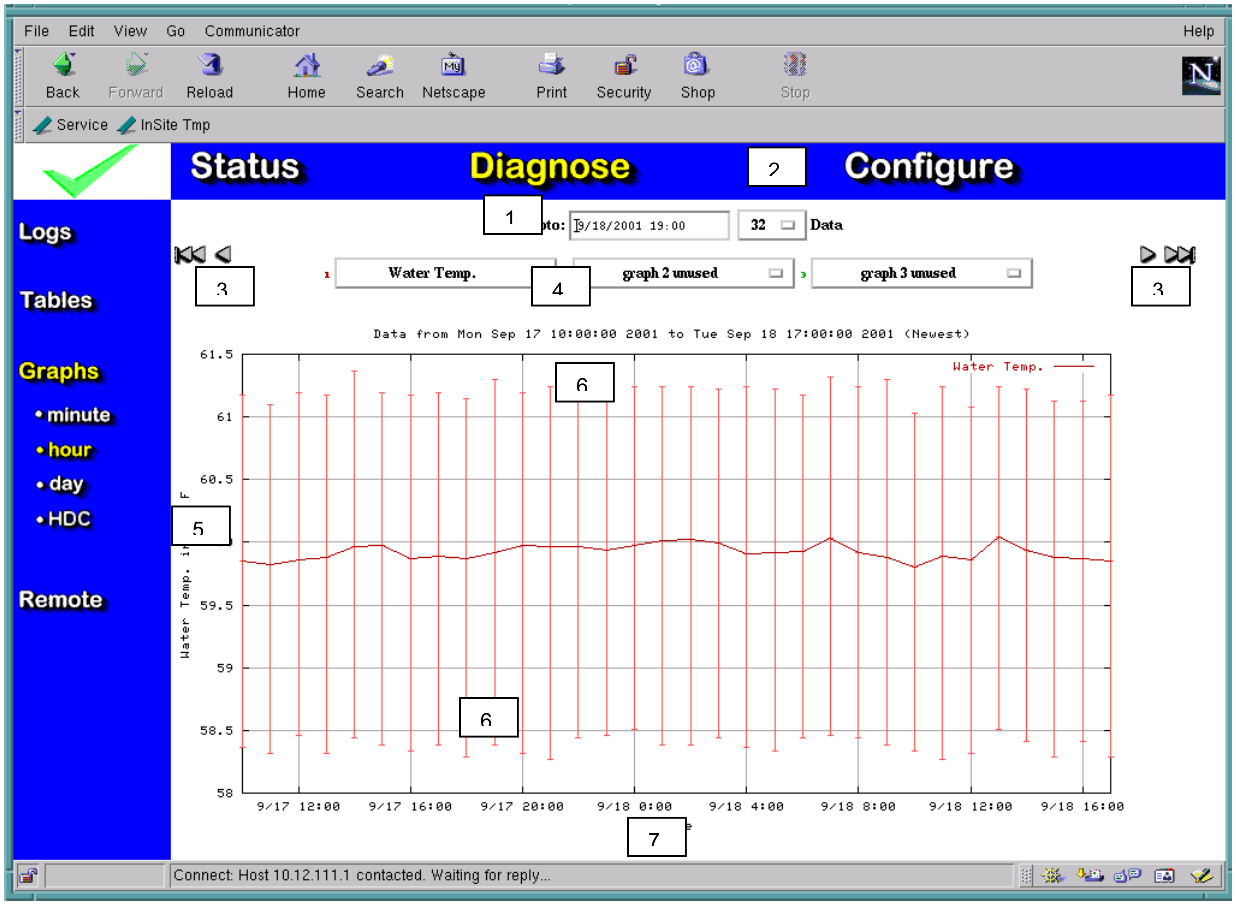


MINUTE GRAPH
ILLUSTRATION 4-12

TABLE 4-10
MINUTE GRAPH

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Data	Number of data points displayed.
3	Up/Down Arrows	As needed to scroll curve.
4	Graph Types	Parameter graph (up to three curves).
5	Magnitude Axis	Data magnitude.
6	Time Axis	Date and time of data.

Hour Graph

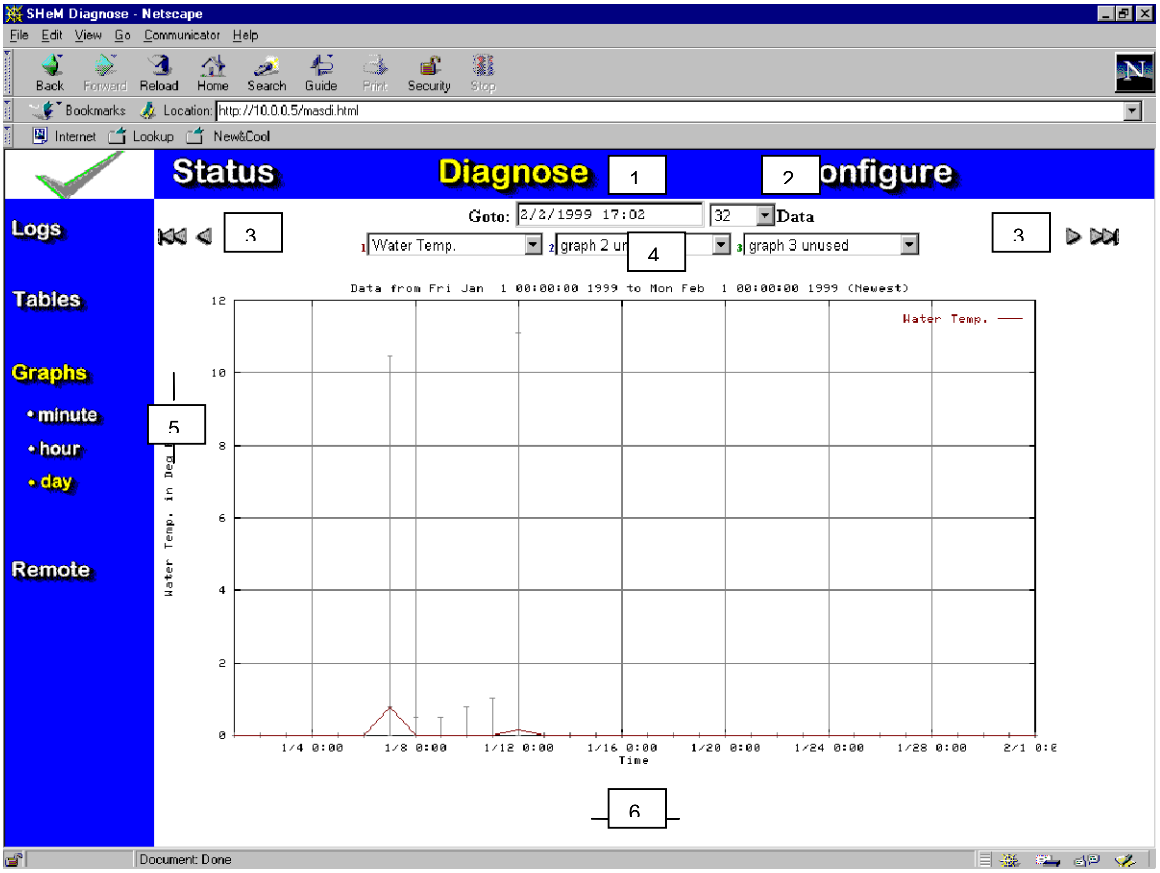


HOUR GRAPH
ILLUSTRATION 4-13

TABLE 4-11
HOUR GRAPH

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Data points	Select to make greater/lesser data points available.
3	Right/left Arrows	As needed to scroll curve.
4	Graph Types	Parameter graph (up to three curves).
5	Magnitude Axis, average.	Data magnitude averaged over an hour
6	Magnitude Axis, Min and Max	Min and Max data for the particular hour.
7	Time Axis	Date and time of data.

Day Graph

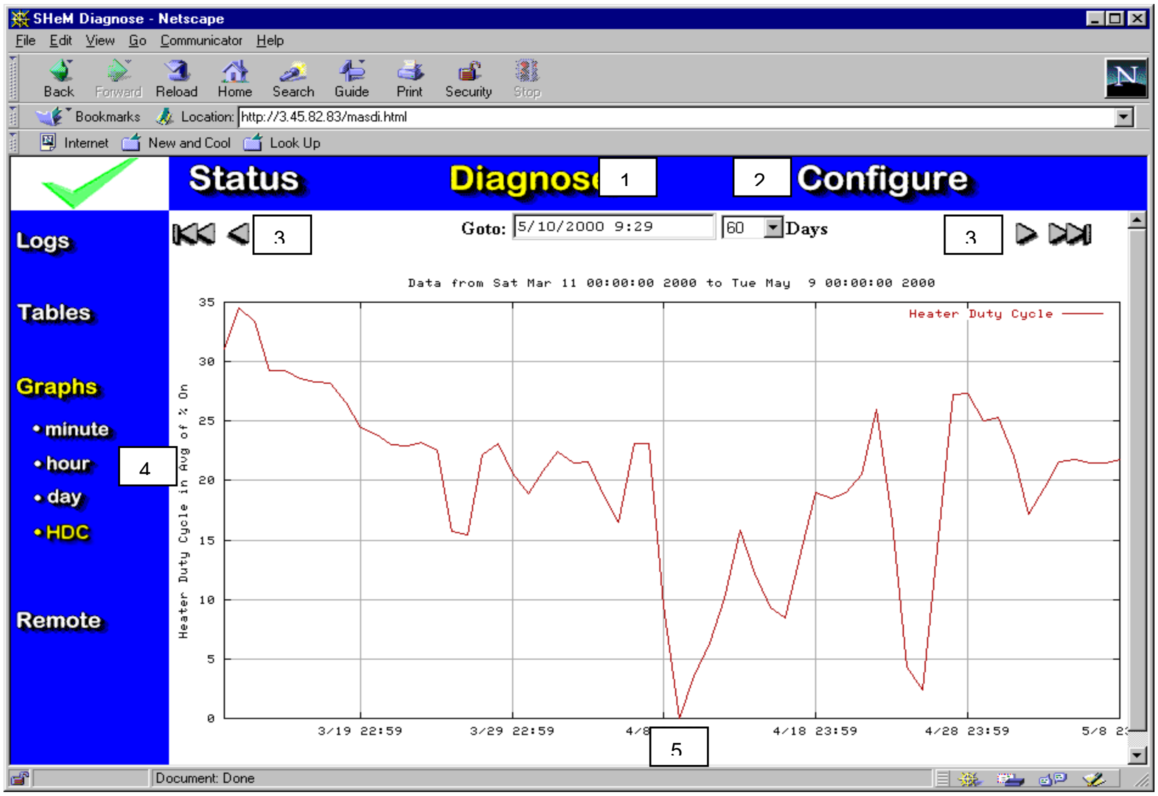


DAY GRAPH
ILLUSTRATION 4-14

TABLE 4-12
DAY GRAPH

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Data Points	Select to make greater/lesser Data Points available.
3	Right/left Arrows	As needed to scroll curve.
4	Graph Types	Parameter graph (up to three curves).
5	Magnitude Axis	Data magnitude.
6	Time Axis	Date and time of data.

Heater duty cycle graph

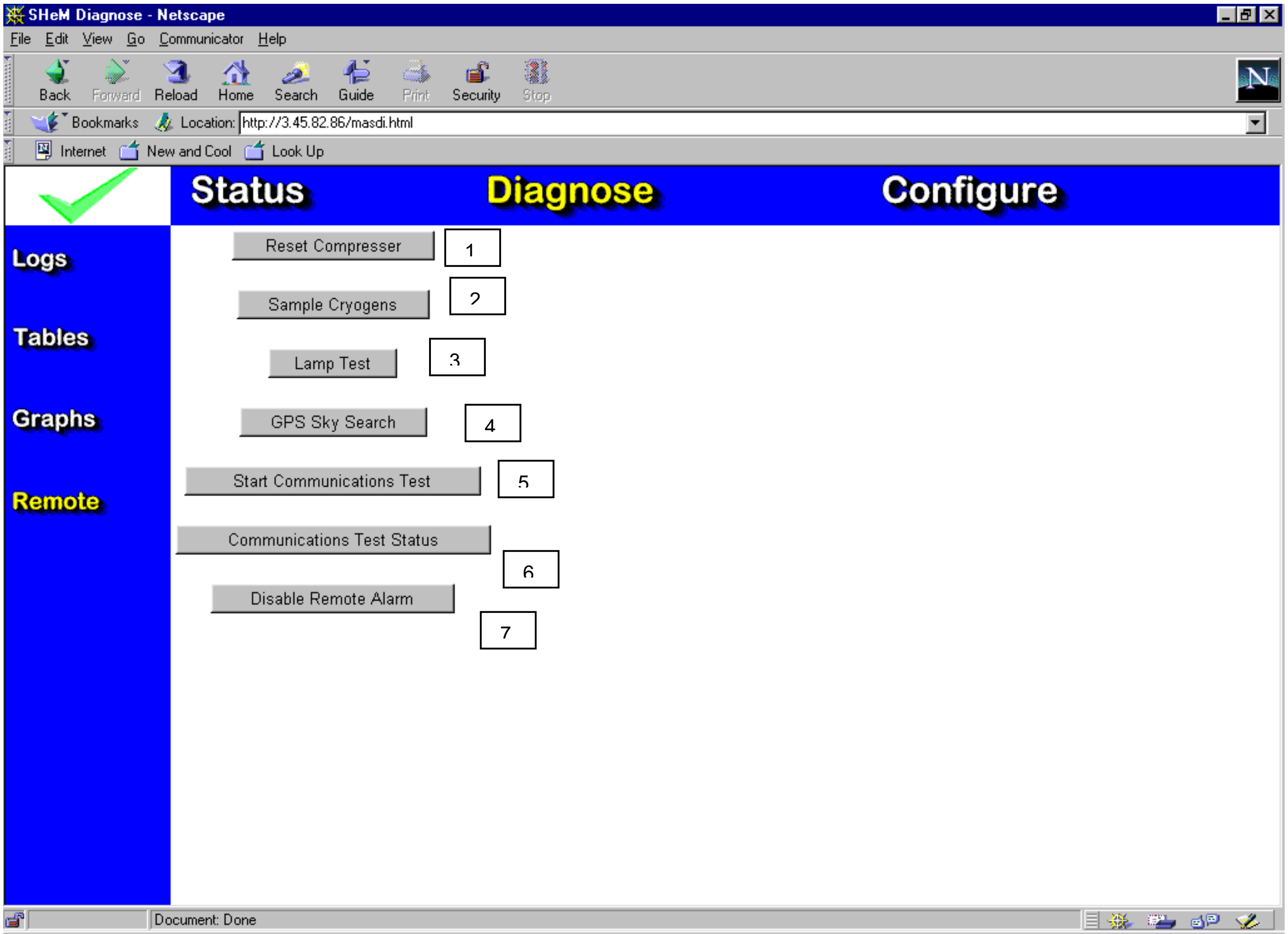


COMPRESSOR DUTY CYCLE GRAPHDAY GRAPH
ILLUSTRATION 4-15

TABLE 4-13
DAY GRAPH

No.	Field	Description
1	Date	Select and type in to view particular dates.
2	Logs	Select to make greater/lesser logs available.
3	Right/left Arrows	As needed to scroll curve.
4	Magnitude Axis	Data magnitude.
5	Time Axis	Date and time of data.

4-4-4 Remote Screen



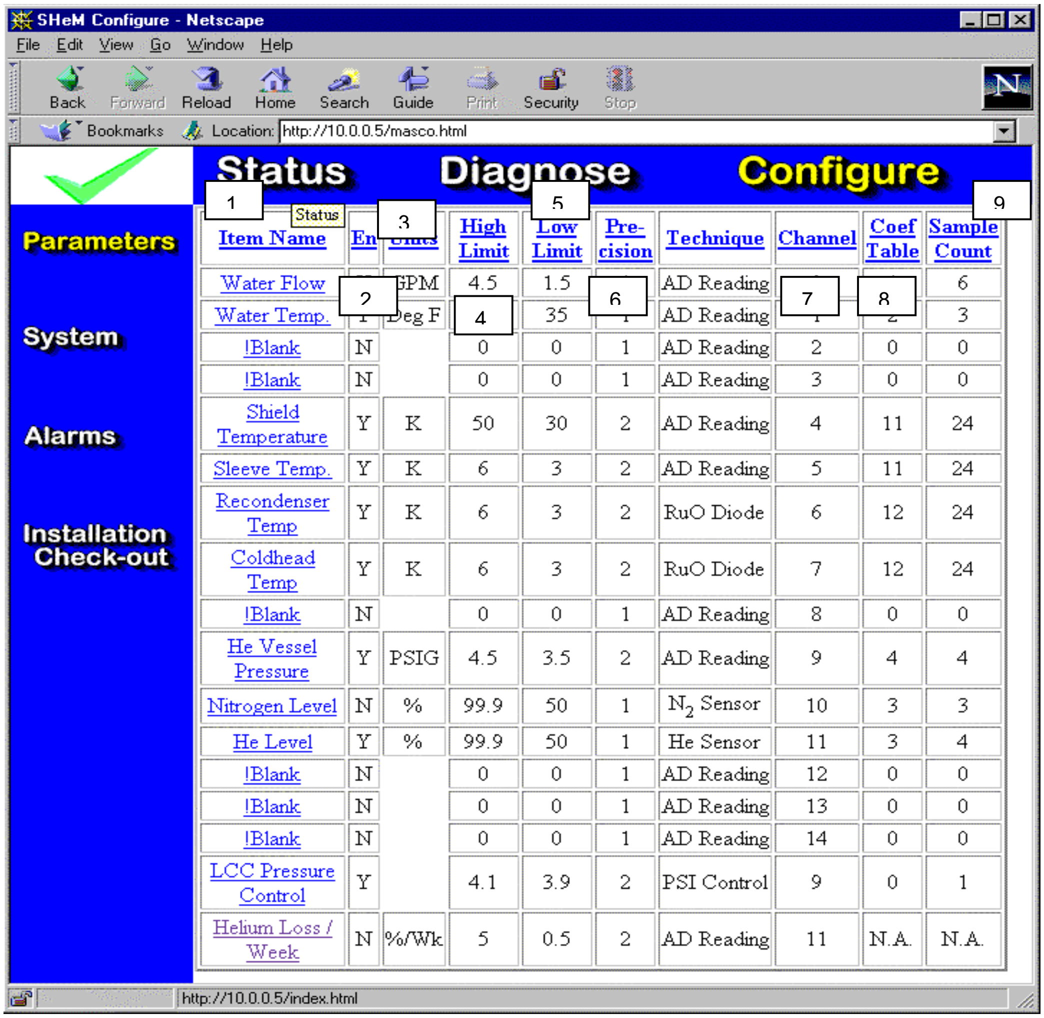
REMOTE SCREEN
ILLUSTRATION 4-16

TABLE 4-14
REMOTE SCREEN

No.	Button	Description
1	[Reset Compressor]	Resets compressor from support center location.
2	[Sample Cryogens]	Initiates a cryogen sample from support center location.
3	[Lamp Test]	Activates MM local display (all segments).
4	[GPS Sky Search]	Not currently functional.
5	[Start Communications Test]	Initiate communications check.
6	[Communications Test Status]	Display results.
7	Remote alarm disable	Used for HFO only

4-5 Configure Screens

4-5-1 Parameters Screen



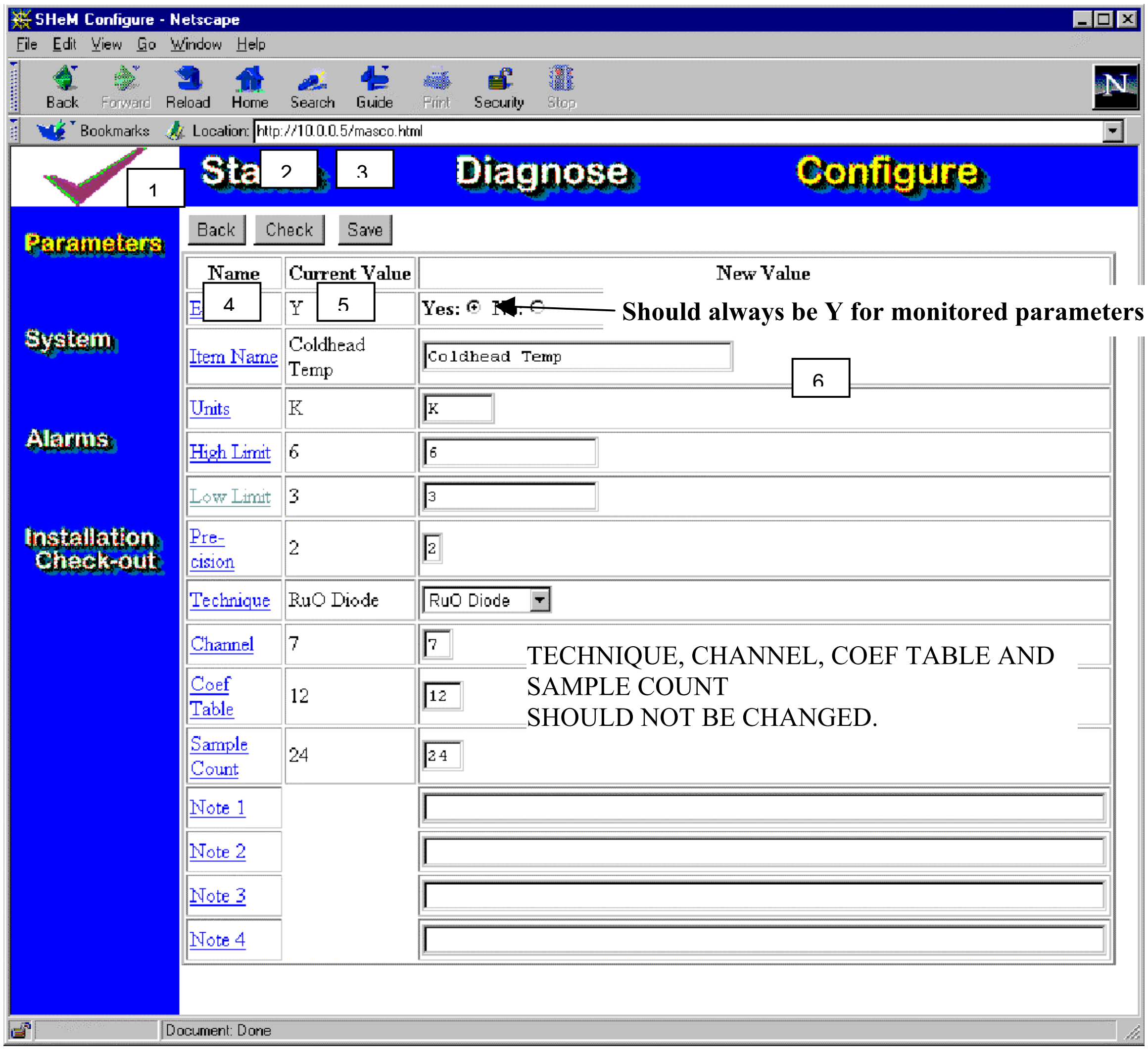
PARAMETERS SCREEN
ILLUSTRATION 4-17

TABLE 4-15
PARAMETERS SCREEN

No.	Column	Description
1	Item Name	Name of the monitored parameter.
2	Enabled	Enable indicator. The display shows `Y` when parameter monitoring is enabled and `N` when disabled. **This should always remain Y for monitored parameters.**
3	Units	Units of measure.
4	High Limit	Value above which causes an alarm condition.
5	Low Limit	Value below which causes an alarm condition.
6	Precision	Displayed precision of data (X.X, X.XX…).
7	Channel	Analog channel assignment (A/D Conv).
8	Coef Table	Coefficient Table. Indicates which coefficient table is used for error compensation.
9	Sample Count	Number of samples to record before averaging.

You can edit each value for each parameter by clicking on the Item Name. This opens a single parameter screen with edit fields for each value for the selected parameter. An example of a single parameter screen is shown below.

Individual Parameter Screen

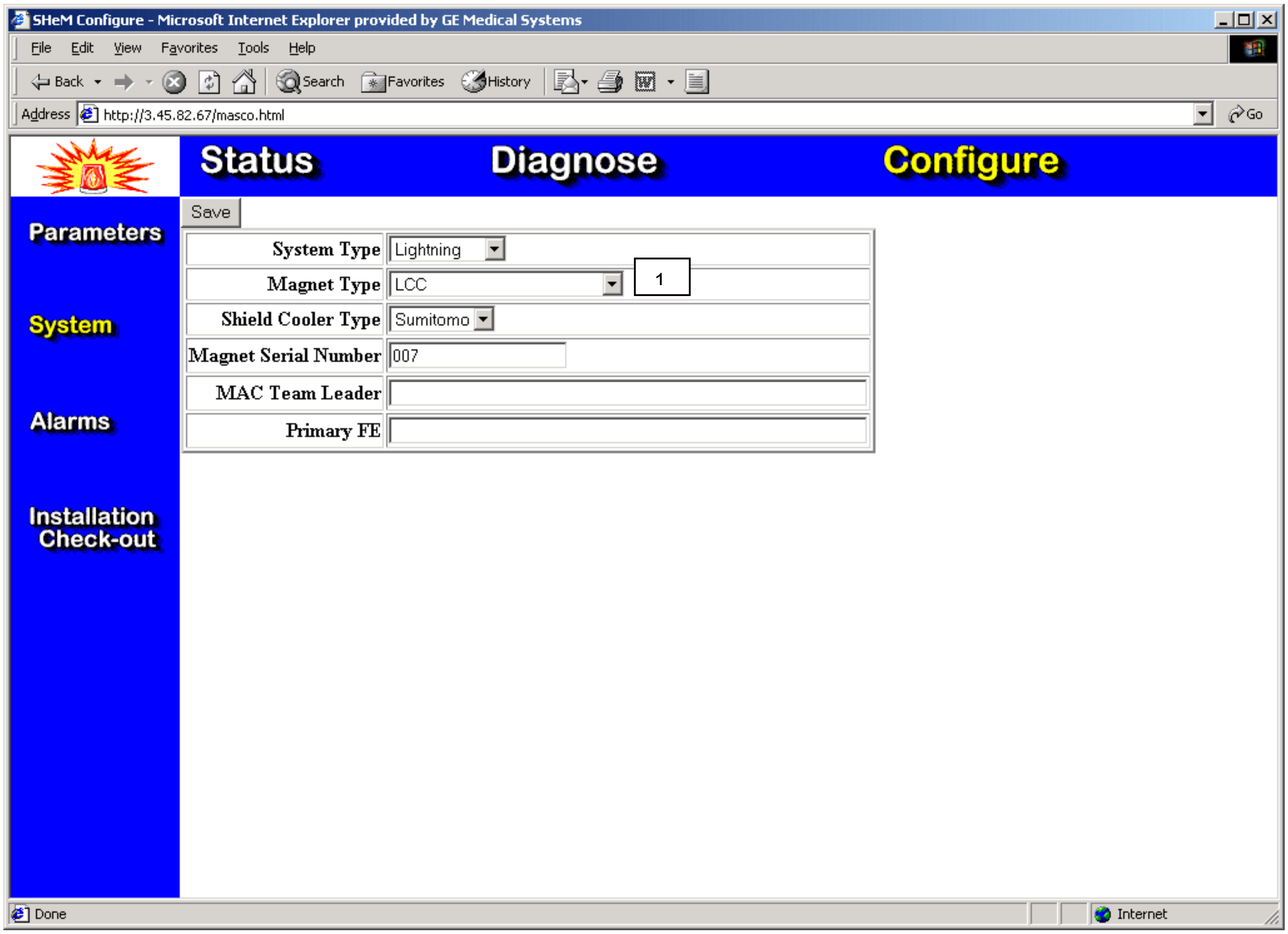


SINGLE PARAMETER SCREEN
ILLUSTRATION 4-18

TABLE 4-16
SINGLE PARAMETER SCREEN

No.	Name	Description
1	[Back] button	Returns you to the Configure Parameters screen.
2	[Check] button	
3	[Save] button	Saves the new parameter values.
4	Name field	Name of the parameter whose values are being modified.
5	Current Value field	Displays the current parameter value settings.
6	New Value field/toggle	Fields and toggle for typing in new values or toggling to a different setting.

4-5- 2 System Screen

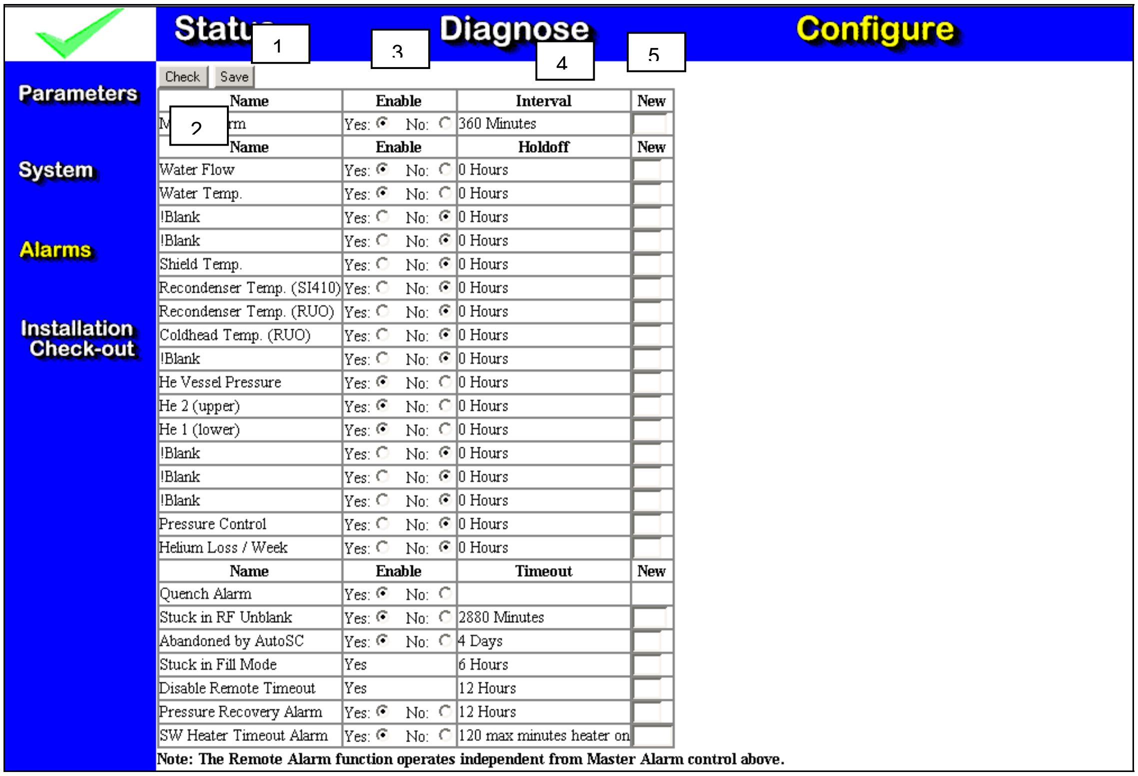


SYSTEM SCREEN
ILLUSTRATION 4-19

TABLE 4-17
SYSTEM SCREEN

No.	Name	Description
1	System Information	Reflects data entered during the install process.

Alarm Screen

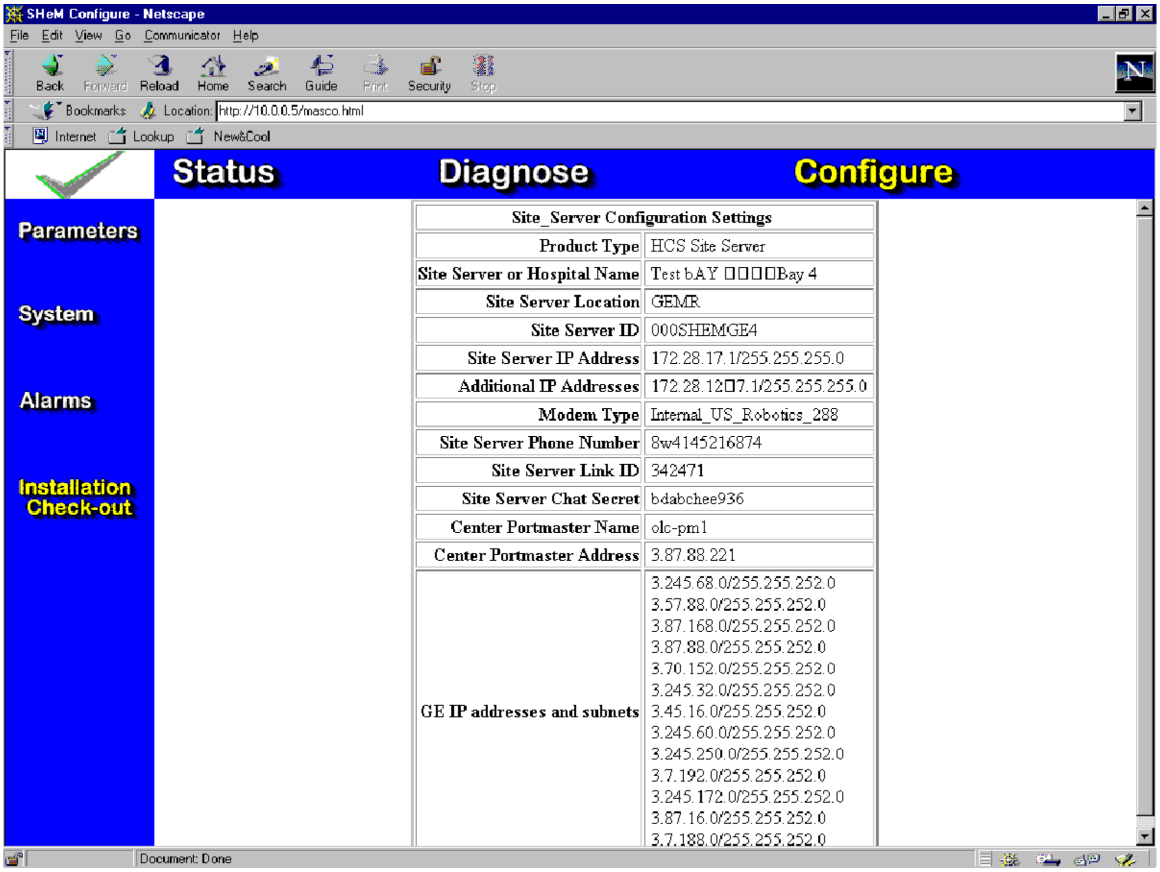


ALARM SCREEN
ILLUSTRATION 4-20

TABLE 4-19
ALARMS SCREEN

No.	Name	Description
1	[Save] button	Saves the modified alarm settings.
2	Name field	Name of the alarm parameter.
3	Enable	Enables or disables the alarm for the selected parameter.
4	Interval/Holdoff	Interval: time for holding off master alarm in minute values between 1 and 720 minutes(12 hours). Holdoff: Set time to delay before enabling an alarm in hours between 1 and 99. Clicking on the *Check* button will display the time left for alarm hold-off.
5	New	Allows new values for Interval and Holdoff to be entered.

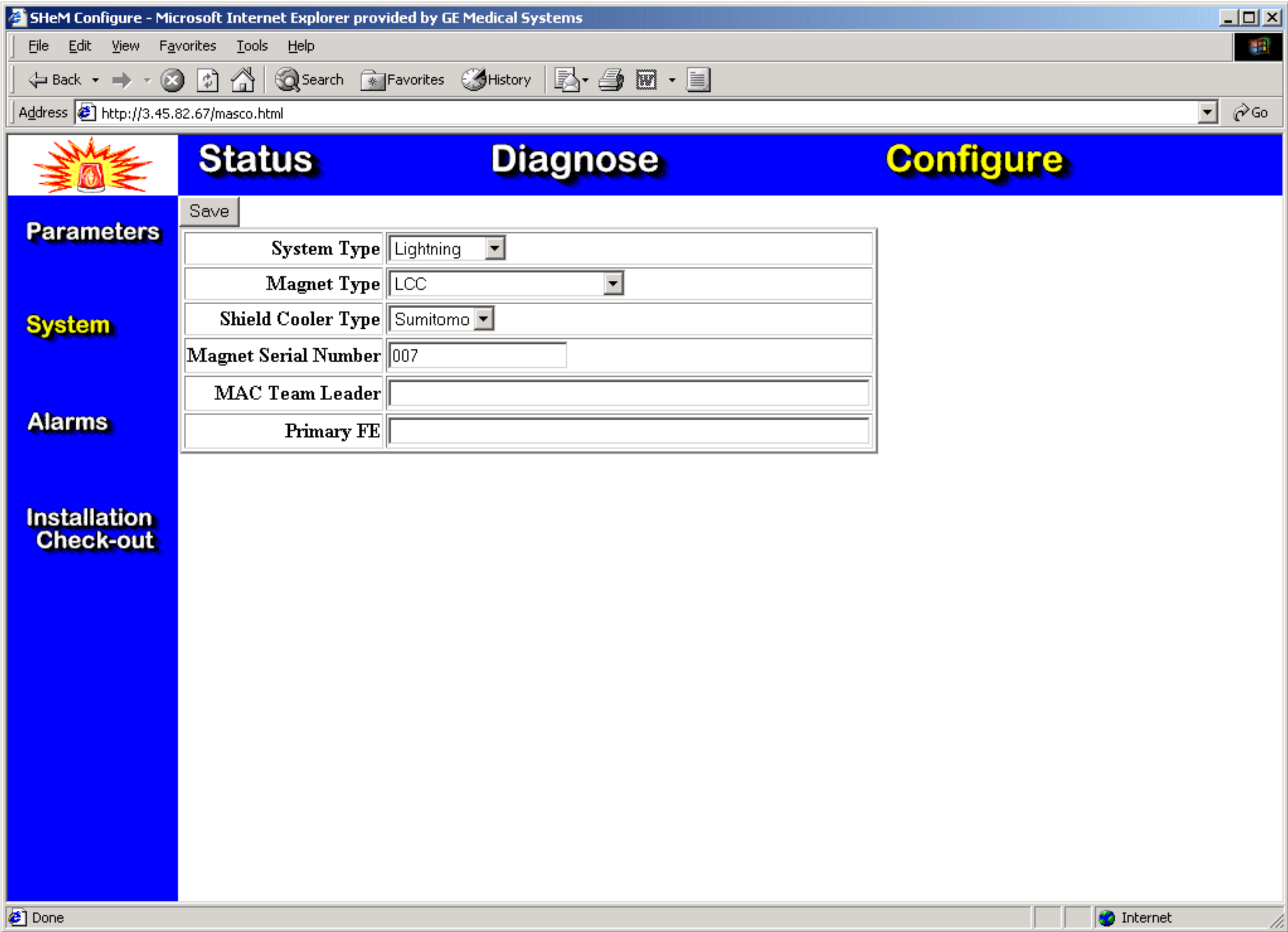
4-5-4 Installation Check-out Screen



INSTALLATION CHECK-OUT SCREEN
ILLUSTRATION 4-20

This screen reflects data downloaded to the MM by the OLC after initial check-out.

5- SYSTEM CONFIGURATION



SYSTEM CONFIGURATION SCREEN
ILLUSTRATION 5-1

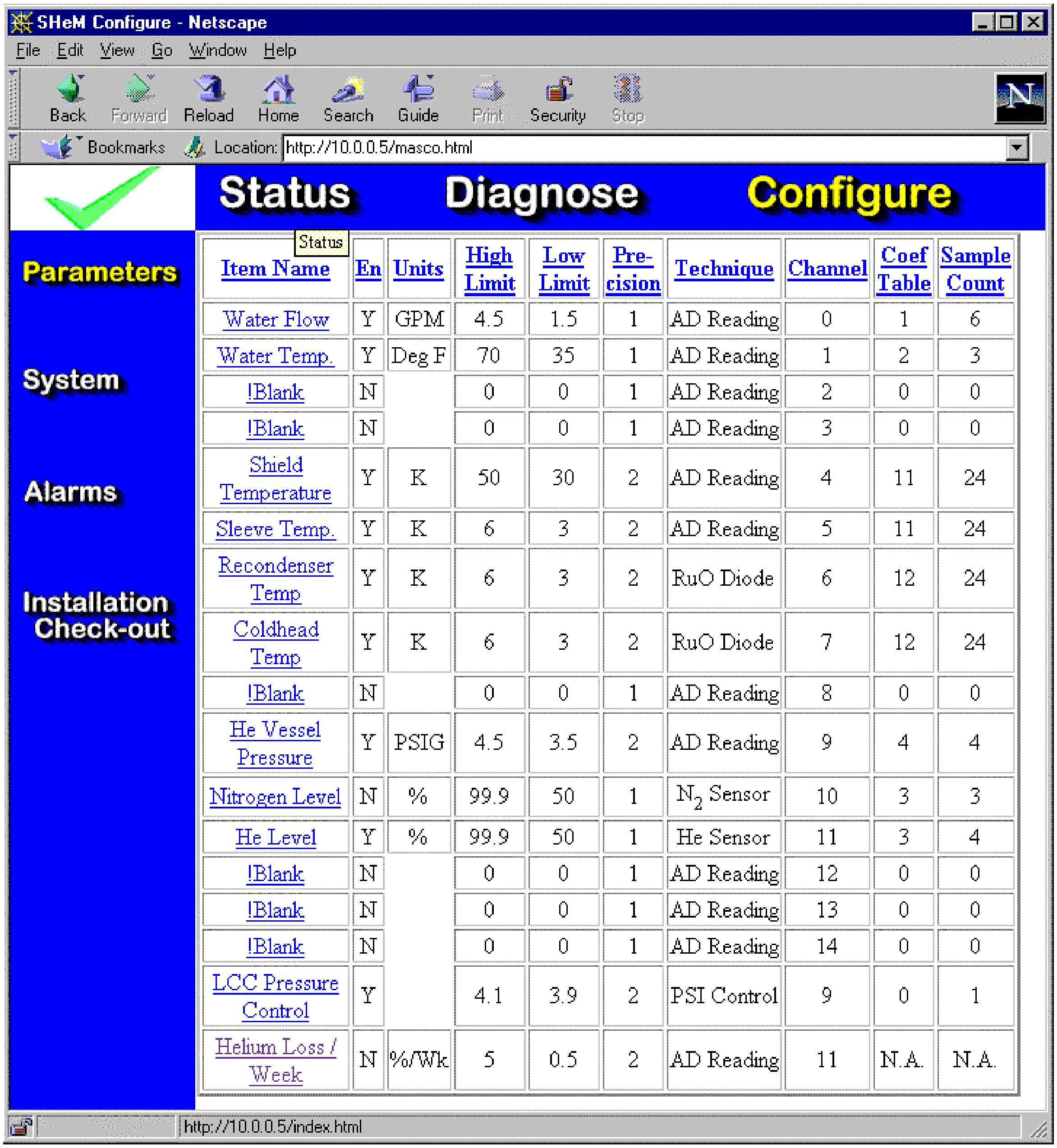
Go to the Configure System screen.

The information displayed here is from the original setup script.

You can change the following:

- Magnet type
- Magnet Serial number
- MAC team leader
- Primary FE

6- PARAMETER SETUP



Item Name	En	Units	High Limit	Low Limit	Pre-cision	Technique	Channel	Coef Table	Sample Count
Water Flow	Y	GPM	4.5	1.5	1	AD Reading	0	1	6
Water Temp.	Y	Deg F	70	35	1	AD Reading	1	2	3
!Blank	N		0	0	1	AD Reading	2	0	0
!Blank	N		0	0	1	AD Reading	3	0	0
Shield Temperature	Y	K	50	30	2	AD Reading	4	11	24
Sleeve Temp.	Y	K	6	3	2	AD Reading	5	11	24
Recondenser Temp	Y	K	6	3	2	RuO Diode	6	12	24
Coldhead Temp	Y	K	6	3	2	RuO Diode	7	12	24
!Blank	N		0	0	1	AD Reading	8	0	0
He Vessel Pressure	Y	PSIG	4.5	3.5	2	AD Reading	9	4	4
Nitrogen Level	N	%	99.9	50	1	N_2 Sensor	10	3	3
He Level	Y	%	99.9	50	1	He Sensor	11	3	4
!Blank	N		0	0	1	AD Reading	12	0	0
!Blank	N		0	0	1	AD Reading	13	0	0
!Blank	N		0	0	1	AD Reading	14	0	0
LCC Pressure Control	Y		4.1	3.9	2	PSI Control	9	0	1
Helium Loss / Week	N	%/Wk	5	0.5	2	AD Reading	11	N.A.	N.A.

CONFIGURE PARAMETERS SCREEN
ILLUSTRATION 6-1

1. Go to the Configure Parameters screen.
2. Click on an Item Name to edit the fields for that parameter. See table 6-1 for default values for each parameter for the different magnet types.
3. While it is possible, using this method, to alter various parameter settings, it is recommended that you do not change these settings unless specifically required to do so.

IMPORTANT

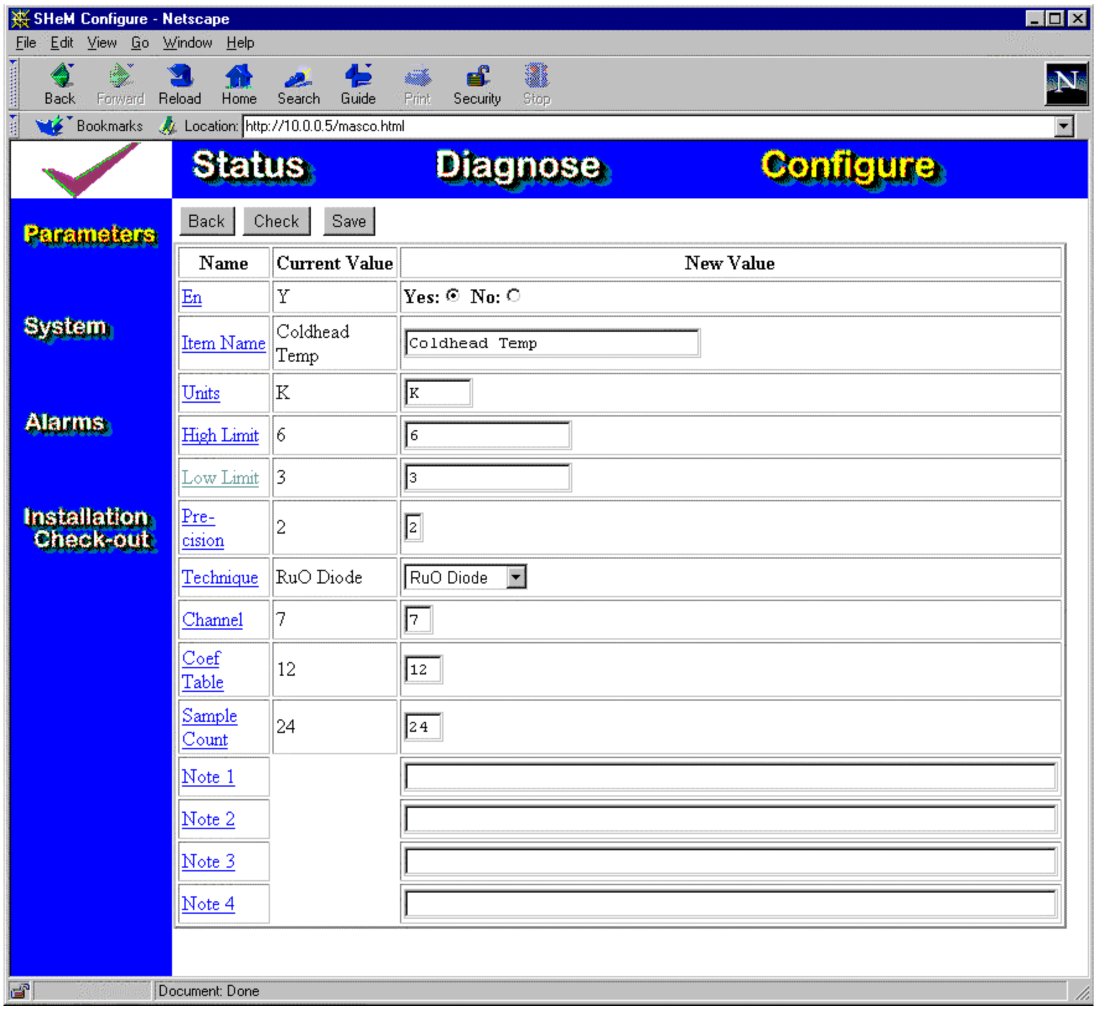
The enable value should always be Y for monitored parameters. Use the Alarm's page to turn off or disable individual alarms.

TABLE 6-1
PARAMETER VALUES

LCC 1.5T & 1.0T magnets			
Parameter name	**High limit**	**Low Limit**	**Precision**
Water flow	4.5	1.3	1
Water temp	75 degrees F	40 degrees F	1
Shield temp(SI-410)	50 K	30K	2
Sleeve temp(SI-410)	6 K	3 K	2
Recondenser temp(RUO)	5 K	3 K	2
Coldhead temp(RUO)	5 K	3 K	2
Magnet vessel pressure	4.50 psi	3.75psi	2
Helium level	110 %	* see note	1
LCC pressure control	4.1psi	3.9psi	1

* For LCC magnets we recommend setting the helium low level to about 5% below the helium level at the time of magnet monitor checkout. The magnet monitor would then alarm if the magnet would develop a slow leak. The absolute minimum helium level for the LCC magnet is 55%.

INDIVIDUAL PARAMETER SCREEN



INDIVIDUAL PARAMETER SCREEN
ILLUSTRATION 6-2

This is the screen you would see if you clicked on an Item Name from the configure Parameters screen(Illustration 6-1).

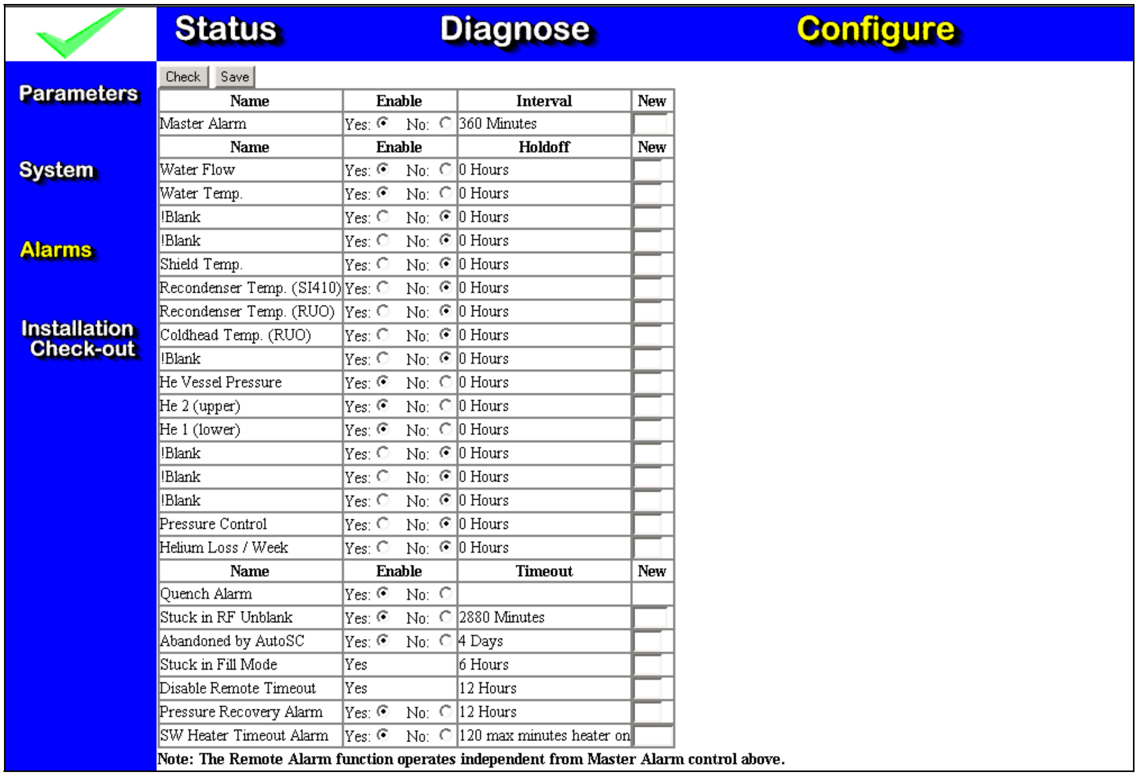
These parameters can be altered by typing a new value into a field and pressing [Save]. It is not recommended that these selections be modified unless specifically required to do so.

IMPORTANT

The enable value should always be Y for monitored parameters. Use the Alarm's page to turn off or disable individual alarms.

7- ALARMS

7-1 ALARM SETUP



Name	Enable	Interval	New
Master Alarm	Yes: ● No: ○	360 Minutes	
Name	**Enable**	**Holdoff**	**New**
Water Flow	Yes: ● No: ○	0 Hours	
Water Temp.	Yes: ● No: ○	0 Hours	
!Blank	Yes: ○ No: ●	0 Hours	
!Blank	Yes: ○ No: ●	0 Hours	
Shield Temp.	Yes: ○ No: ●	0 Hours	
Recondenser Temp. (SI410)	Yes: ○ No: ●	0 Hours	
Recondenser Temp. (RUO)	Yes: ○ No: ●	0 Hours	
Coldhead Temp. (RUO)	Yes: ○ No: ●	0 Hours	
!Blank	Yes: ○ No: ●	0 Hours	
He Vessel Pressure	Yes: ● No: ○	0 Hours	
He 2 (upper)	Yes: ● No: ○	0 Hours	
He 1 (lower)	Yes: ● No: ○	0 Hours	
!Blank	Yes: ○ No: ●	0 Hours	
!Blank	Yes: ○ No: ●	0 Hours	
!Blank	Yes: ○ No: ●	0 Hours	
Pressure Control	Yes: ○ No: ●	0 Hours	
Helium Loss / Week	Yes: ○ No: ●	0 Hours	
Name	**Enable**	**Timeout**	**New**
Quench Alarm	Yes: ● No: ○		
Stuck in RF Unblank	Yes: ● No: ○	2880 Minutes	
Abandoned by AutoSC	Yes: ● No: ○	4 Days	
Stuck in Fill Mode	Yes	6 Hours	
Disable Remote Timeout	Yes	12 Hours	
Pressure Recovery Alarm	Yes: ● No: ○	12 Hours	
SW Heater Timeout Alarm	Yes: ● No: ○	120 max minutes heater on	

Note: The Remote Alarm function operates independent from Master Alarm control above.

Configure Alarms Screen
Illustration 7-1

1. Go to the Configure Alarms screen.

2. Set the following parameters to the specified values:

   - Enable (Yes/No)

   Normally all parameters that are being monitored must be enabled.

   - New Interval/Holdoff

3. Click on the [Save] button.

IMPORTANT

The value for SW Heater Timeout Alarm should be set to 2880 minutes. If the magnet heater is on longer that the value for the SW Heater Timeout Alarm, the magnet heater will turn off, and an alarm will be sent to OLC, if a fill has not been performed in the last 24 hours.
2880 minutes is equal to 48 hours.

7-2 Alarm Activation

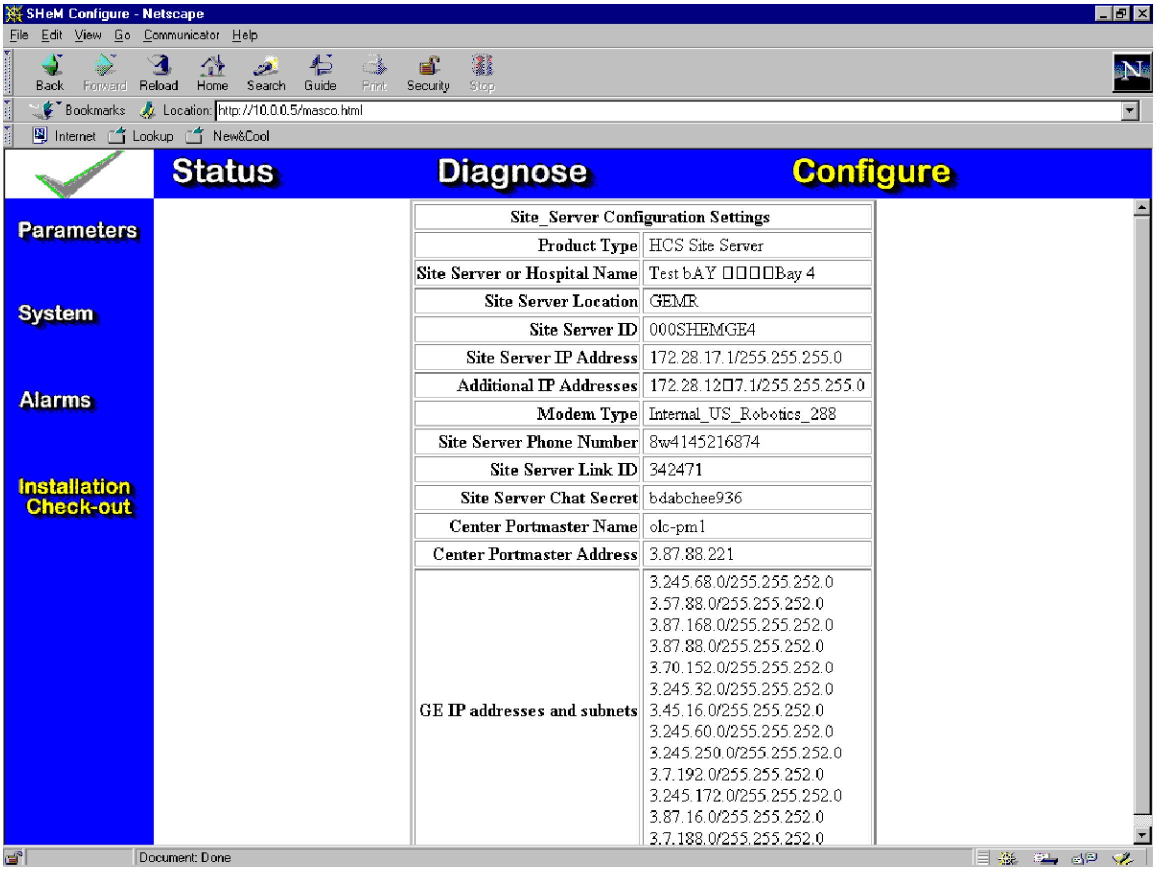
All alarms noted in Illustration 7-1 can be Enabled (select **Yes**) or Disabled (select **No**).

The “digital” alarms listed in Table 7-1 will report to the OLC.

TABLE 7-1
ALARM STATUS

Alarm	Alarm to OLC?	Can Disable	Explanation
Quench Alarm	Y	Y	Activates when magnetic reed switch does not detect magnetic field
Stuck in RF Unblank	Y	Y	Activates when RF unblank signal is activated for greater than 6 hours **and** System cabinet power is detected
Abandoned by AutoSC	Y	Y	Activates when connection from AutoSC to magnet monitor has failed. Mag Mon can possibly still dial out to AutoSC
Stuck in Fill mode	Y	Y	Activates when key is in “Fill” position has been on for greater than number of minutes in the **Timeout** section.
Disable remote timeout	N	N	Used for HFO only
SW heater timeout	Y	Y	Activates when magnet heater has been on for greater than number of minutes in the **Timeout** section. Value for LCC magnet should be 5000.
Unable to reset compressor	Y	N	Automatic restart of compressor failed. Compressor restart setup is part of magnet monitor configuration setup.
Fill Notification	Y	N	Sends fill data to OLC
Latched pressure detected	Y	N	Activates if magnet monitor reports the same pressure reading for 60, 1 minute samples.
UPS Power Failure	Y	N	Activates when magnet monitor is on UPS power.
Helium meter error	Y	N	Activates when magnet monitor does not detect proper current going out to magnet.
Touch	Y	N	Used for HFO Only.

8- SYSTEM FUNCTIONAL TESTS



CONFIGURE INSTALLATION CHECK-OUT SCREEN
ILLUSTRATION 8-1

1. Go to the Configure Installation Check-out screen.

2. View the following parameters:

   - Product Type
   - Magnet Monitor or Hospital Name
   - Magnet Monitor Location
   - Magnet Monitor ID
   - Magnet Monitor IP Address
   - Additional IP Addresses
   - Modem Type
   - Magnet Monitor Phone Number
   - Magnet Monitor Link ID
   - Magnet Monitor Chat Secret
   - Center Portmaster Name
   - Center Portmaster Address
   - GE IP addresses and subnets

3. These parameters are set by the OLC after check-out is successfully performed.

9- TROUBLESHOOTING

9-1 TROUBLESHOOTING TABLES

TABLE 9-1
MAGNET MONITOR FAULT ISOLATION GUIDELINES

Symptom	Suggested Actions
Your local support center is unable to call into the Magnet Monitor	Check phone connections between the Magnet Monitor and the customer phone line. Make sure the phone line is a direct connection (no switch board), and is not shared with other devices Try bypassing the multiplexer box. Make sure the support center has the correct phone number for this site, including the multiplexer code if applicable. Run the COMMUNICATION TEST command and make sure the Magnet Monitor can dial out. The test can be found in the Status/Remote field. Verify the site phone line connection by plugging in a telephone. Try calling this telephone from outside the site. If using an external modem, verify that it is configured to auto-answer incoming calls.
The Magnet Monitor does not call PPADS when an alarm is detected.	Verify that the Master Alarm time has elapsed, alarms are not reported immediately, they must wait by this amount of time, check in Configure/Alarms. Verify proper dial-out prefix (9 or 9,1 etc). Verify that the parameter in question does not have a hold-off time set, check in Configure/Alarms Verify that the MASTER ALARM in the status/alarm field is set to enable. Also verify that all pertinent alarms are enabled. Check the COMMS field to see if there is any error messages that might assist you in determining the problem. The COMMS field is found in Status/Log/Comms. Run the **Toucholc.pl** command and make sure the Magnet Monitor can dial out. The test can be found in the Status/Remote field. Verify the phone line can make outgoing calls by plugging a telephone into the line and calling PPADS, or a different 800 number. The PPADS server may be inoperative, try again later. The phone system at the site may be trying to block toll free numbers. Work with the customer to change the phone system programming.
The Magnet Monitor does not properly reset.	Unless specifically requested to boot from a floppy disk, verify that there is no floppy disk in the drive.
Spike noise in the image	Check all connectors for tight mechanical connections, jackscrews and other hardware must be secure. Verify that none of the connectors are laying against the magnet, or other conductive surfaces.

TABLE 9-2
PARAMETER FAULT ISOLATION

Symptom	Suggested Actions
Shield temperature/s out of range.	If temperature reading is either 0K or 592K, check cabling between Magnet Monitor and sensors. This type of reading can indicate an open circuit. Check the shield temperature/s using a lakeshore thermometer. If the measured temperatures differ check the continuity of the cables between the Magnet Monitor and the sensors. Verify the 10 ua current source by placing a microamp meter in series between pins 1 & 2 or 11 & 12 or 15 &16 or 22 & 23, of the J8 connector at the Magnet Monitor. The first pin is positive, the second is negative. Disconnect the cable while performing test.
RuO temps out of range	If temperature reading is either 1.5K or 24K, check cabling between Magnet Monitor and sensors. This type of reading can indicate an open circuit. Also verify that the RuO Buffer amplifier, located at the magnet, is properly connected. Check the RuO temperature/s using an RUO meter. If the measured temperatures differ check the continuity of the cables between the Magnet Monitor and the sensors. Verify the 10 ua current source by placing a microamp meter in series between pins 1 & 2 or 11 & 12 or 15 &16 or 22 & 23, of the J8 connector at the Magnet Monitor. The first pin is positive, the second is negative. Disconnect the cable while performing test.
Vessel pressure out of range	If the vessel pressure reading is close to zero, check the actual pressure in the magnet using the mechanical gauge. You may open the V2 valve to verify that there is positive pressure in the magnet. The plumbing will frost up in about 10 seconds if V2 is opened and there is positive pressure in the magnet. If you disconnect the cable at the pressure transducer at the magnet, the Magnet Monitor should indicate a magnet pressure of zero. The cabling between the pressure transducer and the Magnet Monitor can be checked. Disconnect the cable at the pressure transducer. Place a 1.5vdc battery on pin 7 positive side and pin 6 negative side of J8 (Scan Room) at the Magnet Monitor. The Magnet Monitor should then read about 1.8 psi.
Compressor Water flow or water temperature out of range.	Check water source to compressor. Check connections to water flow meter. Verify that water is flowing.
System Power always active or inactive	Verify that the “Line trigger” signal from the PDU is correct. This is an AC 60Hz 6.3 Vrms signal. This signal should be seen at the connector coming into J10 of the Magnet Monitor, on pins 10 (+) and 11 whenever the system cabinet is powered up.

TABLE 9-2 (Continued)
PARAMETER FAULT ISOLATION

Symptom	Suggested Actions
Helium level out of range	If you suspect there is a problem with the helium reading, try using a different helium meter to verify the helium level in the magnet. If the readings differ do the helium meter calibration, located in the "Magnet Monitor Hardware Installation" manual, section 4-4. If the helium reading is consistently around 100% check the cabling between the Magnet Monitor and the helium sensors in the magnet. Also verify that a current of 75 ma is measured between pins 1 & 7, J7 of the Magnet Monitor when the sample button is pressed. Disconnect the cable while performing this test. **Caution: The voltage at these pins can be as high as 100 VDC. Use appropriate safety measures, and do NOT allow either pin to contact any other point other than your meter leads. Damage will result.**
RF-Unblank always active or inactive	If the RF-Unblank is always active, a logic low at pin 6 (+) and pin 7 (-) of the cable coming into J10 on the Magnet Monitor, check cabling at system cabinet. The cable used for the Unblank signal should connect to J9 (RF UNBLNK 1) of the IPG board in the Systems cabinet. On Signa 3X systems, verify SMB/BNC connector is plugged into J113 UNBLANK. On Signa 4X and 5X systems, verify the cable is plugged into J4 TR/DR UNBLK on the exciter board. Check continuity of the cabling between the UNBLANK connection in the systems cabinet and the Magnet Monitor.
No Field is active.	Verify the reed switch is functioning. Disconnect the wires at the reed switch and short them together. The NO FIELD message should appear at the Magnet Monitor. An open circuit at the Switch indicates magnetic field. Change the orientation of the reed switch while in the magnetic field to try and activate the reed switch properly.
"Low Helium Level" message appears at the console	Jumper pins 4 and 6 together on connector J5(cryo mon) on the back of the TYME board in the SC cabinet.
Message **ER1** appears on the seven segment display of an Advantech Magnet Monitor	This means that there is an open circuit in the helium cable between the Magnet Monitor and the magnet, or the Magnet Monitor is not sending the proper current to the magnet for an accurate helium reading. Check cables between magnet and Magnet Monitor. If continuity is OK check for 75ma coming out of the magnet monitor
Message **ER2** appears on the seven segment display of an Advantech Magnet Monitor	This means that the magnet heater is stuck on, possibly due to a software hang or other circumstance. Latched pressure was detected and a reboot of the magnet monitor has already occurred. If problem occurs again the Magnet Monitor must be replaced. Latched pressure indicates that the PC in the magnet monitor could have a software hang.
Message **ER3** appears on the seven segment display of an Advantech Magnet Monitor	Heater timeout. The magnet heater was being held on for an extended period. Possible cause might be a slow leak in the magnet, and vessel pressure would never reach the upper limit. In this case the heater would turn off to conserve helium.

9-2 Boot-up Timeline

The following will describe the sequential events that will occur as a magnet monitor is re-booted or powered up. The times noted are approximate.

10 sec The HDD light located on the left side of the unit will come on as the hard drive is accessed. It will flicker during the remainder of the boot-up cycle

60 sec The display will come on with **8.8.8.8.** displayed.

70 sec The display will go blank.

80 sec The helium reading will activate and display the helium level in the magnet.

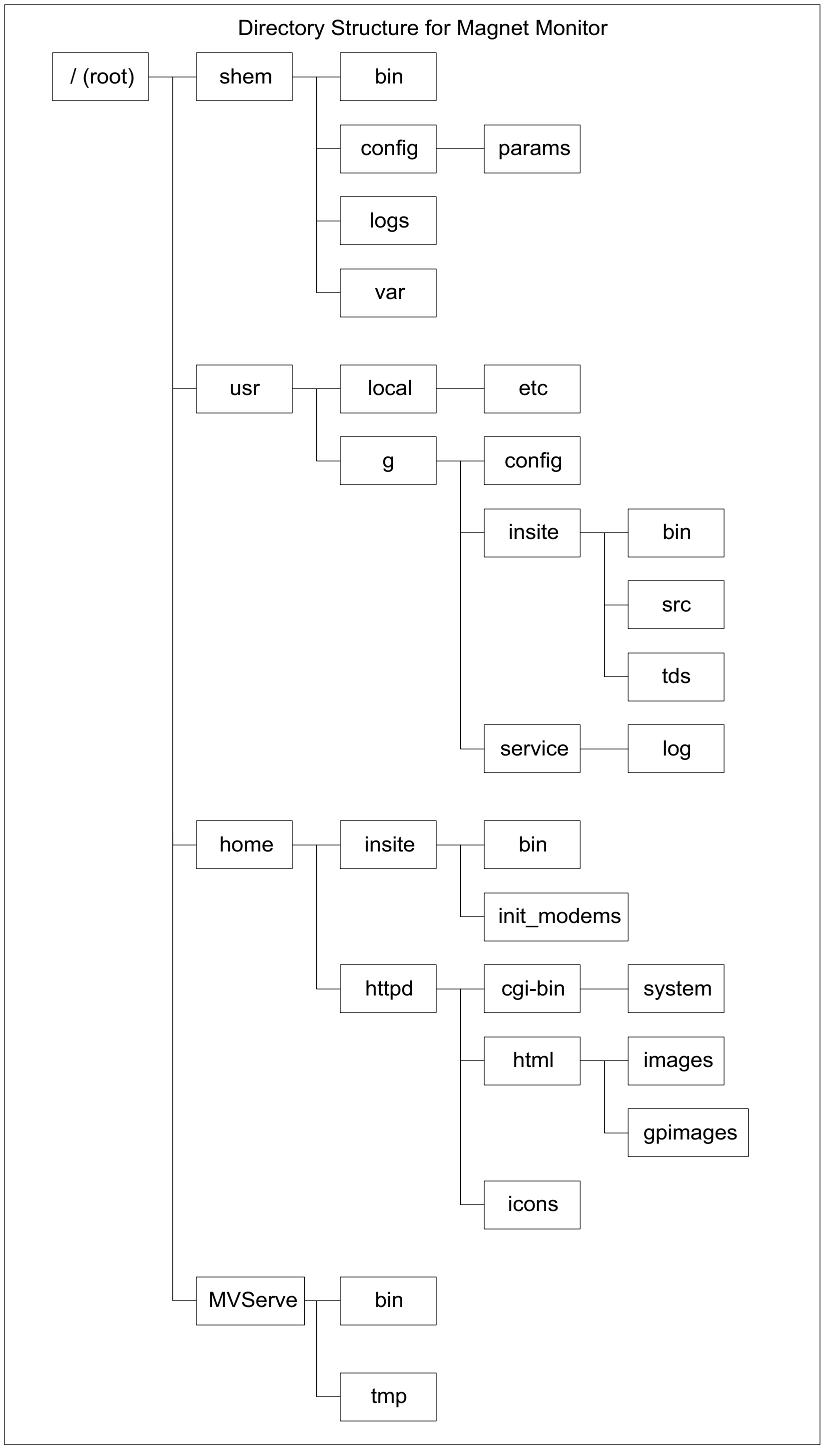
120 sec to 180 sec The magnet monitor will cycle between helium level and vessel pressure, indicating that the magnet monitor is fully booted.

Note

Do not reset or power down the magnet monitor during the boot-up process. Software may be corrupted during this time.

9-3 Software Structure

Four new branches have been added to the standard Linux directory tree. The shem branch contains all of the files for the Magnet Monitor pertaining to reading parameter data, logging the data, reporting alarms, etc. The /usr/g branch contains all of the files used by the IP/TDS system that is used to report alarm conditions to the AutoSC. These files are also used during remote access of the Magnet Monitor from the Online Center. The /home/httpd branch contains the files used by the Web Server. The /MVServe branch contains the InitConfig and ConfigLink scripts that are used to do the Insite Checkout procedure. See Illustration 9-2.



SOFTWARE DIRECTORY STRUCTURE
ILLUSTRATION 9-2

9-4 Files on Proprietary Disk

There are 5 files on the proprietary floppy.

TABLE 9-2
PROPRIETARY SOFTWARE FILES

Filename	Size	Description
Install.tgz	67K	
Installn	2K	Executes non-proprietary install over proprietary mode
Installp	1K	Executes proprietary software install
Installp.tgz	332K	
Mdmcbx.inf	4K	Null modem file needed for laptop communication

9-5 Dumping/Loading Magnet Monitor Data to a Floppy, ie Saveinfo

If needed you can store the log files from a magnet monitor onto a floppy using the following steps. The data files are in binary format and cannot be viewed. You would have to transfer the files to another magnet monitor in order to view the files.

1. Connect a laptop to the magnet monitor using a 9 pin female to 9 pin female null modem cable.

2. Open a hyperterminal window. See section 2 of this manual for the details.

3. At login prompt type **root**, password is **mvserve**. Some older sites may have a different login **ac79-E38** and password **RvmL&Z2pFc**.

4. Mount the floppy: **mount -t msdos /dev/fd0 /mnt/floppy**
5. To copy the files from Magnet Monitor to floppy do the following:
   cp /shem/logs/*.dat /mnt/floppy
   cp /shem/logs/*.logs /mnt/floppy
6. To copy the files from floppy to Magnet Monitor do the following:

   cp /mnt/floppy/* /shem/logs

7. Un-mount the floppy: **umount /dev/fd0**

9-6 Archiving Insite Information

You can archive checkout information to a floppy. In the event that a magnet monitor is replaced, the checkout information can be restored. This step eliminates the need for an Insite Checkout after magnet monitor replacement.

1. Insert a blank floppy into the magnet monitor floppy drive.

2. Open a C-shell on the console. Telnet to the Magnet Monitor by typing **telnet x.xx.xx.xx** where x.xx.xx.xx is the IP address of the magnet monitor.

Login name is **root** (case sensitive)
Password is **mvserve** (case sensitive)

Note

You can also connect your laptop to the Magnet Monitor and use Hyperterminal to communicate. See Section 2 for proper settings.

3. Mount the floppy by typing **mount –t msdos /dev/fd0 /mnt/floppy**

4. Copy the files by typing **cp /usr/g/insite/sclink.cfg /mnt/floppy**
 cp /shem/config/alarm.cfg /mnt/floppy
 cp /shem/config/param.cfg /mnt/floppy
 cp /shem/logs/.shemtype /mnt/floppy/shemtype (must be exact, file is not present on version 1.6 software)

5. Unmount the floppy by typing **umount /dev/fd0**

6. Remove floppy from the drive, label as **Insite Information** and store in a safe location.

9-7 Changing time and date

If needed you may change the time and date of the magnet monitor using the following steps.

4. Connect a laptop to the magnet monitor using a 9 pin female to 9 pin female null modem cable.

5. Open a hyperterminal window. See section 2 of this manual for the details.

6. At login prompt type **root**, password is **mvserve**. Some older sites may have a different login **ac79-E38**and password **RvmL&Z2pFc**.

7. At the prompt type in **date MMDDhhmm** where MM is the 2 digit month
 DD is the 2 digit day
 hh is the 2 digit hour in military time
 mm is the 2 digit minute

 Example is **date 08161320** for August 16 at 1:20pm.

8. To write the date into firmware type **/sbin/clock --systohc** If you omit this step the magnet monitor will loose the time and date when rebooted.

9-8 Displaying Network Information

There are two methods for displaying Network Information.

1. Log in to the magnet monitor using Hyperterminal. Login is **root** password is **mvserve**
   Change directory by typing **cd /MVServe/bin <enter>**
   Display data by going into network config **Net_Config <enter>**
   Do you want to continue with configuring the magnet monitor's network? **Y <enter>**
   Answer **Y** to maintain default settings.
   Answer **N** to start/restart the network.

2. You can execute the following commands to view network information.
   Log in to the magnet monitor using Hyperterminal. Login is **root** password is **mvserve**

Type **more /etc/sysconfig/network-scripts/ifcfg-eth0 <enter>** This will display Mag Mon network info, see below

BOOTPROTO=static	
IPADDR=3.45.82.67	**Mag Mon Address**
NETMASK=255.255.252.0	**Netmask**
GATEWAY=3.45.80.252	**Gateway**
ONBOOT=yes	
FE_IPADD=10.250.24.6	**Laptop Port IP Address**
[root@10 /root]#	

Type **more /home/syscon/.rhosts <enter>** to display host Network info

3.45.82.206 sdc	**Host Address**
[root@10 /root]#	

Type **ifconfig <enter>** to display Ethernet config info currently being used.

```
eth0      Link encap:Ethernet  HWaddr 00:D0:C9:34:05:42
          inet addr:3.45.82.67  Bcast:3.45.83.255  Mask:255.255.252.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3162719 errors:0 dropped:0 overruns:0 frame:7
          TX packets:15528 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:100
          Interrupt:11 Base address:0x6800

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:3924  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
```

10- RETURNING TO NON-PROPRIETARY CONFIG

The following process describes how to install Non-Proprietary software onto a Magnet Monitor. The process assumes that your Magnet Monitor is operating.
After completing this procedure inform the On-Line Center in your pole that the Magnet Monitor software was set to Non-Proprietary mode so the site can be removed from the PPADS database.

1. Using a serial connection (9-pin female to 9-pin female), and a laptop computer, open a terminal window (Hyperlink connection, refer to section 5 in the Hardware Installation Manual (2230681)).
2. At the Login prompt, type in **root**
3. At the password prompt, type in **mvserve**

CAUTION

VERIFY THAT YOU HAVE THE PROPER PROPRIETARY DISKETTE INSTALLED. FAILURE TO USE THE PROPER DISK WILL DESTROY THE HART DRIVE ON THE MAGNET MONITOR.

- Use disk 2252032 if you have the Granite Magnet Monitor, <u>without</u> the Pacemaker FMI installed.
- Use disk 2252032-3 if you have the Granite Magnet Monitor <u>with</u> the Pacemaker FMI installed.
- Use disk 2252032-3 if you have the Advantech Magnet Monitor.

4. Insert the proper Proprietary floppy disk into the floppy drive on the Magnet Monitor.
5. Mount the drive by typing the following:
   mount –t msdos /dev/fd0 /mnt/floppy (case sensitive, don't ignore spaces)
6. Press [Enter].
7. Copy files to /tmp directory: **cp /mnt/floppy/* /tmp**
8. Press [Enter]
9. After the copy function is completed, unmount the floppy drive: **umount /dev/fd0**
10. Press [Enter]
11. Remove floppy disk from drive
12. Execute the install script: **/tmp/installn**
13. Press [Enter].

14. When the install script completes, the Magnet Monitor will re-boot. This will cause a disconnect in your laptop connection. Remove the floppy from the magnet monitor.

15. Log back into the magnet monitor via Hyperterminal, login is **root** password is **mvserve.**

16. Remove the proprietary software by executing the following commands:

    cd /tmp <enter>
    rm installp
    This removes the proprietary software. The magnet monitor will automatically reboot.

17. Once the re-boot has completed, using the Hyperlink terminal, Login as **setup**, and use the password **monitor**.

18. Select the appropriate configuration file by answering **y** then select the number of the file.

19. If you want to change the settings for the Automatic Compressor Reset feature, answer **y**, and enter values for the five variables. (This feature is automatically enabled to default values if you bypass this step.)

    If either of the above were changed, the system will again re-boot at completion.
    After re-boot is complete the Magnet Monitor will be in the non-proprietary mode.

11- RENEWAL PARTS

11-1 Renewal Parts for Granite Magnet Monitor

TABLE 11-1
MAGNET MONITOR

Item	Description	Part Number	FRU Code	Qty
1	Granite Magnet Monitor Unit	2209000	1	1

11-2 Renewal Parts for Advantech Magnet Monitor

TABLE 11-1
MAGNET MONITOR

Item	Description	Part Number	FRU Code	Qty
1	Advantech Magnet Monitor Unit	2219400	1	1
2	Global Modem	2245794	1	1
3	Cable between Magnet Monitor and Modem	2276568	1	1

11-3 Related Parts

Table 11-3 contains some of the unique parts that may be required for Magnet Monitor.

TABLE 11-3
UNIQUE PARTS LIST FOR MAGNET MONITOR

Description	Part Number	FRU code
4 line multiplexer with 115VAC	46-328475P1	1
8 line multiplexer with 115VAC	46-328475P2	1
4 line multiplexer with 220VAC	46-328475P3	1
8 line multiplexer with 220VAC	46-328475P4	1
Water flow and temp sensor (new style)	2333825	1
Pressure transducer for non-LCC magnets	46-320640P1	1
Leybold compressor adapter cable	2116875	N
Desiccant bag for water flow/temp sensor	2135979	1
Jumper for Systems cabinet to eliminate console helium reading	2239626	
Proprietary Software Floppy	2252032-3	1

12- GLOSSARY

Cryogen
: A coolant in the form of a liquefied gas such as helium or nitrogen used to cool the magnet to superconducting temperatures.

Gauss
: Unit of measurement for the strength of a magnetic field.

Generic Back Plane
: A circuit board that has only interconnection hardware (such as an ISA bus) and no electronic circuitry.

Industry Standard Architecture (ISA) Bus
: A series of common connection points used to connect ISA compatible boards.

LCC
: Low cryogen consumption magnet.

Magnet Interface Board
: Provides a connection between the Magnet Monitor and the magnet and associated support hardware and electronics such as compressors and cooling shields. The board also drives the display board and conditions analog signals before sending them to the analog to digital converter board (ADC).

Modem
: A device used to allow computers to communicate over telephone lines.

Network Interface Card (NIC)
: Provides circuitry for connecting a computer to a network.

OLC
: GE OnLine Center

Penetration Panel
: An connection panel used to isolate the magnet room’s electrical cables from external cables.

Single Board Computer (SBC)
: A single board that contains the main computing hardware such as the central processing unit (CPU), memory, controllers for the disk drives, etc. This board differs from computer “mother boards” in that it does not have a built-in bus for connecting other boards.

APPENDIX A - LAPTOP SERIAL CONNECTION SETUP

This connection, and description, is to be used when first installing a Magnet Monitor. The Serial connection can be used at other times, with the same or different Logins and Passwords. This information will be provided for each case that it is needed. This type of connection to the Magnet Monitor is generally intended to launch script files, copy new files onto the internal hard drive or to interact directly with the Linux operating system.

TABLE A-1
PARAMETERS MONITORED BY MAGNET MONITOR

Step	Description	Comments
1	Connect Laptop to the Magnet Monitor using a Null Modem serial cable.	Connect to an available COM port on the Laptop, and to the "Laptop" connection on the Magnet Monitor.
2	Use the Hyperterminal connection in the laptop to connect to the Magnet Monitor as explained in Hardware Installation Manual, Section 5 (2230681)	The Hyperterminal screen comes up with the Linux prompt for Login.
3	Give login name as **setup** and password as **monitor**.	A menu asking the Service Engineer to select the type of magnet used appears.
4	Select the appropriate option for your Magnet type. The default is an LCC type magnet.	A message showing the type of magnet appears and the machine logs out the user and Hyperterminal connection is lost. After a while the LEDs on the magnet monitor panel that correspond to Helium level, Nitrogen level and Helium Vessel Pressure start lighting up in cycles, with corresponding values for them being displayed on the display panel.
5	Disconnect the Laptop and cable from the Magnet Monitor	

APPENDIX B - NULL MODEM DIAL-UP SETUP FOR LAPTOP

In order to properly connect your laptop, via the Null Modem cable, using the Dial-up Networking feature, you will need to install a "Modem" for this purpose. The modem to which I refer is not a hardware modem, although Windows will treat it that way.

B-1 Step For Windows 95 or 98

1. Open Start/Settings/Control Panel.
2. Select [Add new hardware].
3. Press [Next], to begin install wizard.
4. Press [No], to prevent Windows from searching for your modem.
5. Press [Next], to continue.
6. Select [Modem], to add a new modem.
7. Press [Next], to continue.
8. Select [Don't detect my modem, I will select from list].

Note

If you do not see a Null Modem Type selection in the "Manufacturers" column, place your Proprietary software install disk in your laptop, and select "Have Disk" when prompted. A window should appear saying "copy manufactures files from" A:\. Select OK.

9. Select [Generic Null Modem] (or equivalent). The A drive should now be accessed.
10. Press [Next].
11. Select the COM port that will be attached to your null modem cable. You might have to select country, area code and dial out prefix.
12. Select an available COM port.
13. Press [Finish].

This ends the Modem install process.

Note

If you later have problems with this connection you may need to inspect the "Properties" of this new modem by viewing it in Start/Settings/Control Panel, open [Modems], find your "Null Modem" and check settings. All settings should be at the default values.

Now you'll need to create a new "Dial-up" connection in order to use this Modem.

14. Open Start/Programs/Accessories/Dial-up Networking

Note

If the above selection does not exist, add it to the Windows Settings found in Start/Settings/Control Panel, “Add/Remove Programs”, “Windows Setup”, “Communications”. Verify that the “Dial-up Networking” box is selected.

15. In the “Dial-up Networking” window, select [Make New Connection].

16. In the “Select a Modem” box, select your Null Modem.

17. Type a Name for this connection, example would be Magnet Monitor.

18. Press [Configure].

    The settings should be as follows;

    - General
        - Communications Port (COM X), whatever you previously selected.
        - Maximum Speed, 115200. Do NOT select “Connect only at this speed”.
    - Connection
        - Use all default settings(8 data bits, no parity, 1 stop bit). Select Advance, use flow control, click on software.
    - Options
        - Select [Bring up terminal window after dialing].
    - All other settings leave as default.

19. Press [Next].

    A phone number is required here, although this number is not actually used. Type in “521-0000”. An area code is not needed.

20. Press [Next].

21. Press [Finish].

    Your Null Modem connection is now ready for use.

Note

Refer to Appendix C, “Browser Setup”, to learn how to use this connection to view the Magnet Monitor Status, Remote and Configuration screens.

APPENDIX C - BROWSER SETUP

Either Netscape Navigator or Microsoft Internet Explorer can be used as a browser. The setup is a little different for each.

C-1 Microsoft Internet Explorer

Note

It is recommended that you complete the Y2K upgrade on your PC prior to using Internet Explorer as the browser interface.

1. Open Start/Settings/Control Panel/Internet.

2. Open the tab “Connection”.

3. Select [Connect to the Internet using a Modem].

4. Press [Settings].

5. Select the Dial-up connection you made for the Null Modem in appendix B.

6. Select OK to get out of the Dial-up Settings window, then select OK to get out of the Internet Properties window.

7. Open Start/programs/Accessories/Dial-up Networking. Then select the connection you made in Appendix B of this procedure.

8. Using the backspace key, remove any characters in the *Login* and *Password* fields.

9. Select the connect option.

10. You must get only the *login* prompt on the window that opened up automatically after selecting the connect option. If the login prompt is not present, try pressing the enter key twice. If the “status dialing” box remains on the screen until it times out, there is something wrong with the modem set-up. See table C-1 for possible causes

11. You may need to press the “enter” key to get the login prompt. For Login, type in **pppuser**.

12. For Password, type in **insite**.

13. You should then see characters scroll down the window. Select F7 to exit.

14. Open Internet Explorer. Shortly after Internet Explorer is opened select the **stop** button to halt any communication attempt. Then enter the internet address for the magnet monitor which is **10.0.0.5**

Note

If you do not get a response at this point you might need to reload Internet Explorer. Refer to table C-1 for instructions.

C-2 Netscape Navigator 2.02

1. Open Netscape Navigator.
2. In menu, open Options/Network Preferences.
3. Select [No Proxies].
4. Select [OK].

C-3 Netscape Navigator 4.01

1. Open Netscape Navigator.
2. In menu, open Edit/Preferences.
3. Expand the “Advanced” selection.
4. Select [Proxies].
5. Select [Direct connection to the Internet].
6. Select [OK].

C-4 Connecting your Browser to the Magnet Monitor

1. Using a Null Modem cable, connect your laptop to the Magnet Monitor. Use the “Laptop” connector on the Magnet Monitor.
2. Make sure the Magnet Monitor is powered up. Hit the reset switch on the magnet monitor. It will take 2 minutes for the magnet monitor to boot up.
3. Find your Dial-up for the Null Modem connection, activate it.

   When the connection is made, you will get a pop up window.
4. If the terminal window is blank, press [Enter] twice. A prompt should appear `Login`.

Note

If this attempt to login fails, re-boot the MM and repeat from C-4, step 1.

5. Type in **pppuser**, press [Enter].
6. Type in **insite** at the password prompt.
7. A string of characters will show up. Press F7 to continue.
8. Open your browser.

9. Whichever browser you've selected, type in the URL IP address **10.0.0.5** and activate the search. You will be prompted for login, the login name is **MMService** password is **MagnetMonitor** (both login and password is case sensitive).

10. Momentarily you should see the Magnet Monitor screen come up. If you do not see this, check all your settings and connections.

TABLE C-1

Troubleshooting tips

Problem	Possible cause
You cannot connect to the magnet monitor, after selecting the icon in Dial-up Networking. The laptop and magnet monitor will not establish communication.	In the dial-up networking box, highlight the connection icon by single clicking on it. Then hit the right mouse button and select properties. Select *configure* , then *options.* Make sure "bring up a terminal window after dialing" is selected.
The option "connect to the internet using a modem" cannot be selected and you have not done the Y2K upgrade. To verify go into Internet Explorer, select the *view* pull-down menu, Select *Internet options*, then *connections.* You must be able to select the option "Connect to the internet using a modem".	Internet Explorer software will need to be reloaded using the following procedure. You must have the Blue Disaster Recovery disk dated 1999 to do this procedure. 1. Go to My Computer/Control Panel/Add Remove Programs. 2. Highlight Microsoft Internet Explorer 4.0. 3. Select Add/Remove, highlight un-install Internet Explorer 4.0 and all its components, and select OK. If asked, **DO NOT** remove any personal files or links You may have to select OK two more times before the program files are removed. You must close all programs if prompted. It takes at about 7 minutes for all files to be removed. 4. Reboot your laptop. 5. Go to the D drive or CD-ROM, select Add-ons, Ie4,Iesetup.exe. 6. Go to Explorer/Support/Internet Explorer 4.01/Iesetup.exe 7. Run setup. 8. Re-install Ie4. It takes about 17 minutes for Internet Explorer to install. 9. Follow the procedure to connect to magnet monitor.

APPENDIX D - COEF.CFG FILE

The following listing is a copy of the coef.cfg files used in the Magnet Monitor for the purpose of converting measured voltages into the displayed measured values with appropriate units. Refer to Appendix E for information on how to use these coefficients.

```
# COEFDATA.CFG -- Coefficient Data File Rev. 1.02  10 Sept, 1998..Modified He Level
coefficients
#               Coefficient Data File Rev. 1.01  9 Sept, 1998...modified RuO data for
gain 65*2 and 1.5K to 24K range, see "New_limited_25.mcd"
#                                                                      and
"Vendor_Data_RuO.xls". See "coef_9_10_98.cfg"
#               Coefficient Data File Rev. 1.00  22 July, 1998..see "Coef_9_98.cfg for
this version
#   Based on the EDM file coefdata.cfg Rev. 3.0 May 15, 1995
#  Added tables for SI-410 Diode and Ruthinium Oxide Diode.
#
# Curve #0 A to D Tics
# Used for debugging
0 0 0 4095 0.000 1.000E+3 0.000 0.000 0.000 0.000
#
# English Units
#
# Curve #1 0-5 scale
# Used for He Flow and Water Flow (and Voltage for debugging)
# Units: Use "SCFH" for He Flow, and "GPM" for water flow.
1 0      0 4095 0.000 1.221001 0.000 0.000 0.000 0.000
#
# Curve #2 0-100 scale
# Used for Water Temp
# Units: Use "DEG F"
2 0      0 4095 0.000 2.442002E+1 0.000 0.000 0.000 0.000
#
# Curve #3 0-100 scale
# Used for He Level
# Units: Use "%"
3 0      0 4095 1.000E+2 -2.442002E+1 0.000 0.000 0.000 0.000
#
# Curve #4 0-15 scale
# Used for He Pressure
# Units: Use "PSIG"
4 0        0  819  0.000    0.000    0.000 0.000 0.000 0.000
4 1 820 4095 -3.755725 4.580153 0.000 0.000 0.000 0.000
#
# Curve #5
# Used for DT500D Diode Temp
# Units: Use "K"
5 0    0 1406 +4.457076E+2 -2.277978E+2  0.000        0.000        0.000        0.000
5 1 1407 1685 +5.929520E+4 -1.917673E+5 +2.492882E+5 -1.621930E+5 +5.276006E+4 -6.865506E+3
5 2 1686 1920 -1.775261E+5 +3.343076E+5 -1.927495E+5 +7.476784E+3 +2.547142E+4 -5.857463E+3
5 3 1921 3467 +6.370002E+2 -1.073492E+3 +7.577936E+2 -2.691138E+2 +4.774339E+1 -3.389978
5 4 3468 4095 +4.917566E+1 -1.158421E+1  0.000        0.000        0.000        0.000
#
# Metric Units
#
# Curve #6
# Used for Water Flow
# Units: Use "LPM"
6 0 0 4095 0.000 4.621978 0.000 0.000 0.000 0.000
#
# Curve #7
# Used for Water Temp
# Units: Use "DEG C"
7 0 0 4095 -1.7777778E+1 1.356668E+1 0.000 0.000 0.000 0.000
#
# Curve #8
# Used for He Pressure
# Units: Use "KPA"
8 0   0 819   0.000       0.000       0.000 0.000 0.000 0.000
```

```
8 1 820 4095 -2.587269E+1 3.155206E+1 0.000 0.000 0.000 0.000
#
# Curve #9
# Used for He Flow
# Units: Use "SLPM"
9 0 4095 0.000 5.762320E-1 0.000 0.000 0.000 0.000
#
# Curve #10
# Spare
10 0 0 4095 0.000 1.000E+3 0.000 0.000 0.000 0.000
#
# Curve #11
# Used for SI410 Diode Temp
# Units: Use "K"
11 0 0000 1450 +5.919119E+2 -3.044092E+2  0.000        0.000        0.000        0.000
11 1 1451 1778 +2.843001E+4 -8.599055E+4 +1.051892E+5 -6.441759E+4 +1.969781E+4 -2.409274E+3
11 2 1779 1926 -5.637970E+5 +9.241173E+5 -3.167606E+5 -2.145678E+5 +1.664921E+5 -3.014900E+4
11 3 1927 2816 +1.938138E+2 -6.084114E+2 +8.611591E+2 -5.592035E+2 +1.665666E+2 -1.860774E+1
11 4 2817 4096  0.000        0.000        0.000        0.000        0.000        0.000
#
# Curve #12
# Used for Ruthinium Oxide Diode Temp
# Units: Use "K"
12 0    0 1163 +24.00      0.000      0.000       0.000      0.000      0.000
12 1 1163 1194  +9.98206E+5 -3.35393E+6 +4.22848E+6 -2.37063E+6 +4.98639E+5 0.000
12 2 1194 1216  -9.10448E+5 +3.04221E+6 -3.81026E+6 +2.12016E+6 -4.42245E+5 0.000
12 3 1216 1250  +4.67292E+6 -1.51569E+7 +1.84361E+7 -9.96647E+6 +2.02042E+6 0.000
12 4 1250 1308  -1.34338E+4 +4.53571E+4 -5.66914E+4 +3.12002E+4 -6.39349E+3 0.000
12 5 1308 1933  +9.02581E+2 -2.05478E+3 +1.76578E+3 -6.75149E+2 +9.66482E+1 0.000
12 6 1933 4095 +1.500       0.000      0.000       0.000      0.000      0.000
```

APPENDIX E - CALCULATING WITH THE COEF.CFG FILE DATA

The MM utilizes multiple polynomials to facilitate curve fitting of the measured voltage to the displayed value. Some sensors have simple straight line transfer functions, others are non-linear curves. Each measured parameter can have up to 5 separate fifth order polynomials to characterize different ranges of the value. In the cases where a straight line is all that is required, a single polynomial is used, with only the first 2 coefficients non-zero. In more complex functions, all 5 polynomials may be used, each having up to 5 separate coefficients.
Refer to Appendix D Coef.cfg file, to see a copy of the coefficient table used in the MM. The table below is a copy of one of these coefficient sections. The one shown is for the RuO sensors, Coefficient table 12.
The subsequent equations demonstrate how to use the coefficient tables to calculate what a displayed value would be, given a measured voltage from an RuO sensor.
The actual RuO sensors produce an output voltage of approximately 10 to 30 millivolts. Before making it to the MM, this signal is amplified by a gain of 65 in the RuO Buffer amplifier (Located at the magnet). On entry into the MM, the signal is further multiplied by 2.
The MM routes this signal to an Analog-to-Digital converter (ADC). This converter is a 12 bit conversion process. The ADC is scaled to read uni-polar 0 to 5 Volts. The following equations use ADC counts (0 to 4095), rather than actual voltage. Another important fact to keep in mind, the MM software also divides this count by 1000 prior to applying a value to the polynomials. Therefore, as an example, if you read 2.5 volts with an external meter, the number you would plug into the equations (all coefficient tables are handled the same way) would be as follows:

$$\mathrm{Vmeasured} := 2.5$$

$$\mathrm{AD\_counts} := 2.5 \cdot \frac{2^{12}}{5} \quad \text{therefore...} \quad \mathrm{AD\_counts} = 2048$$

$$\mathrm{Equation\_input} := \frac{\mathrm{AD\_counts}}{1000} \quad \text{therefore...} \quad \mathrm{Equation\_input} = 2.048$$

Keep this in mind when experimenting with the equations below…

		A/D Count reading		**Coefficients**					
Curve #	**Coef Set**	**Lower**	**Upper**	**$Coef_0$**	**$Coef_1$**	**$Coef_2$**	**$Coef_3$**	**$Coef_4$**	**$Coef_5$**
12	0	0	1163	24	0	0	0	0	0
12	1	1163	1194	9.98206E+5	-3.35393E+6	+4.22848E+6	-2.37063E+6	+4.98639E+5	0.000
12	2	1194	1216	-9.10448E+5	+3.04221E+6	-3.81026E+6	+2.12016E+6	-4.42245E+5	0.000
12	3	1216	1250	+4.67292E+6	-1.51569E+7	+1.84361E+7	-9.96647E+6	+2.02042E+6	0.000
12	4	1250	1308	-1.34338E+4	+4.53571E+4	-5.66914E+4	+3.12002E+4	-6.39349E+3	0.000
12	5	1308	1933	+9.02581E+2	-2.05478E+3	+1.76578E+3	-6.75149E+2	+9.66482E+1	0.000
12	6	1933	4095	+1.500	0.000	0.000	0.000	0.000	0.000

This demonstration will be done using only Coef set #5, from the above table. The equation using this set of coefficients can only be used for voltage readings that result in ADC counts in the range of 1308 to 1933. This correlates to a voltage range as shown below, as would be measured at the RuO sensor.

$Vlower := 1308 \cdot \frac{5}{2^{12}} \cdot \frac{1}{65 \cdot 2}$ therefore... $Vlower = 0.012$

$Vupper := 1933 \cdot \frac{5}{2^{12}} \cdot \frac{1}{65 \cdot 2}$ therefore... $Vupper = 0.018$

For the demonstration, we first need to create our "Input Voltage", for simplicity I'll assume you can do that calculation and I'll just create a number that ranges from 1308 to 1933, that variable is AD.

$x := 0, 1 .. (1933 - 1308)$

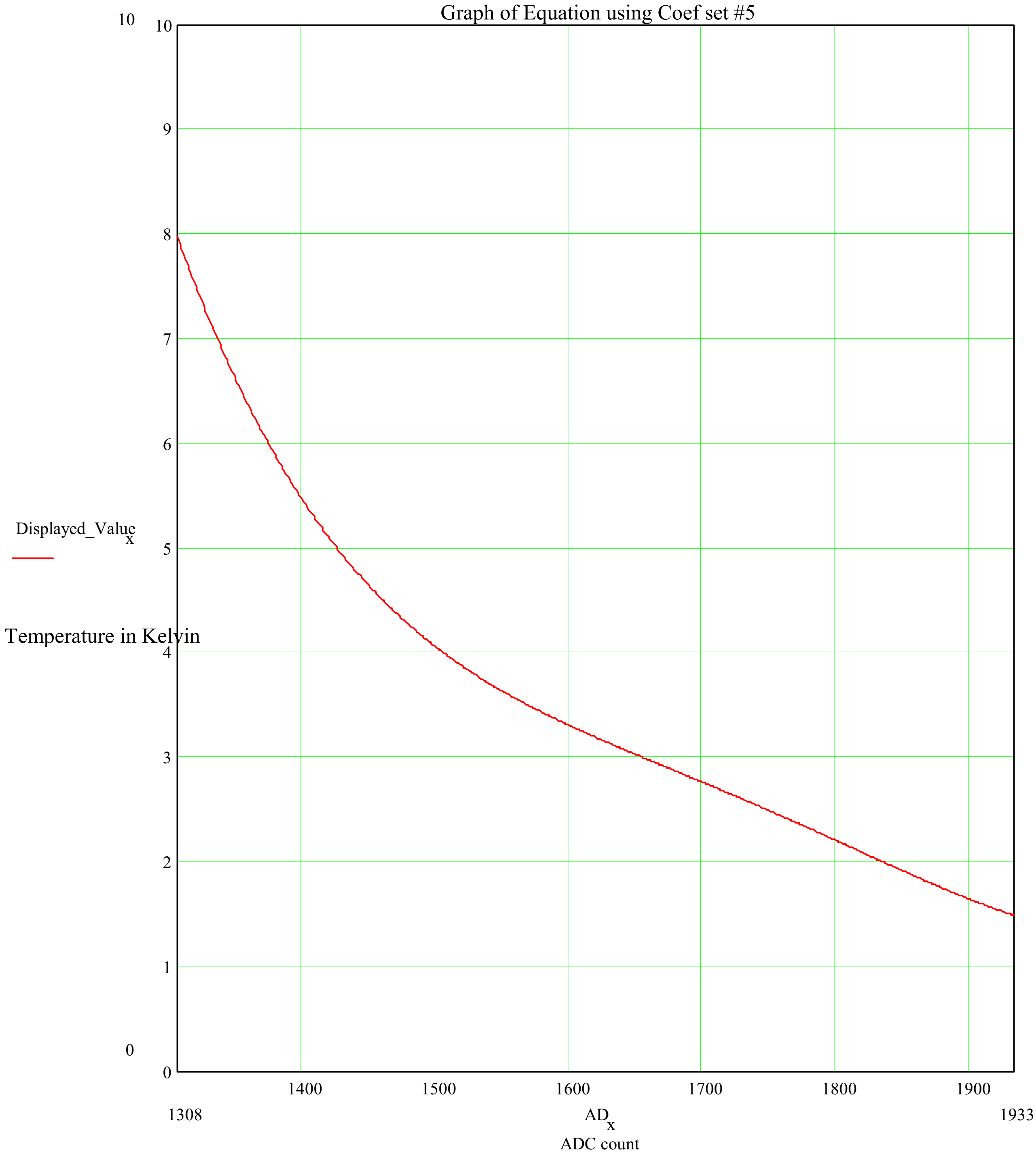
$AD_x := x + 1308$

Next define the coefficients (taken from the table above)…
…and now perform the calculation for all values of "x" (i.e. 1308 to 1933)…

$Coef_0 := 9.02581 \cdot 10^2$ $Coef_1 := -2.05478 \cdot 10^3$ $Coef_2 := 1.76578 \cdot 10^3$ $Coef_3 := -6.75149 \cdot 10^2$ $Coef_4 := 9.66482 \cdot 10^1$ $Coef_5 := 0$

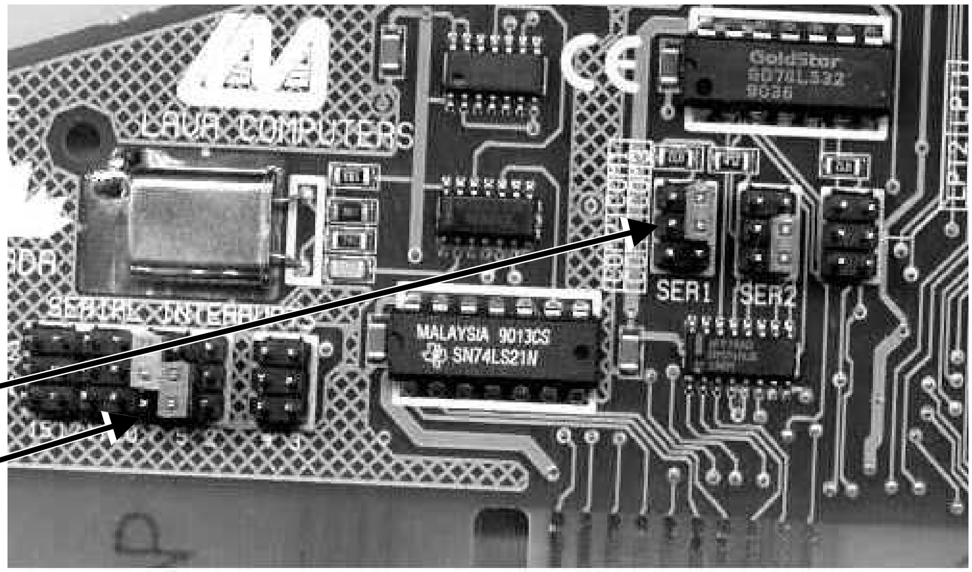
$$Displayed\_Value_x := Coef_0 + Coef_1 \cdot \frac{AD_x}{1000} + Coef_2 \cdot \left(\frac{AD_x}{1000}\right)^2 + Coef_3 \cdot \left(\frac{AD_x}{1000}\right)^3 + Coef_4 \cdot \left(\frac{AD_x}{1000}\right)^4 + Coef_5 \cdot \left(\frac{AD_x}{1000}\right)^5$$

The following graph demonstrates the computed value "Displayed Value", given the various values of AD…



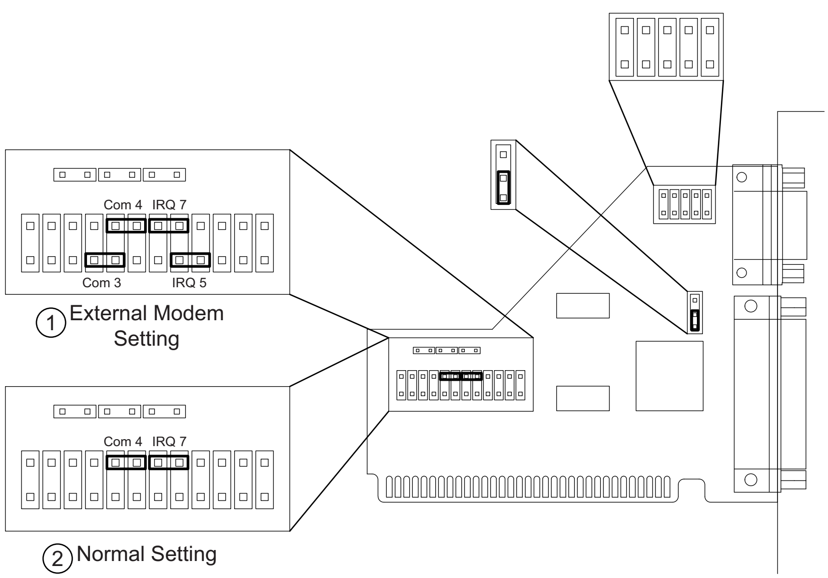
APPENDIX F - GENERAL I/O BOARD JUMPER SETTINGS FOR LATER GRANITE (POST JANUARY 2001) UNITS

Added jumpers for external Modem



GENERAL I/O BOARD
ILLUSTRATION F-1

GENERAL I/O BOARD JUMPER SETTINGS FOR EARLY GRANITE (PRE JANUARY 2001) UNITS



GENERAL I/O BOARD
ILLUSTRATION F-2

APPENDIX G - PPADSCMD EMAIL

G-1 Subscribing to a site via the intranet

1. To subscribe to sites via the intranet, go to the magnet monitoring home page at: http://3.7.192.186/cgi_bin/ss/ppads/home.cgi

2. Select **Site Subscription Control** at the left hand of the screen.
3. Select Class (or country set) for which you want to subscribe.
   Nam is North America
   Eur is Europe
   Jap is Japan
   Aus is Australia
   MRT is for MRT magnets only

4. Click on the **Submit** icon.
5. Scroll down to the bottom and enter your E-mail address in the **Use a New Address** option. Then click on **Submit** which is under the New Address.
6. Re-enter your E-mail address and click on **Submit**.
7. You can then subscribe to any site. There is no limitation as to how many sites you can subscribe to. If you would like to subscribe to system ID **402572MRC**, first click on the 4 (or the first letter of the system ID). You will then get a list with all system ID's that begin with 4. Check the system ID's that you want to subscribe to, then click on **Subscribe** on the top or bottom of the "list of sites". You will receive a message indicating that you are now subscribed to the selected sites.

G-2 Overview

The following sections provide the procedures necessary to receive Magnet Monitoring information by sending email to PPADSCMD.

- SECTION G-2, RECEIVING MAGNET MONITORING DATA THROUGH EMAIL - This section provides the following information.
  - How to address email to PPADSCMD from the various types of GE email accounts.
  - What the general guidelines are for sending email to PPADSCMD.
  - How to use the online help files to find the correct command syntax.
  - When to use the BEGIN and END commands to prevent PPADSCMD from incorrectly processing information embedded in the email message.
- SECTION G-3, FINDING SITE SPECIFIC INFORMATION - This section describes some common combinations of commands for keeping track of what is happening (alarms and fills) at specific sites and how to begin troubleshooting problems at specific sites with Magnet Monitoring.
- SECTION G-4, GETTING INFORMATION ABOUT ALL MAGNET MONITORING SITES - This section describes some common combinations of commands for checking the revisions, connectivity, and settings at all of the sites with Magnet Monitoring.

- SECTION G-5, RECEIVING MAGNET MONITORING PROGRAM INFORMATION - This section provides procedures for receiving current Magnet Monitoring information, including Magnet Monitoring part numbers, toll free phone numbers, and computer usage. You can also address specific questions to the PPADS administrator.
- SECTION G-6, ENTERING AND RETRIEVING TEAM INFORMATION - This section gives the list of commands to use to enter your contact information, join the email distribution list for a team, and find the information to contact another member of a team.
- SECTION G-7, PPADSCMD COMMAND SUMMARY - This section provides an alphabetical list of the available types of PPADS commands.

G-2 Receiving Magnet Monitoring Data Through Email

To access Magnet Monitoring infomation, you need the following.

- To have a GE email account: CC:Mail, Macintosh/Microsoft Mail, Mainframe (Wiz Mail), Microsoft Exchange Mail, or UNIX.
- To meet the PPADSCMD guidelines.
- To know the correct command syntax. For example, you must provide the information inside the angle brackets, <>, (such as, LIST DISPATCHES <*System ID*>); you may optionally provide the information in the square brackets, [ ], (such as, REPORT [SHORT or LONG]). HELP for the specific commands (for example, HELP SHOW) provides the most current information on command syntax.
- To use the BEGIN and END commands, when it is necessary to stop the PPADS System from trying to process any header or footer information embedded in your email messages.

G-2-1 Address Email

To address email to PPADSCMD, use one of the addresses in Table G-1. Note that only Mainframe/Wizz mail requires the Subject field be completed. PPADS does not process the Subject field; however, if the subject field is completed (in any of the GE email systems), PPADS' reply identifies the entry in the Subject field.

TABLE G-1
ADDRESSES FOR EMAIL TO PPADSCMD

EMAIL TYPE	EMAIL ADDRESS
CC:Mail/Laptop	To: PPADSCMD
Macintosh/ Microsoft mail	To: ppadscmd@iscmed.med.ge.com
Mainframe/Wiz mail	To: PPADSCMD Subject: XXX (NOTE: This field must be filled.)
Microsoft exchange mail[1]	To: ppadscmd@iscmed.med.ge.com
PROFS mail[1]	To: ppadscmd (NOTE: Use the GEMS-Am email directory, User ID: I000216)
UNIX/Internet	To: ppadscmd@iscmed.med.ge.com
YMSNET	To: ppadscmd
1 To send commands for Magnet Monitoring information from Microsoft Exchange Mail and PROFS, start the	

list of PPADS commands with the BEGIN command and stop the list with the END command. Microsoft Exchange Mail and PROFs place information in the body of the message that can cause an error unless the BEGIN and END commands are used.

G-2-2 PPADSCMD Guidelines

Use the following guidelines to access Magnet Monitoring data.

- Place all commands to the database in the body of the email message.
- Place only commands in the body of a PPADSCMD email message. The PPADS system expects only commands in the body of a PPADSCMD Email message, so any other information causes an error. If the PPADS list server does not recognize a command, all following commands are ignored.
- Keep each command on its own line.
- Send email to PPADSCMD with the NEWS command every couple of months. Additional commands are available occasionally. Also, use the HELP command periodically to see an overview of all commands.

G-2-3 Use the HELP Command to See Correct Command Syntax

The HELP command without an argument sends the most current version of the general guidelines for sending email to PPADSCMD. To get detailed "help" on a specific command, use the command name as the argument.

1. Address email to PPADSCMD (Refer to Section G-2-1, Address Email.)
2. Place the HELP [ADMIN, BEGIN, END, LIST, NEWS, REPORT, SET, SHOW, SUBSCRIBE, or UNSUBSCRIBE] command on a new line in the body of the message.
3. Send the message. See Illustration G-1.

Command: HELP ADMIN
Reply: ------------------------PPADS EMAIL COMMAND PROCESSOR RESULTS
Command: help admin
Help for the PPADS ADMIN Command
PPADS Information Server
Magnet Monitoring
MR Global Readiness
Advanced Service Methods

WARNING
Permission for the use of this software by persons other than GE Medical Systems employees is NOT GRANTED. A license to use Advanced Service Material does NOT extend to or cover this material. Also, a license to use Advanced Service Material does NOT extend to or cover this material.

If you are not a GE Medical Systems employee, delete this document immediately.

ADMIN COMMAND CAPABILITIES

The ADMIN command sends a copy of your PPADSCMD Email to the PPADS administrator. When the PPADS system recognizes the ADMIN command, the system ignores all the text after the command. Use ADMIN to send messages to the administrator.

USING THE PPADS ADMIN COMMAND

Use the following procedure to send messages to the PPADS administrator.

1) Address e mail to PPADSCMD. (Refer to the main PPADSCMD Help for information on addressing e mail messages.
2) Place the ADMIN command on a new line in the body of a new e mail message; then enter the message for the PPADS administrator. (Place no other commands in a message with the ADMIN command.)
3) Send the message.

Example: admin
For the last week, I haven't been getting e mail alarm notices for the sites that I'm subscribed to. Could you check on it? Thanks, Kate
Example Reply: Sent to the list server administrator!

EXAMPLE HELP COMMAND
ILLUSTRATION G-1

G-2-4 Use the BEGIN and END Commands to Prevent Incorrect Processing

The BEGIN command marks the beginning of a list of PPADS commands. Use this command to prevent PPADS from trying to process header information that is embedded in the message. Use this command if you are entering commands (email) from Microsoft Exchange Mail or from PROFS.

The END command marks the end of your PPADS commands. Use this command to prevent PPADS from trying to process footer information that is embedded in the message. Use this command if you are entering commands (mail) from Microsoft Exchange Mail or PROFS or if you use a .signature.

1. Address email to PPADSCMD (Refer to Section G-2-1, Address Email.)
2. Place the BEGIN command on a new line in the body of the message before the other command(s).
3. Place the END command on a new line after the other PPADS command(s).
4. Send the message. See Illustration G-2.

Command: begin
show db 219879 mrc
end

Reply:

```
------------ PPADS EMAIL COMMAND PROCESSOR RESULTS ------------

Command:  show db 219879mrc

System ID:  219879MRC
ST ANTHONY HOSPITAL/CRS
Customer:  390820446
MTL RUBEN GAITAN, XXXX, BALZER, 5x, MUX=4
Class:  nam
Country:  usa
Room ID:  CRYOGENS
System Type:  SIGNA34
EDM REV:  4.50
EDM Phone:  2198771522w44
Customer Phone:  (219)874-0520
Primary FE:  LEE, JAMES, E
Magnet Type:  SX
Magnet S/N:  XXXX

Individuals subscribed to site:

Last Contact:  Tue  Nov 19  05:20:24  CST  1996
Last Alarm:    Thu  Oct 24  13:09:09  CST  1996
Last Fill:     Mon  Aug 12  20:59:26  CST  1996
Last Checkin:  Tue  Nov 19  05:20:23  CST  1996
Last Fetch:    Mon  Jun 17  20:56:22  CDT  1996
Last Test:     Wed  Oct 11  17:31:23  CDT  1995
```

EXAMPLE BEGIN AND END COMMANDS
ILLUSTRATION G-2

G-3 Finding Site Specific Information

PPADS offers a great deal of site specific information. For a complete list of applicable commands, send email to PPADSCMD with the HELP SHOW and HELP SUBSCRIBE commands.

If you are responsible for a specific site or a group of sites, you may want to do the following tasks:

- To receive email when a specific site has an alarm or fill , send email to PPADSCMD with the following command. (See Illustration G-3.)
  SUBSCRIBE EMAIL <system ID>

```
Command:   subscribe 215431CRY

Reply:     ------------ PPADS EMAIL COMMAND PROCESSOR RESULTS ------------

Command:    subscribe email 215431cry

       You have been added to the Email list for 215431CRY
       You will be notified by email in the event of an alarm or fill at
       this site.

       In the event that you wish to be removed from this site's list,
       send an unsubscribe command to this list server.
```

EXAMPLE SUBSCRIBE EMAIL COMMAND
ILLUSTRATION G-3

- To check that a specific site is able to connect to PPADS, send email to PPADSCMD with the following command:
  SHOW CALLS <system ID> 15

  Note that the “15” in the previous command refers to the number of lines to display from the call.log.
- To view the most current parameter values at the site, send email to PPADSCMD with the following command. (See Illustration G-4.)
  SHOW CURRENT <system ID>

G-3 Finding Site Specific Information (continued)

Command: show current 209435cryo2

Reply:
```
------------ PPADS EMAIL COMMAND PROCESSOR RESULTS -------------

Command:  show current  209435cryo2
      The last data to be reported to the PPADS sytem was on_97/04/09!10:22
      The He level was    88.8
      The 1st stage temp was    47.4
      The 2nd stage temp was    11.4
      The He pressure was    0.93
      The water flow rate was  3.04
      The inlet water temp was    57.09
```

EXAMPLE SHOW CURRENT COMMAND
ILLUSTRATION G-4

- To see the site's name, location, telephone numbers, primary field engineer, time of last contact with PPADS, and who is subscribed to receive email when the site has an alarm or fill, send email to PPADSCMD, with the following command: SHOW DB <system ID>
- To troubleshoot a problem at a site, send email to PPADSCMD with one or more of the following commands. Note that you may need to see the results of a message before sending the other commands. For an example of the LIST DISPATCHES command, see Illustration G-5.
  LIST DISPATCHES <system ID>
  SHOW DISPATCH <system ID> <dispatch #>
  SHOW CURRENT <system ID>
  SHOW MAGALARM <system ID> 25
  SHOW EDM <system ID> 25
  SHOW MAGDATA <system ID>

  Note that the you can import the data from the SHOW MAGDATA into spreadsheet software.

  Note that the "25" in the previous commands refers to the number of lines to display from either the magalarm.log or the edm.log.

G-3 Finding Site Specific Information (continued)

Command: list dispatches 214247rmrc

Reply:

```
------------ PPADS EMAIL COMMAND PROCESSOR RESULTS ------------

Command:  list dispatches 214247rmrc

     Mar  24  1995 DISP0270175003
     Mar  24  1995 DISP0270178335
     Mar  24  1995 DISP0270178755
     Mar  24  1995 DISP0270180556
     Mar  24  1995 DISP0270183897
     Mar  24  1995 DISP0270174710
     Mar  24  1995 DISP0270174965
     Mar  24  1995 DISP0270183627
     Mar  24  1995 DISP0270183705
     Sep  18 07:44 DISP0270206439
```

EXAMPLE LIST COMMAND
ILLUSTRATION G-5

G-4 Receiving Information About All Magnet Monitoring Sites

PPADS offers information about the sites with Magnet Monitoring. For a complete list of applicable commands, send email to PPADSCMD with the HELP COUNT, HELP LIST, and HELP REPORT commands.

If you are responsible for a group of sites, you may want to do the following tasks:

- To see the total number of sites with Magnet Monitoring divided into geographic location, send email to PPADSCMD with the following command. (See Illustration G-6.)
  REPORT SHORT

Command: report short

Reply:
```
------------ PPADS EMAIL COMMAND PROCESSOR RESULTS ------------

Command:    report short

*** PPADS Usage ***

Total number of registered sites:   868

Total number of dispatches:    23115

Total number of calls:         97343

Registered American Sites:        797
Registered European Sites:         48
Registered Asian Sites:             5
Registered Australian Sites         2

***PPADSCMD EMAIL SERVER USAGE ***

Number of Users:    404
Number of Command Requests:    18392
```

EXAMPLE REPORT SHORT COMMAND
ILLUSTRATION G-6

G-4 Receiving Information About All Magnet Monitoring Sites (continued)

- To check how many sites have been able to connect to PPADS recently (in this case, within 15 days), send email to PPADSCMD with the following command (see Illustration G-7): **COUNT FRESH 15 CONTACT**

```
Command:     count fresh 99 check
             count fresh 99 test

Reply:   ------------ PPADS EMAIL COMMAND PROCESSOR RESULTS ------------

Command:  count fresh 99 check

          314 sites have had a check in last 99 days

Command:  count fresh 99 test
          201 sites have had a test in last 99 days
```

EXAMPLE COUNT FRESH COMMAND
ILLUSTRATION G-7

G-4 Receiving Information About All Magnet Monitoring Sites (continued)

- To check how many sites have been unable to connect to PPADS recently (in this case, 45 days), send email to PPADSCMD with the following command:
  COUNT STALE 45 CONTACT

  If you want to troubleshoot why these sites are unable to connect, first list the system IDs of these sites. Send email to PPADSCMD with the following command:
  LIST STALE 45 CONTACT

  Alternatively, you may want to print a list of each site with Magnet Monitoring in order according to the last date and time that the sites had contact with PPADS. To see a list of the sites according to their last contact with PPADS, send email to PPADSCMD with the following command:
  LIST LAST CONTACT DATE

  Once you have the list of system IDs, send email to PPADSCMD with the following command for each of the system IDs. (Note that as long as you put each of these commands on a new line, they do not have to be in different messages.)
  SHOW DB <system ID>

  For example,
  SHOW DB 1073CRYO
  SHOW DB 1100CRYO
  SHOW DB 1196CRYO
  SHOW DB 1325CRYO
  etc.

  Now that you have the modem telephone number for each of these sites, you can try to connect directly with the site and check its .log files, messages from field engineers (these would be left in the Note fields of the Error Reporting Setup screen), and the site's modem configuration file.

- To monitor sites that may have lost connectivity recently (in this case in the last 10 days), send email to PPADSCMD with the following command:
  LIST DIFF FRESH 10 CONTACT FRESH 20 CONTACT
- To list the sites according to MAC (Magnets And Cryogens) Team Leader, if the sites are in North America, send email to PPADSCMD with the following command:
  LIST MTL

  Note that the COUNT MTL command just gives the number of sites assigned to each MAC Team Leader, if applicable.
- To generate a list of every site with Magnet Monitoring, send email to PPADSCMD with the following command:
  LIST SITES

G-5 Receiving Magnet Monitoring Program Information

PPADS offers information about the Magnet Monitoring program. For a complete list of applicable commands, send email to PPADSCMD with the HELP ADMIN, HELP LIST, HELP SUBSCRIBE, HELP NEWS, and HELP REPORT commands.

If you are installing Magnet Monitoring equipment at new sites or are working with existing sites, you may want to do the following tasks:

- To ask questions of the PPADS administrator or notify him of possible problems, send email to PPADSCMD with the following command (see Illustration J-8): ADMIN <your message>

```
Command:   admin

       Could you please send me a copy of the current MRT pager list?  Chris

Reply: Sent to the list server administrator!
```

EXAMPLE ADMIN COMMAND
ILLUSTRATION G-8

- To see a list of all the email distributions (and their purposes), send email to PPADSCMD with the following command:
  LIST DISTRIBUTIONS
- To be notified by email when information becomes available that pertains to a specific distributions, send email to PPADSCMD with the following command:
  SUBSCRIBE DISTRIBUTION <distribution name>
- To list all of the available Magnet Monitoring parts and tools, send email to PPADSCMD with the following command:
  NEWS PARTS
- To get the current list of international toll free numbers (for EDMs to call PPADS), send email to PPADSCMD with the following command:
  NEWS PHONES
- To receive information on registering unusual sites (such as, MRT sites), send email to PPADSCMD with the following command:
  NEWS REGISTRATION
- To see site statistics, email server usage, the time of the last calls to the PPADS machines, the time and date that the PPADS machines were last started, the status of the PPADS machines, and statistics about processes and data storage on the PPADS machines, send email to PPADSCMD with the following command:
  REPORT LONG
- To see the number of alarms waiting to have dispatches cut, send email to PPADSCMD with the following command:
  COUNT ALARMQUEUE

G-6 Entering And Retrieving Team Contact Information

PPADS offers information about entering and retrieving team information. For a complete list of applicable commands, send email to PPADSCMD with the HELP LIST, HELP SHOW, HELP SET, HELP SUBSCRIBE, and HELP UNSUBSCRIBE commands.

These tasks let you either enter information into a Magnet Monitoring “address book” or check existing information in that address book. If you want to join or quit teams or view the contact information for a team member, you may want to do the following tasks:

- To see the exact names of the teams, send email to PPADSCMD with the following command: **LIST TEAMS**

- To put your contact information into PPADS and join one or more teams, send email to PPADSCMD with the following commands. (Note that you can set additional information; refer to the HELP SET command.)
  SET FIRSTNAME <your first name>
  SET LASTNAME <your last name>
  SET ADDRESS <your address>
  SET ASPEN <your 5-digit Aspen mailbox number>
  SET CPHONE <your cellular phone number>
  SET OPHONE <your office phone number>
  SUBSCRIBE TEAM <team name>

- To view the contact information of all the persons on a team, send email to PPADSCMD with the following command: **SHOW TEAM <team name>**

- To remove your contact information from a team list, send email to PPADSCMD with the following command: **UNSUBSCRIBE TEAM <team name>**

G-7 PPADSCMD Command Summary

Use PPADS commands to receive information about Magnet Monitoring. You must provide the information inside the angle brackets, <>, (for example, <*Dispatch#*>); you may optionally provide the information in the square brackets, [ ], (for example, HELP [ADMIN, BEGIN, END, LIST, NEWS, REPORT, SET, SHOW, SUBSCRIBE, or UNSUBSCRIBE]).

The following commands are used most frequently.

- ADMIN - Use to notify of problems or send requests to the PPADS administrator.
- BEGIN and END - Use to prevent PPADSCMD from trying to process embedded header or footer information. Always use these commands when sending mail from Microsoft Exchange Mail or from PROFS.
- COUNT - Use to see the number of number of dispatches at a site; the number of Magnet Monitoring sites per MAC Team Leader; or the number of alarms, checks, contacts, fills, or tests within or before a specified number of days.
- HELP - Use to see the most recent commands and general guidelines for using those commands. To see specific help on an individual command, send the HELP <command type> message to PPADSCMD. For example, the HELP SHOW command shows you how to use each of the SHOW commands and provides example responses for each command (such as, SHOW CONFIG, SHOW DB, SHOW EDM, SHOW MAGALARM, and SHOW MAGDATA).
- LIST - Use to list all the Magnet Monitoring sites; all the dispatches for a specific site; all the sites according to MAC Team Leader; the sites that have had alarms, checks, contacts, fills, or tests within or before a specified number of days; or the sites' last alarms, checks, contacts, fills, or tests either according to cryogen system ID or by date .
- NEWS - Use to see the most recent toll free numbers, the most recent part numbers, or the most recent software information.
- REPORT - Use to get statistics on both the Magnet Monitoring sites and the PPADS System.
- SET - Use to provide contact information prior to subscribing to a team.
- SHOW - Use to see a site's EDM and magnet configurations, contact information, status of a specific dispatch, EDM log, magnet alarm log, or magnet statistics.
- SUBSCRIBE and UNSUBSCRIBE - Use to place or remove your email address on a list that causes you to receive email when the site you "subscribed" to has an alarm or fill.

APPENDIX H – PACEMAKER THEORY

The Pacemaker is designed to monitor the status of the magnet heater. There are several situations that can cause the magnet heater to remain on. Two significant failure modes are:

- Magnet Monitor software is hung
- Hardware is causing heater to remain either on or off.

There are three major parts to the Pacemaker circuit:

- Magnet Monitor Software, revision 2.0
- Pacemaker box
- Remote Alarm

The Remote Alarm will activate when either the Magnet Monitor or Pacemaker determines that the magnet heater is being activated when it should not be.

The latest magnet monitor software (rev 2) now pulses the heater output. The Pacemaker box is looking for the pulse. If the Pacemaker does not see the pulse from the Magnet Monitor, it will turn off the heater and activate the remote alarm.

There are additional circuits in the Pacemaker that are in series with the relays for the magnet heater circuit and the helium sample circuit. These circuits greatly reduce the number of relay failures for the Magnet Monitor.

There are four LED's on the Pacemaker:

- **POWER IN** Indicates power to the Pacemaker. Should illuminate when the magnet heater is active.
- **HEAT IN** Should be lit when the Magnet Monitor is applying power to the magnet heater.
- **HEAT OUT** Should illuminate when the Magnet Monitor is applying heat to the magnet heater. If the HEATER IN is illuminated, and the HEATER OUT is not illuminated, the Pacemaker has manually turned the heater off
- **ALARM** Should illuminate when either the Pacemaker or Magnet Monitor determines that the magnet heater is activated when it should not be. The Remote Alarm Should activate then the ALARM LED is on.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Jan , 1999	P. Kargard	Initial Release
1	Nov, 1999	P. Kargard	Updates after initial release feedback
2	Feb, 2000	P. Kargard	Updates for changes made to software
3	July, 2000	P. Kargard	Minor modifications for software load procedure
4	Jan, 2001	P. Kargard	Updated user interface section
5	Oct, 2002	P. Kargard	Updates for Advantech magnet monitor and for revision 2 proprietary software
6	Jan 2003	P. Kargard	Added Alarm information, updated troubleshooting section
7	Oct 2003	P. Kargard	Info for 2.3 proprietary software